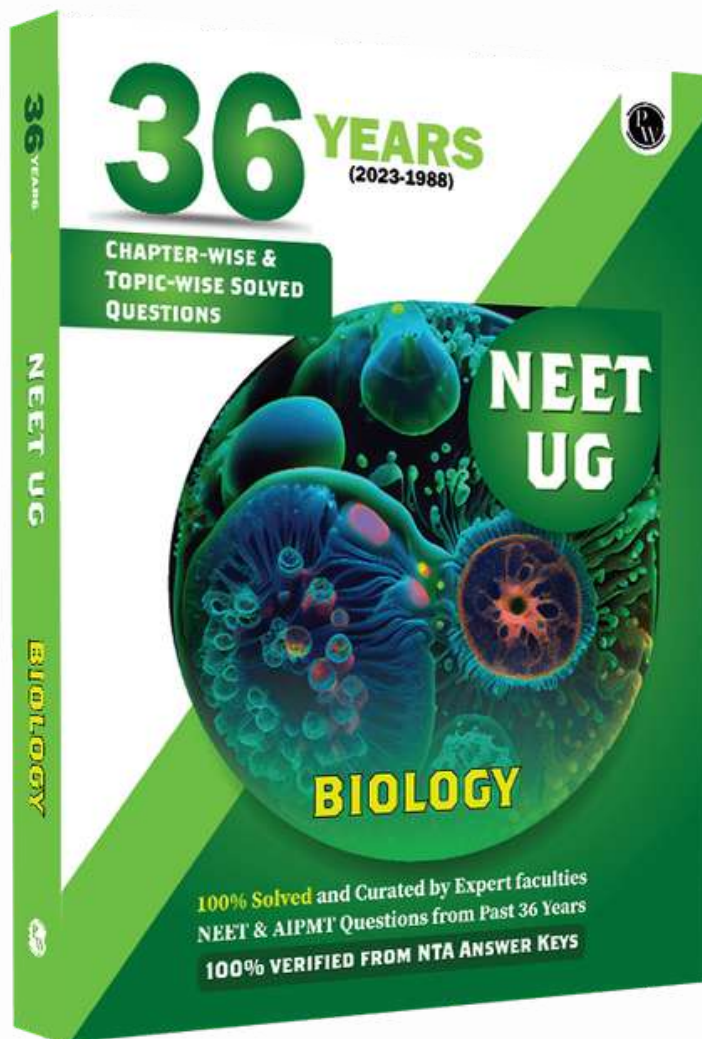




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**CHAPTER-WISE &
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The background features a large circular graphic. The left side shows a fiery, orange and yellow celestial body, possibly a sun or a nebula. The right side shows a blue and white Earth with magnetic field lines. The entire graphic is set against a dark, textured background with concentric circles.

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CHAPTER & TOPIC-WISE SOLUTIONS



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What Makes this Book your Prep Better than Others ?



Most Accurate Book - Through multiple proofreading this book gives you most correct answers with appropriate explanations.

There are a few MCQs that were asked previously, but the concepts or facts of those Questions are not covered under the current NEET syllabus so we have omitted those questions as solving them will be the waste of time.



SYLLABUS



A few NEET exams were held twice due to Covid, Paper Leakage, or other reasons. In order to provide more Relevant Questions for your Preparation, we have covered those exams' MCQs in this book.

Class - XI

Systems of Units

- The angle of 1' (minute of arc) in radian is nearly equal to, (2020-Covid)
 - 4.85×10^{-4} rad
 - 4.80×10^{-6} rad
 - 1.75×10^{-2} rad
 - 2.91×10^{-4} rad
- The unit of thermal conductivity is (2019)
 - $\text{W m}^{-1} \text{K}^{-1}$
 - J m K^{-1}
 - $\text{J m}^{-1} \text{K}^{-1}$
 - W m K^{-1}
- The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are: (2012 Pre)
 - kgms^{-1}
 - kg/ms
 - kgs^{-1}
 - kgs
- The density of a material in CGS system of units is 4 g/cm^3 . In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be: (2011 Mains)
 - 0.4
 - 40
 - 400
 - 0.04
- If $x = at + bt^2$, where x is the distance travelled by the body in kilometres while t is the time in seconds, then the units of b is: (1989)
 - km/s
 - kms
 - km/s^2
 - kms^2

Dimensions of Physical Quantities

- Plane angle and solid angle have: (2022)
 - Both units and dimension
 - Units but no dimensions
 - Dimensions but no units
 - No units and no dimensions
- The dimension $[\text{MLT}^{-2}\text{A}^{-2}]$ belong to the: (2022)
 - electric permittivity
 - magnetic flux
 - self inductance
 - magnetic permeability
- If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of: (2021)
 - $[\text{M}][\text{L}^{-1}][\text{T}^{-1}]$
 - $[\text{M}][\text{L}^0][\text{T}^0]$
 - $[\text{M}^2][\text{L}^{-2}][\text{T}^{-1}]$
 - $[\text{M}^2][\text{L}^{-1}][\text{T}^0]$
- Dimensions of stress are: (2020)
 - $[\text{ML}^2\text{T}^{-2}]$
 - $[\text{ML}^0\text{T}^{-2}]$
 - $[\text{ML}^{-1}\text{T}^{-2}]$
 - $[\text{MLT}^{-2}]$
- The dimensions of $(\mu_0 \epsilon_0)^{-1/2}$ are: (2012 Mains)
 - $[\text{L}^{1/2}\text{T}^{-1/2}]$
 - $[\text{L}^{-1}\text{T}]$
 - $[\text{LT}^{-1}]$
 - $[\text{L}^{1/2}\text{T}^{1/2}]$
- The dimension of $\frac{1}{2}\epsilon_0 E^2$, where ϵ_0 is permittivity of free space and E is electric field, is: (2010 Pre)
 - MLT^{-1}
 - ML^2T^{-2}
 - $\text{ML}^{-1}\text{T}^{-2}$
 - ML^2T^{-1}
- If the dimensions of a physical quantity are given by $\text{M}^a\text{L}^b\text{T}^c$, then the physical quantity will be: (2009)
 - Velocity if $a = 1, b = 0, c = -1$
 - Acceleration if $a = 1, b = 1, c = -2$
 - Force if $a = 0, b = -1, c = -2$
 - Pressure if $a = 1, b = -1, c = -2$
- Which two of the following five physical parameters have the same dimensions? (2008)
 - Energy density
 - Refractive index
 - Dielectric constant
 - Young's modulus
 - Magnetic field
 - (A) and (E)
 - (B) and (D)
 - (C) and (E)
 - (A) and (D)
- Dimensions of resistance in an electrical circuit, in terms of dimension of mass M , of length L , of time T and of current I , would be: (2007)
 - ML^2T^{-2}
 - $\text{ML}^2\text{T}^{-1}\text{I}^{-1}$
 - $\text{ML}^2\text{T}^{-3}\text{I}^{-2}$
 - $\text{ML}^2\text{T}^{-3}\text{I}^{-1}$
- The dimensions of universal gravitational constant are: (2004)
 - ML^2T^{-1}
 - $\text{M}^{-2}\text{L}^3\text{T}^{-2}$
 - $\text{M}^{-2}\text{L}^2\text{T}^{-1}$
 - $\text{M}^{-1}\text{L}^3\text{T}^{-2}$

16. The dimension of Planck constant equals to that of: (2001)
 a. Energy b. Momentum
 c. Angular momentum d. Power
17. Which pair have not equal dimensions? (2000)
 a. Energy and torque
 b. Force and impulse
 c. Angular momentum and Planck's constant
 d. Elastic modulus and pressure
18. The dimensions of impulse are equal to that of: (1996)
 a. Pressure b. Linear momentum
 c. Force d. Angular momentum
19. Which of the following dimensions will be the same as that of time? (1996)
 a. $\frac{L}{R}$ b. $\frac{C}{L}$
 c. LC d. $\frac{R}{L}$
20. Which of the following is a dimensional constant? (1995)
 a. Relative density b. Gravitational constant
 c. Refractive index d. Poisson ratio
21. The dimensions of RC is: (1995)
 a. Square of time b. Square of inverse time
 c. Time d. Inverse time
22. Which of the following has the dimensions of pressure? (1994, 90)
 a. $[MLT^{-2}]$ b. $[ML^{-1} T^{-2}]$
 c. $[ML^{-2} T^{-2}]$ d. $[M^{-1} L^{-1}]$
23. The dimensional formula of permeability of free space μ_0 is (1991)
 a. $[MLT^{-2} A^{-2}]$ b. $[M^0 L^{-1} T]$
 c. $[M^0 L T^{-1} A^2]$ d. None of these
24. According to Newton, the viscous force acting between liquid layers of area A and velocity gradient $\Delta v/\Delta Z$ is given by $F = -\eta A \frac{\Delta v}{\Delta Z}$, where η is constant called coefficient of viscosity. The dimensional formula of η is: (1990)
 a. $[ML^{-2} T^{-2}]$ b. $[M^0 L^0 T^0]$
 c. $[ML^2 T^2]$ d. $[ML^{-1} T^{-1}]$
25. Which of the following quantities, which one has dimensions different from the remaining three? (1989)
 a. Energy per unit volume
 b. Force per unit area
 c. Product of voltage and charge per unit volume
 d. Angular momentum
26. Dimensional formula of self inductance is: (1989)
 a. $[MLT^{-2} A^{-2}]$ b. $[ML^2 T^{-1} A^{-2}]$
 c. $[ML^2 T^{-2} A^{-2}]$ d. $[ML^2 T^2 A^{-1}]$
27. The dimensional formula of torque is: (1989)
 a. $[ML^2 T^{-2}]$ b. $[MLT^{-2}]$
 c. $[ML^{-1} T^{-2}]$ d. $[ML^{-2} T^{-2}]$
28. If C and R denote capacitance and resistance, the dimensional formula of CR is: (1988)
 a. $[M^0 L^0 T^1]$
 b. $[M^0 L^0 T^0]$
 c. $[M^0 L^0 T^{-1}]$
 d. Not expressible in terms of MLT
29. The dimensional formula of angular momentum is: (1988)
 a. $[ML^2 T^{-2}]$ b. $[ML^{-2} T^{-1}]$
 c. $[MLT^{-1}]$ d. $[ML^2 T^{-1}]$

Application of Dimensions

30. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy. (2021)
 a. $[F] [A] [T^2]$ b. $[F] [A] [T^{-1}]$
 c. $[F] [A^{-1}] [T]$ d. $[F] [A] [T]$
31. A physical quantity of the dimensions of length hat can be formed out of c, G and $\frac{e^2}{4\pi\epsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]: (2017-Delhi)
 a. $c^2 \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{\frac{1}{2}}$ b. $\frac{1}{c^2} \left[\frac{e^2}{G 4\pi\epsilon_0} \right]^{\frac{1}{2}}$
 c. $\frac{1}{c^2} G \frac{e^2}{4\pi\epsilon_0}$ d. $\frac{1}{c^2} \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{\frac{1}{2}}$
32. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length? (2016 - II)
 a. $\sqrt{\frac{hc}{G}}$ b. $\sqrt{\frac{Gc}{h^{3/2}}}$
 c. $\frac{\sqrt{hG}}{c^{3/2}}$ d. $\sqrt{\frac{hG}{c^{5/2}}}$
33. If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be: (2015)
 a. $[EV^{-1} T^{-2}]$ b. $[EV^{-2} T^{-2}]$
 c. $[E^{-2} V^{-1} T^{-3}]$ d. $[EV^{-2} T^{-1}]$
34. If dimension of critical velocity of liquid flowing through a tube are expressed as $v_c \propto [\eta^x \rho^y r^z]$ where η , ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by: (2015 - Re)
 a. 1, 1, 1 b. 1, -1, -1
 c. -1, -1, 1 d. -1, -1, -1
35. If Force (F), Velocity (V) and Time (T) are taken as fundamental units, then the dimensions of mass are: (2014)
 a. $[F V T^{-1}]$ b. $[F V T^{-2}]$
 c. $[F V^{-1} T^{-1}]$ d. $[F V^{-1} T]$

Errors

36. Dimensions of resistance in an electrical circuit, in terms of dimension of mass M , of length L , of time T and of current I , would be (2007)
- $[ML^2T^{-2}]$
 - $[ML^2T^{-1}I^{-1}]$
 - $[ML^2T^{-3}I^{-2}]$
 - $[ML^2T^{-3}I^{-1}]$
37. The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a , b and c are constants. The dimensions of a , b and c are respectively: (2006)
- $[LT^{-2}]$, $[L]$ and $[T]$
 - $[L]$, $[T]$ and $[LT^2]$
 - $[L^2T^2]$, $[LT]$ and $[L]$
 - $[L]$, $[LT]$ and $[T^2]$
38. The ratio of the dimensions of Planck's constant and that of the moment of inertia is the dimension of: (2005)
- Frequency
 - Velocity
 - Angular momentum
 - Time
39. An equation is given here $\left(P + \frac{a}{V^2}\right) = b \frac{\theta}{V}$ where P = Pressure, V = Volume and θ = Absolute temperature. If a and b are constants, the dimensions of a will be: (1996)
- $[ML^{-5}T^{-1}]$
 - $[ML^5T^{-1}]$
 - $[ML^5T^{-2}]$
 - $[ML^{-1}L^5T^2]$
40. Turpentine oil is flowing through a tube of length l and radius r . The pressure difference between the two ends of the tube is P . The viscosity of oil is given by $\eta = \frac{P(r^2 - x^2)}{4vl}$ where v is the velocity of oil at a distance x from the axis of the tube. The dimensions of η are: (1993)
- $[M^0L^0T^0]$
 - $[MLT^{-1}]$
 - $[ML^2T^{-2}]$
 - $[ML^{-1}T^{-1}]$
41. The time dependence of a physical quantity p is given by $p = p_0 \exp(-\alpha t^2)$, where α is constant and t is the time. The constant α : (1993)
- Is dimensionless
 - Has dimensions $[T^{-2}]$
 - Has dimensions $[T^2]$
 - Has dimensions of p
42. P represents radiation pressure, c represents speed of light and S represents radiation energy striking per unit area per sec. The non-zero integers x , y , z such that $P^x S^y c^z$ is dimensionless are: (1992)
- $x = 1, y = 1, z = 1$
 - $x = -1, y = 1, z = 1$
 - $x = 1, y = -1, z = 1$
 - $x = 1, y = -1, z = -1$
43. The frequency of vibration f of a mass m suspended from a spring of spring constant k is given by a relation $f = am^x k^y$, where a is a dimensionless constant. The values of x and y are: (1990)
- $x = \frac{1}{2}, y = \frac{1}{2}$
 - $x = -\frac{1}{2}, y = -\frac{1}{2}$
 - $x = \frac{1}{2}, y = -\frac{1}{2}$
 - $x = -\frac{1}{2}, y = \frac{1}{2}$
44. The intervals measured by a clock given the following readings: 1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s. What is the percentage relative error in the observations? (2020-Covid)
- 4%
 - 16%
 - 1.6%
 - 2%
45. In an experiment, the percentage of error occurred in the measurement of physical quantities A , B , C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement of X , where $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$ will be (2019)
- $\left(\frac{3}{13}\right)\%$
 - 16%
 - 10%
 - 10%
46. In an experiment four quantities a , b , c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. Quantity P is calculated as follows $P = \frac{a^3 b^2}{cd}$. % error in P is: (2013)
- 4%
 - 14%
 - 10%
 - 7%
47. A student measures the distance traversed in free fall of a body, initially at rest in a given time. He used this data to estimate g , the acceleration due to gravity. If the maximum percentage errors in measurement of the distance and the time are e_1 and e_2 respectively, the percentage error in the estimation of g is: (2010 Mains)
- $e_2 - e_1$
 - $e_1 + 2e_2$
 - $e_1 + e_2$
 - $e_1 - 2e_2$
48. If the error in the measurement of radius of a sphere is 2%, then the error in the determination of volume of the sphere will be: (2008)
- 2%
 - 4%
 - 6%
 - 8%
49. The error in measurement of radius of a sphere is 0.1% then error in its volume is: (1999)
- 0.3%
 - 0.4%
 - 0.5%
 - 0.6%
50. The density of a cube is measured by measuring its mass and length of its sides. If the maximum error in the measurement of mass and lengths are 3% and 2% respectively, the maximum error in the measurement of density would be: (1996)
- 12%
 - 14%
 - 7%
 - 9%
51. Percentage errors in the measurement of mass and speed are 2% and 3% respectively. The error in the estimate of kinetic energy obtained by measuring mass and speed will be: (1995)
- 8%
 - 2%
 - 12%
 - 10%
52. A certain body weighs 22.42 g and has a measured volume of 4.7 cc. The possible error in the measurement of mass and volume are 0.01 g and 0.1 cc. Then maximum error in the density will be: (1991)
- 22%
 - 2%
 - 0.2%
 - 0.02%

Significant Figures

53. The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is : (2022)
- 14×10^2
 - 138×10^1
 - 1382
 - 1382.5
54. Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$? (2020)
- 9.98 m
 - 9.980 m
 - 9.9 m
 - 9.9801 m

Measuring Instruments

55. A screw gauge gives the following readings when used to measure the diameter of a wire
Main scale reading: 0 mm
Circular scale reading: 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is:

- 0.026 cm
 - 0.26 cm
 - 0.052 cm
 - 0.52 cm
56. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale.
The pitch of the screw gauge is: (2020)
- 0.25 mm
 - 0.5 mm
 - 1.0 mm
 - 0.01 mm
57. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of -0.004 cm , the correct diameter of the ball is (2018)
- 0.053 cm
 - 0.525 cm
 - 0.521 cm
 - 0.529 cm

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	a	c	b	c	b	d	d	c	c	c	d	d	c	d	c	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	a	b	c	b	a	d	d	c	a	a	d	a	d	c	b	b
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
d	c	a	a	c	d	b	c	d	c	b	b	b	c	a	d	a
52	53	54	55	56	57											
b	a	a	c	b	d											

Explanations

1. (d) 1 minute of arc $= 1' = \left(\frac{1}{60}\right)^0 = \frac{1}{60} \times \frac{\pi}{180} \text{ radian}$
 $= 2.91 \times 10^{-4} \text{ radian}$

2. (a) $K = \frac{Qx}{A(T_1 - T_2)t}$, where Q is the amount of heat flow, x is the thickness of the slab, A is the area of cross-section, and t is the time taken.

$$K = \frac{\text{J m}}{\text{m}^2 \text{ K s}} = \frac{\text{J}}{\text{s}} \cdot \frac{\text{m}}{\text{m}^2} \cdot \frac{1}{\text{K}} = \text{W m}^{-1} \text{ K}^{-1}$$

3. (c) Let $F = kv$ (k : constant of proportionality)

$$[k] = \left[\frac{F}{v}\right] = \left[\frac{\text{MLT}^{-2}}{\text{LT}^{-1}}\right] = [\text{MT}^{-1}]$$

Hence units of k is kg s^{-1} .

4. (b) $\therefore n_1 u_1 = n_2 u_2$

Where, n_1 and n_2 are numerical values u_1 and u_2 are units.

$$4 \frac{\text{g}}{\text{cm}^3} = n_2 \frac{100 \text{ g}}{(10 \text{ cm})^3} \Rightarrow n_2 = 40$$

5. (c) $\therefore$ Unit of x = unit of bt^2

$$\therefore \text{Units of } b = \frac{x}{t^2} = \frac{\text{km}}{\text{s}^2} = \text{km/s}^2$$

6. (b) Plane angle and solid angle are dimensionless physical quantities but have units.

7. (d) $[\text{MLT}^{-2}\text{A}^{-2}]$ is the dimensional formula for Magnetic permeability

8. (d) For energy, the dimensional formula is $[\text{ML}^2\text{T}^{-2}]$

For gravitational constant, the dimensional formula is $[\text{M}^{-1}\text{L}^3\text{T}^{-2}]$

$$\therefore \frac{[E]}{[G]} = \frac{[\text{M}^1\text{L}^2\text{T}^{-2}]}{[\text{M}^{-1}\text{L}^3\text{T}^{-2}]} = [\text{M}^{1+1}\text{L}^{2-3}\text{T}^{-2+2}]$$

$$\Rightarrow \frac{[E]}{[G]} = [\text{M}^2\text{L}^{-1}\text{T}^0]$$

9. (c) Stress = $\frac{\text{Force}}{\text{Area}}$

$$\text{Dimension of stress} = \frac{[\text{M}^1\text{L}^1\text{T}^{-2}]}{[\text{L}^2]}$$

$$\text{Dimension of stress} = [\text{M}^1\text{L}^{-1}\text{T}^{-2}]$$

10. (c) Speed of light, $c = \frac{1}{\sqrt{\mu_0\epsilon_0}} \Rightarrow [\text{LT}^{-1}]$

11. (c) $\left[\frac{1}{2}\epsilon_0 E^2\right]$ = Energy density

$$\therefore \text{Dimension} = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{L}^3]} = [\text{ML}^{-1}\text{T}^{-2}]$$

12. (d) $[\text{Force}] = [\text{MLT}^{-2}]$; $a = 1, b = 1, c = -2$

$$[\text{Pressure}] = [\text{ML}^{-1}\text{T}^{-2}]; a = 1, b = -1, c = -2$$

$$[\text{Velocity}] = [\text{M}^0\text{LT}^{-1}]; a = 0, b = 1, c = -1$$

$$[\text{Acceleration}] = [\text{M}^0\text{LT}^{-2}], a = 0, b = 1, c = -2$$

13. (d)

$$\text{A. Energy density} = \left[\frac{\text{Energy}}{\text{Volume}}\right] = \left[\frac{\text{ML}^2\text{T}^{-2}}{\text{L}^3}\right]$$

$$\text{Energy density} = [\text{ML}^{-1}\text{T}^{-2}]$$

$$\text{B. [Refractive index]} = [\text{M}^0\text{L}^0\text{T}^0]$$

$$\text{C. [Dielectric constant]} = [\text{M}^0\text{L}^0\text{T}^0]$$

$$\text{D. Young's modulus} = \left[\frac{\text{Stress}}{\text{Strain}}\right] = [\text{ML}^{-1}\text{T}^{-2}]$$

$$\text{E. Magnetic field (B)} = \left[\frac{F}{qv}\right] = [\text{ML}^0\text{T}^{-2}\text{A}^{-1}]$$

So, [(A) and (D)] and [(B) and (C)] have same dimension

14. (c) We can find the resistance by $R = \frac{V}{I}$

$$R = \frac{W}{q \times I} \left[\begin{array}{l} \because W = \text{Work done} \\ q = \text{Charge} \end{array} \right]$$

$$R = \frac{W}{It \times I} (\because q = It)$$

$$[R] = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{I}^2\text{T}]} = [\text{ML}^2\text{T}^{-3}\text{I}^{-2}]$$

Note: In electrostatics, if you will remember the dimension of V (Potential difference) then, it will be easy to find the dimensions of other quantities.

$$[V] = [\text{ML}^2\text{T}^{-2}\text{I}^{-1}]$$

15. (d) $\therefore F = \frac{Gm_1m_2}{r^2}$

$$\therefore [G] = \left[\frac{Fr^2}{m^2}\right] = \frac{[\text{MLT}^{-2}\text{L}^2]}{[\text{M}^2]} = [\text{M}^{-1}\text{L}^3\text{T}^{-2}]$$

16. (c) Planck constant, $h = \frac{\text{Energy}}{\text{Frequency}}$

$$\therefore [h] = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{T}^{-1}]} = [\text{ML}^2\text{T}^{-1}]$$

Angular momentum = Moment of inertia $\times$ Angular velocity

$$L = I \omega$$

$$\therefore [L] = [\text{ML}^2][\text{T}^{-1}] = [\text{ML}^2\text{T}^{-1}]$$

Hence dimension of Plank's constant and angular momentum are same.

17. (b) Dimensions of force = $[\text{MLT}^{-2}]$

$$\text{Dimensions of impulse} = [\text{MLT}^{-1}]$$

So, dimension of force and impulse are not equal.

Whereas,

$$\text{Dimension of energy and torque} = [\text{ML}^2\text{T}^{-2}]$$

$$\text{Dimension of angular momentum and plank's constant} = [\text{ML}^2\text{T}^{-1}]$$

$$\text{Dimension of elastic molecules and pressure} = [\text{ML}^{-1}\text{T}^{-2}]$$

18. (b) Impulse = Force $\times$ Time

$$[\text{Impulse}] = [\text{Force}] \times [\text{Time}]$$

$$= [\text{MLT}^{-2}][\text{T}] = [\text{MLT}^{-1}]$$

$$\text{Linear momentum} = \text{Mass} \times \text{velocity}$$

$$[\text{Linear momentum}] = [\text{Mass}] \times [\text{velocity}]$$

$$= [\text{M}][\text{LT}^{-1}]$$

$$= [\text{MLT}^{-1}]$$

$$\text{So, } [\text{Impulse}] = [\text{Linear momentum}]$$

19. (a) L/R is known as the time constant of the LR series circuit, as its dimension is $[\text{T}]$.

20. (b) Relative density, refractive index and Poisson ratio all the three are ratios, therefore they are dimensionless constants. Gravitational constant has dimension so it is dimensional constant.

21. (c) Units of $RC = \text{ohm} \times \text{ohm}^{-1} \times \text{second} = \text{second}$.
Therefore dimensions of $RC = \text{Time}$.

22. (b) $\text{Pressure} = \frac{\text{Force}}{\text{Area}}$

Therefore dimensions of pressure $= \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

23. (a) We know that,

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow \mu_0 = \frac{F \cdot 2\pi d}{I_1 I_2 L}$$

$$\therefore [\mu_0] = \frac{[F][d]}{[I_1][I_2][L]}$$

$$[\mu_0] = \frac{[MLT^{-2}][L]}{[A][A][L]} = [MLT^{-2}A^{-2}]$$

24. (d) Dimensions of force $F = [MLT^{-2}]$

Dimensions of velocity gradient $\frac{\Delta v}{\Delta Z} = \frac{[LT^{-1}]}{[L]} = [T^{-1}]$

Dimensions of area $A = [L^2]$

Given $F = -\eta A \frac{\Delta v}{\Delta Z}$

Dimensional formula for coefficient of viscosity

$$[\eta] = \frac{[F]}{[A] \left[\frac{\Delta v}{\Delta Z} \right]} = \frac{[MLT^{-2}]}{[L^2][T^{-1}]} = [ML^{-1}T^{-1}]$$

25. (d) Dimensions of energy $E = [ML^2 T^{-2}]$

Dimensions of volume $V = [L^3]$

Dimensions of force $F = [MLT^{-2}]$

Dimensions of area $A = [L^2]$

Dimensions of voltage $V' = [ML^2 T^{-3} A^{-1}]$

Dimensions of charge $q = [AT]$

Dimensions of angular momentum $L = [ML^2 T^{-1}]$

$$\therefore \text{Dimensions of } \frac{E}{V} = \frac{[ML^2 T^{-2}]}{[L^3]} = [ML^{-1} T^{-2}]$$

Dimensions of $\frac{F}{A} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1} T^{-2}]$

Dimensions of $\frac{V'q}{V} = \frac{[ML^2 T^{-3} A^{-1}][AT]}{[L^3]} = [ML^{-1} T^{-2}]$

Dimensions of $\frac{Vq}{V} = \frac{[ML^2 T^{-3} A^{-1}][AT]}{[L^3]} = [ML^{-1} T^{-2}]$

Dimensions of angular momentum is $[ML^2 T^{-1}]$ while other three has dimensions $[ML^{-1} T^{-2}]$

26. (c) Induced $\text{emf} |\mathcal{E}| = L \frac{dI}{dt}$

47 L is the self inductance and $\frac{dI}{dt}$ is the rate of change of current.

$$\therefore \text{Dimensional formula of } L = \frac{[\mathcal{E}]}{\frac{dI}{dt}} = [\mathcal{E}] \cdot \frac{dt}{dI}$$

$$= \frac{W}{q} \cdot \frac{dt}{dI} = \frac{[ML^2 T^{-2}][T]}{[AT][A]} = [ML^2 T^{-2} A^{-2}]$$

27. (a) Torque (τ) = Force $\times$ distance

Dimensional formula for $\tau = [MLT^{-2}][L] = [ML^2 T^{-2}]$

28. (a) Capacitance $C = \frac{\text{Charge}}{\text{Potential difference}}$

$$[C] = \frac{[q]}{[V]}$$

Resistance $R = \frac{\text{Potential difference}}{\text{Current}}$

$$[R] = \frac{[V]}{[I]}$$

$$[CR] = [C][R] = \frac{[q]}{[I]} = \frac{[AT]}{[A]} = [T] = [M^0 L^0 T^1]$$

29. (d) Angular momentum (L)

= Moment of inertia (I) $\times$ Angular velocity (ω).

$\therefore$ Dimensional formula (L) = $[ML^2][T^{-1}]$

= $[ML^2 T^{-1}]$

30. (a) According to the question;

$E \propto F^a A^b T^c$

$$\Rightarrow [M^1 L^2 T^2] \propto [M^1 L^1 T^{-2}]^a [L^2]^{-b} [T]^c$$

$$\Rightarrow [M^1 L^2 T^2] \propto [M^a] [L^{a+b}] [T^{-2a-2b+c}]$$

Comparing both sides,

$$\Rightarrow a = 1 \Rightarrow a + b = 2$$

$$\Rightarrow 1 + b = 2 \Rightarrow b = 2 - 1 = 1 \Rightarrow b = 1 \Rightarrow -2a - 2b + c = -$$

$$\Rightarrow -2(1) - 2(1) + c = -2 \Rightarrow -2 - 2 + c = -2$$

$$\Rightarrow -4 + c = -2 \Rightarrow c = -2 + 4 \Rightarrow c = 2$$

$$E \propto [F][A][T^2]$$

31. (d) $L = [c]^a [G]^b \left[\frac{e^2}{4\pi\epsilon_0} \right]^c$

$$= [LT^{-1}]^a [M^{-1} L^3 T^{-2}]^b [ML^3 T^{-2}]^c$$

$$= L^{a+3b+3c} T^{-a-2b-2c} M^{-b+c}$$

$$a + 3b + 3c = 1; -a - 2b - 2c = 0; -b + c = 0$$

$$b = \frac{1}{2} \quad c = \frac{1}{2} \quad \downarrow$$

$$a = -2$$

$$L = c^{-2} G^{1/2} \left[\frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$$

$$L = \frac{1}{c^2} \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$$

32. (c) $\ell \propto h^x G^y c^z$

$$M^0 L^1 T^0 = (ML^2 T^{-1})^x (M^{-1} L^3 T^{-2})^y (LT^{-1})^z$$

$$= M^{x-y} L^{2x+3y+z} T^{-x-2y-z}$$

Equating:

$$\left. \begin{aligned} x - y &= 0 \\ 2x + 3y + z &= 1 \\ -x - 2y - z &= 0 \end{aligned} \right\} \Rightarrow x = \frac{1}{2}; y = \frac{1}{2}; z = -\frac{3}{2}$$

$$\Rightarrow \ell \propto \frac{\sqrt{hG}}{c^{3/2}}$$

33. (b) Surface tension (S.T.) $\propto [E]^a [V]^b [T]^c$

$$[\text{S.T.}] \propto [\text{ML}^2\text{T}^{-2}]^a [\text{LT}^{-1}]^b [\text{T}]^c$$

$$[\text{MT}^{-2}] \propto \text{M}^a \text{L}^{2a+b} \text{T}^{-2a-b+c}$$

On comparing both sides

$$2a + b = 0, -2a - b + c = -2$$

$$a = 1, b = -2, c = -2$$

we get

$$\text{S.T.} = \text{EV}^{-2} \text{T}^{-2}$$

34. (b) $v_c \propto [\eta^x \rho^y r^z]$

$$[\text{L}^1\text{T}^{-1}] \propto [\text{M}^1\text{L}^{-1}\text{T}^{-1}]^x [\text{M}^1\text{L}^{-3}]^y [\text{L}^1]^z$$

$$[\text{L}^1\text{T}^{-1}] \propto [\text{M}^{x+y}] [\text{L}^{-x-3y+z}] [\text{T}^{-x}]$$

Taking comparison on both size

$$x + y = 0, -x - 3y + z = 1, -x = -1$$

$$x = 1, y = -1, z = -1$$

35. (d) $[\text{Mass}] = \left[\frac{\text{Force}}{\text{Acceleration}} \right] = \left[\frac{\text{Force}}{\text{Velocity/time}} \right]$

$$= [\text{FV}^{-1}\text{T}]$$

36. (c) According to Ohm's law,

$$V = RI \text{ or } R = \frac{V}{I}$$

$$\text{Dimensions of } V = \frac{[W]}{[q]} = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{IT}]}$$

$$\therefore [R] = \frac{[\text{ML}^2\text{T}^{-2} / \text{IT}]}{[I]} = [\text{ML}^2\text{T}^{-3}\text{I}^{-2}]$$

37. (a) $v = at + \frac{b}{t+c}$

As we know that only similar quantities can be added or subtracted so $[c] = T$

By Principle of homogeneity

$$[v] = [at] \Rightarrow a = \frac{[v]}{[t]} = \text{LT}^{-2}$$

$$[b] = VT = L$$

Hence, dimensions of a, b and c are LT^{-2} , L and T respectively.

38. (a) $\frac{\text{Planck constant}}{I(\text{M.O.I})} = \frac{h}{I}$

$$= \frac{E\lambda}{c \times I} = \frac{[\text{ML}^2\text{T}^{-2}][\text{L}]}{[\text{LT}^{-1}][\text{ML}^2]}$$

$$= [\text{T}^{-1}] = \text{Frequency}$$

39. (c) Equation $\left(P + \frac{a}{V^2} \right) = b \frac{\theta}{V}$

Since $\frac{a}{V^2}$ is added to the pressure, therefore dimensions of

$\frac{a}{V^2}$ and pressure (P) will be the same. $\frac{a}{V^2} = P \Rightarrow \frac{a}{[\text{L}^3]^2} = [\text{ML}^{-1}\text{T}^{-2}]$

or $[a] = [\text{ML}^5\text{T}^{-2}]$.

40. (d) Dimensions of $P = [\text{ML}^{-1}\text{T}^{-2}]$

Dimensions of $r = [L]$

Dimensions of $v = [\text{LT}^{-1}]$

Dimensions of $l = [L]$

$$\therefore \text{Dimensions of } \eta = \frac{[P][r^2 - x^2]}{[4vl]} = \frac{[\text{ML}^{-1}\text{T}^{-2}][\text{L}^2]}{[\text{LT}^{-1}][L]}$$

$$= [\text{ML}^{-1}\text{T}^{-1}]$$

41. (b) Given $p = p_0 e^{-\alpha t^2}$

αt^2 is a dimensionless

$$\therefore [\alpha] = \left[\frac{1}{t^2} \right] = \frac{1}{[\text{T}^2]} = [\text{T}^{-2}]$$

42. (c) Let $k = P^x S^y c^z$

...(i)

k is a dimensionless

Dimensions of $k = [\text{M}^0\text{L}^0\text{T}^0]$

$\therefore$ Dimensions of P

$$= \frac{\text{Force}}{\text{Area}} = \frac{[\text{MLT}^{-2}]}{[\text{L}^2]} = [\text{ML}^{-1}\text{T}^{-2}]$$

Dimensions of

$$S = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{L}^2][\text{T}]} = [\text{MT}^{-3}]$$

Dimensions of $c = [\text{LT}^{-1}]$

Substituting these dimensions in Eq. (i), we get

$$[\text{M}^0\text{L}^0\text{T}^0] = [\text{ML}^{-1}\text{T}^{-2}]^x [\text{MT}^{-3}]^y [\text{LT}^{-1}]^z$$

Applying the principle of homogeneity of dimensions, we get

$$x + y = 0 \quad \dots(\text{ii})$$

$$-x + z = 0 \quad \dots(\text{iii})$$

$$-2x - 3y - z = 0 \quad \dots(\text{iv})$$

Solving Eqs. (ii), (iii) and (iv), we get

$$x = 1, y = -1, z = 1$$

43. (d) $f = am^x k^y$

...(i)

Dimensions of frequency $f = [\text{M}^0\text{L}^0\text{T}^{-1}]$

Dimensions of constant $a = [\text{M}^0\text{L}^0\text{T}^0]$

Dimensions of mass $m = [M]$

Dimensions of spring constant $k = [\text{MT}^{-2}]$

Putting these value in Eq. (i), we get

$$[\text{M}^0\text{L}^0\text{T}^{-1}] = [M]^x [\text{MT}^{-2}]^y$$

from principle of homogeneity of dimensions,

$$x + y = 0 \quad \dots(\text{ii})$$

$$-2y = -1 \quad \dots(\text{iii})$$

$$\text{Or } y = \frac{1}{2}, x = -\frac{1}{2}$$

44. (c) Mean of given observations

$$= \frac{1.25 + 1.24 + 1.27 + 1.21 + 1.28}{5} = 1.25 \text{ sec}$$

$$\text{Mean of errors} = \frac{0 + 0.01 + 0.02 + 0.04 + 0.03}{5} = \frac{0.1}{5}$$

$$\% \text{ Error} = \frac{0.1 \times 100}{5 \times 1.25} = 1.6\%$$

45. (b) $X = \frac{A^2 B^{\frac{1}{2}}}{C^{\frac{1}{3}} D^3}$

$$\begin{aligned} \% \text{ error, } \frac{\Delta X}{X} \times 100 &= 2 \frac{\Delta A}{A} \times 100 + \frac{1}{2} \frac{\Delta B}{B} \times 100 + \frac{1}{3} \frac{\Delta C}{C} \times 100 + 3 \frac{\Delta D}{D} \times 100 \\ &= 2\% + 1\% + 1\% + 12\% \\ &= 16\% \end{aligned}$$

46. (b) $P = \frac{a^3 b^2}{cd} \Rightarrow \frac{\Delta P}{P} = \pm \left(3 \frac{\Delta a}{a} + 2 \frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d} \right)$

$$= \pm (3 \times 1 + 2 \times 2 + 3 + 4) \Rightarrow \pm 14\%$$

47. (b) $\because h = \frac{1}{2} g t^2 \Rightarrow g = \frac{2h}{t^2}$

$$\text{So, } \frac{\Delta g}{g} \times 100 = \left(\frac{\Delta h}{h} + 2 \frac{\Delta t}{t} \right) \times 100 = e_1 + 2e_2$$

48. (c) Volume of sphere $= \frac{4}{3} \pi r^3$

$$\Delta r = 2\% \Rightarrow \frac{\Delta V}{V} = \frac{3 \Delta r}{r}$$

$$\Rightarrow \frac{\Delta V}{V} = 3 \times 2 = 6\%$$

Error in determination of volume of sphere is equal to 6%.

49. (a) $V = \frac{4}{3} \pi r^3; \frac{\Delta V}{V} = \frac{3 \Delta r}{r}$

$$\% \text{ change in volume} = 3 \times 0.1 = 0.3\%$$

50. (d) Maximum error in mass $\left(\frac{\Delta m}{m} \right) = 3\% = \frac{3}{100}$

$$\text{Maximum error in length } \left(\frac{\Delta l}{l} \right) = 2\% = \frac{2}{100}$$

Maximum error in the measurement of density,

$$\left(\frac{\Delta \rho}{\rho} \right) = \frac{\Delta m}{m} + \left(3 \times \frac{\Delta l}{l} \right) = \frac{3}{100} + \left(3 \times \frac{2}{100} \right) = \frac{3}{100} + \frac{6}{100} = \frac{9}{100}$$

$$= 9\%$$

51. (a) Percentage error in mass $= 2\% = \frac{2}{100}$

$$\text{Percentage error in speed} = 3\% = \frac{3}{100}$$

$$\therefore \text{Kinetic energy (K.E.)} = \frac{1}{2} m v^2$$

Therefore the error in measurement of kinetic energy

$$\frac{\Delta K.E.}{K.E.} = \frac{\Delta m}{m} + 2 \times \frac{\Delta v}{v} = \frac{2}{100} + 2 \times \frac{3}{100} = \frac{8}{100} = 8\%$$

52. (b) Density $(\rho) = \frac{\text{Mass (m)}}{\text{Volume (V)}}$

Take logarithm to take base e on the both sides of Eq. (i), we get

$$\ln \rho = \ln m - \ln V$$

Differentiate Eq. (ii), on both sides, we get

$$\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} - \frac{\Delta V}{V}$$

Errors are always added, Error in the density ρ will be

$$= \left[\frac{\Delta m}{m} + \frac{\Delta V}{V} \right] \times 100\% = \left[\frac{0.01}{22.42} + \frac{0.1}{4.7} \right] \times 100\% = 2\%$$

53. (a) Area = Length $\times$ Breadth

$$= 55.3 \times 25 = 1382.5$$

$$= 14 \times 10^2$$

Area should have 2 significant figures, as the minimum significant figure in the data given is two.

54. (a) In subtraction the number of decimal places in the result should be equal to the number of decimal places of that term in the operation which contain lesser number of decimal places.

$$\begin{array}{r} 9.99 \\ -0.0099 \\ \hline \end{array}$$

$$\begin{array}{r} 9.99 \\ -0.0099 \\ \hline 9.98 \end{array} \rightarrow 3 \text{ significant figures.}$$

55. (c) Pitch of the screw gauge, $P = 1 \text{ mm}$

Number of circular division, $n = 100$

$$\text{Least count, L.C.} = \frac{P}{n} = \frac{1}{100} = 0.01 \text{ mm}$$

$$\text{L.C.} = 0.001 \text{ cm}$$

Observation

MSR = 0 mm & CSR = 52 divisions

So, diameter of the wire,

$$D = \text{MSR} + (\text{CSR} \times \text{LC})$$

$$D = 0 + (52 \times 0.001)$$

$$D = 0.052 \text{ cm}$$

56. (b) Least count = $\frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$

$$\Rightarrow 0.01 \text{ mm} = \frac{\text{Pitch}}{50}$$

$$\Rightarrow \text{Pitch} = 0.5 \text{ mm.}$$

57. (d) Reading = MSR + $(n \times \text{LC})$ + Zero error

$$= 0.5 + (25 \times 0.001) + 0.004$$

$$= 0.529 \text{ cm}$$

Motion in a Straight Line

Distance, Displacement, Speed and Velocity

1. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1 . On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2 . The time taken by her to walk up on the moving escalator will be: (2017-Delhi)

a. $\frac{t_1 t_2}{t_2 - t_1}$ b. $\frac{t_1 t_2}{t_2 + t_1}$
c. $t_2 - t_1$ d. $\frac{t_1 + t_2}{2}$

2. Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $X_p(t) = at + bt^2$ and $X_q(t) = ft - t^2$. At what time do the cars have the same velocity? (2016 - I)

a. $\frac{a+f}{2(1+b)}$ b. $\frac{f-a}{2(1+b)}$
c. $\frac{a-f}{1+b}$ d. $\frac{a+f}{2(b-1)}$

3. If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is: (2016 - I)

a. $\frac{3}{2}A + 4B$ b. $3A + 7B$
c. $\frac{3}{2}A + \frac{7}{3}B$ d. $\frac{A}{2} + \frac{B}{3}$

4. A particle covers half of its total distance with speed v_1 and the rest half distance with speed v_2 . Its average speed during the complete journey is: (2011 Mains)

a. $\frac{v_1 v_2}{v_1 + v_2}$ b. $\frac{2v_1 v_2}{v_1 + v_2}$
c. $\frac{v_1^2 v_2^2}{v_1^2 + v_2^2}$ d. $\frac{v_1 v_2}{2}$

5. A car moves from X to Y with a uniform speed v_u and returns to Y with a uniform speed v_d . The average speed for this round trip is: (2007)

a. $\sqrt{v_u v_d}$
c. $\frac{v_u + v_d}{2}$

b. $\frac{v_d v_u}{v_d + v_u}$
d. $\frac{2v_d v_u}{v_d + v_u}$

6. A car runs at a constant speed on a circular track of radius 100 m, taking 62.8 s for every circular lap. The average velocity and average speed for each circular lap respectively is: (2006)

a. 0, 0 b. 0, 10 m/s
c. 10 m/s, 20 m/s d. 20 m/s, 0

7. A particle moves along a straight line OX. At a time t (in seconds) the distance x (in meters) of the particle from O is given by $x = 40 + 12t - t^3$. How long would the particle travel before coming to rest? (2006)

a. 14 m b. 16 m
c. 56 m d. 40 m

8. The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a , b , α and β are positive constants. The velocity of the particle will (2005)

a. Be independent of β
b. Drop to zero when $\alpha = \beta$
c. Go on decreasing with time
d. Go on increasing with time

9. For a particle displacement time relation is $t = \sqrt{x} + 3$. Its displacement when its velocity is zero: (1999)

a. 2 m b. 4 m
c. 0 d. None of these

10. A bus travelling the first one-third distance at a speed of 10 km/h, the next one-third at 20 km/h and at last one-third at 60 km/h. The average speed of the bus is: (1997)

a. 9 km/h b. 16 km/h
c. 18 km/h d. 48 km/h

11. A car moves a distance of 200 m. It covers the first half of the distance at speed 40 km/h and the second half of distance at speed v . The average speed is 48 km/h. The value of v is: (1991)

a. 56 km/h b. 60 km/h
c. 50 km/h d. 48 km/h

12. A car covers the first half of the distance between two places at 40 km/h and another half at 60 km/h. The average speed of the car is: (1990)

a. 40 km/h b. 48 km/h
c. 50 km/h d. 60 km/h

Acceleration

13. A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$ where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by: (2015)

a. $-2n\beta^2 x^{-4n-1}$ b. $-2\beta^2 x^{-2n+1}$
c. $-2n\beta^2 e^{-4n+1}$ d. $-2n\beta^2 x^{-2n-1}$

14. The motion of a particle along a straight line is described by equation : $x = 8 + 12t - t^3$ where x is in metre and t in second. The retardation of the particle when its velocity becomes zero, is: (2012 Pre)

a. 24 ms^{-2} b. Zero
c. 6 ms^{-2} d. 12 ms^{-2}

15. A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to: (2010 Pre)

a. (Velocity) $^{2/3}$ b. (Velocity) $^{3/2}$
c. (Distance) 2 d. (Distance) $^{-2}$

16. A particle starts its motion from rest under the action of a constant force. If the distance covered in first 10 seconds is S_1 and that covered in the first 20 seconds is S_2 , then: (2009)

a. $S_2 = 3S_1$ b. $S_2 = 4S_1$
c. $S_2 = S_1$ d. $S_2 = 2S_1$

17. The distance travelled by a particle starting from rest and moving with an acceleration $\frac{4}{3} \text{ ms}^{-2}$, in the third second is (2008)

a. $\frac{10}{3} \text{ m}$ b. $\frac{19}{3} \text{ m}$
c. 6 m d. 4 m

18. A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 ms^{-1} to 20 ms^{-1} while passing through a distance 135 m in to second. The value of t is (2008)

a. 12 b. 9
c. 10 d. 1.8

19. A particle moving along x -axis has acceleration f , at time t , given by, $f = f_0 \left(1 - \frac{t}{T}\right)$ where f_0 and T are constants. The particle at $t = 0$ has zero velocity. In the time interval between $t = 0$ and the instant when $f = 0$, the particle's velocity (v_x) is: (2007)

a. $\frac{1}{2} f_0 T^2$ b. $f_0 T^2$
c. $\frac{1}{2} f_0 T$ d. $f_0 T$

20. The position x of a particle with respect to time t along x -axis is given by $x = 9t^2 - t^3$ where x is in metres and t in second. What will be the position of this particle when it achieves maximum speed along the +ve x direction? (2007)

a. 54 m b. 81 m
c. 24 m d. 32 m

21. Motion of a particle is given by equation $S = (3t^3 + 7t^2 + 14t + 8)\text{m}$. The value of acceleration of the particle at $t = 1$ sec. is: (2000)

a. 10 m/s^2 b. 32 m/s^2
c. 23 m/s^2 d. 16 m/s^2

22. A particle starts from rest with constant acceleration. The ratio of average velocity to the time average velocity is: (1999)

a. $\frac{1}{2}$ b. $\frac{3}{4}$
c. $\frac{4}{3}$ d. $\frac{3}{2}$

23. If a car at rest accelerates uniformly to a speed of 144 km/h in 20 sec, it covers a distance of: (1997)

a. 1440 cm b. 2980 cm
c. 20 m d. 400 m

24. The position x of a particle varies with time, (t) as $x = at^2 - bt^3$. The acceleration will be zero at time t is equal to: (1997)

a. $\frac{a}{3b}$ b. Zero
c. $\frac{2a}{3b}$ d. $\frac{a}{b}$

25. The acceleration of a particle is increasing linearly with time t as bt . The particle starts from origin with an initial velocity v_0 . The distance travelled by the particle in time t will be: (1995)

a. $v_0 t + \frac{1}{3} bt^2$ b. $v_0 t + \frac{1}{2} bt^2$
c. $v_0 t + \frac{1}{6} bt^3$ d. $v_0 t + \frac{1}{3} bt^3$

26. The velocity of train increases uniformly from 20 km/h to 60 km/h in 4 hours. The distance travelled by the train during this period, is: (1994)

a. 160 km b. 180 km
c. 100 km d. 120 km

27. A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β and comes to rest. If total time elapsed is t , then maximum velocity acquired by car will be: (1994)

a. $\frac{(\alpha^2 - \beta^2)t}{\alpha\beta}$ b. $\frac{(\alpha^2 + \beta^2)t}{\alpha\beta}$
c. $\frac{(\alpha + \beta)t}{\alpha\beta}$ d. $\frac{\alpha\beta t}{\alpha + \beta}$

28. A particle moves along a straight line such that its displacement at any time t is given by $s = (t^3 - 6t^2 + 3t + 4)$ metres. The velocity when the acceleration is zero is: (1994)

a. 3 m/s
b. 42 m/s
c. -9 m/s
d. -15 m/s

29. A body starts from rest, what is the ratio of the distance travelled by the body during the 4th and 3rd second? (1993)

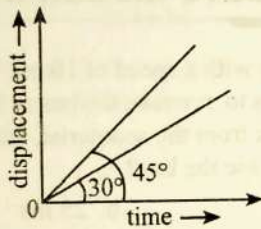
a. $\frac{7}{5}$
b. $\frac{5}{7}$
c. $\frac{7}{3}$
d. $\frac{3}{7}$

30. A car is moving along a straight road with a uniform acceleration. It passes through two points P and Q separated by a distance with velocity 30 km/h and 40 km/h respectively. The velocity of the car midway between P and Q is: (1988)

a. 33.3 km/h
b. $20\sqrt{2}$ km/h
c. $25\sqrt{2}$ km/h
d. 35 km/h

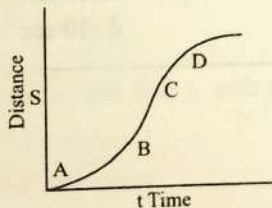
Graphs

31. The displacement time graphs of two moving particle make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is: (2022)



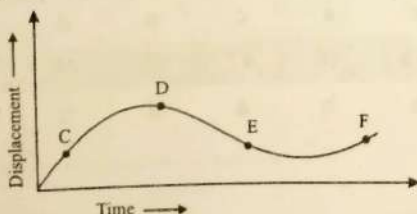
a. $1:\sqrt{3}$
b. $\sqrt{3}:1$
c. $1:1$
d. $1:2$

32. A particle shows distance-time curve as given in this figure. The maximum instantaneous velocity of the particle is around the point (2008)



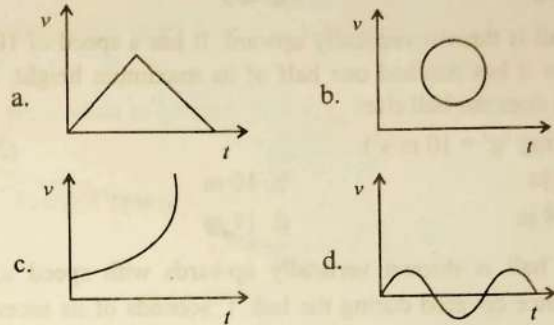
a. D
b. A
c. B
d. C

33. The displacement-time graph of a moving particle is shown below. The instantaneous velocity of the particle is negative at the point: (1994)



a. E
b. F
c. C
d. D

34. Which of the following curve does not represent motion in one dimension? (1992)



Motion under Gravity

35. The ratio of the distance traveled by a freely falling body in the 1st, 2nd, 3rd and 4th second is: (2022)

a. $1:1:1:1$
b. $1:2:3:4$
c. $1:4:9:16$
d. $1:3:5:7$

36. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: ($g = 10 \text{ m/s}^2$) (2020)

a. 340 m
b. 320 m
c. 300 m
d. 360 m

37. A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window? ($g = 10 \text{ m/s}^2$) (2020-Covid)

a. 14.5 m/s
b. 4.5 m/s
c. 20 m/s
d. 15.5 m/s

38. A stone falls freely under gravity. It covers distances h_1 , h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1 , h_2 and h_3 is: (2013)

a. $h_1 = h_2 = h_3$
b. $h_1 = 2h_2 = 3h_3$
c. $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$
d. $h_2 = 3h_1$ and $h_3 = 3h_2$

39. A boy standing at the top of a tower of 20 m height drops a stone. Assuming $g = 10 \text{ m/s}^2$, the velocity with which it hits the ground is: (2011 Pre)

a. 10.0 m/s
b. 20.0 m/s
c. 40.0 m/s
d. 5.0 m/s

40. A ball is dropped from a high rise platform at $t = 0$ starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v . The two balls meet at $t = 18$ s. What is the value of v ? (2010 Pre)

a. 75 m/s
b. 55 m/s
c. 40 m/s
d. 60 m/s

41. Two bodies, A (of mass 1 kg) and B (of mass 3 kg) are dropped from heights of 16 m and 25 m, respectively. The ratio of the time taken by them to reach the ground is: (2006)
- a. 5/4
b. 8/5
c. 5/8
d. 4/5
42. A ball is thrown vertically upward. It has a speed of 10 m/s when it has reached one half of its maximum height. How high does the ball rise? (Taking 'g' = 10 m/s²) (2005)
- a. 6 m
b. 10 m
c. 14 m
d. 18 m
43. If a ball is thrown vertically upwards with speed u, the distance covered during the last 't' seconds of its ascent is: (2003)
- a. ut
b. $\frac{1}{2}gt^2$
c. $ut - \frac{1}{2}gt^2$
d. (u + gt)t
44. A man throws ball with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given g = 9.8 m/s²) (2003)
- a. More than 19.6 m/s
b. At least 9.8 m/s
c. Any speed less than 19.6 m/s
d. Only with speed 19.6 m/s
45. A particle is thrown vertically upward. Its velocity at half of the height is 10 m/s, then the maximum height attained by it: (g = 10 m/s²) (2001)
- a. 8 m
b. 20 m
c. 10 m
d. 16 m
46. A body starts falling from height 'h' and travels distance h/2 during last second of motion then time of flight is (in second): (1999)
- a. $\sqrt{2} - 1$
b. $2 + \sqrt{2}$
c. $\sqrt{2} + \sqrt{2}$
d. $\sqrt{3} + 2$
47. A body dropped from a height h strikes the ground with a velocity 3 m/s. Another body of same mass dropped from the same height h with an initial velocity of 4 m/s. The final velocity of second mass, with which it strikes the ground is: (1996)
- a. 5 m/s
b. 12 m/s
c. 3 m/s
d. 4 m/s
48. The water drop falls at regular intervals from a tap 5 m above the ground. The third drop is leaving the tap at instant the first drop touches the ground. How far above the ground is the second drop at that instant? (1995)
- a. 3.75 m
b. 4.00 m
c. 1.25 m
d. 2.50 m
49. A body dropped from top of a tower fall through 40 m during the last two seconds of its fall. The height of tower is (g = 10 m/s²): (1992)
- a. 60 m
b. 45 m
c. 80 m
d. 50 m
50. What will be the ratio of the distance moved by a freely falling body from rest in 4th and 5th seconds of journey? (1989)
- a. 4 : 5
b. 7 : 9
c. 16 : 25
d. 1 : 1
- Relative Motion in 1-D**
51. A bus is moving with a speed of 10 ms⁻¹ on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus? (2009)
- a. 40 ms⁻¹
b. 25 ms⁻¹
c. 10 ms⁻¹
d. 20 ms⁻¹
52. A train of 150 metre length is going towards north direction at a speed of 10 m/s. A parrot flies at the speed of 5 m/s towards south direction parallel to the railways track. The time taken by the parrot to cross the train is: (1988)
- a. 12 sec
b. 8 sec
c. 15 sec
d. 10 sec

Relative Motion in 1-D

- 51.** A bus is moving with a speed of 10 ms^{-1} on a straight road. A scooterist wishes to overtake the bus in 100 s. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus? (2009)
- a. 40 ms^{-1} b. 25 ms^{-1}
c. 10 ms^{-1} d. 20 ms^{-1}
- 52.** A train of 150 metre length is going towards north direction at a speed of 10 m/s. A parrot flies at the speed of 5 m/s towards south direction parallel to the railways track. The time taken by the parrot to cross the train is: (1988)
- a. 12 sec b. 8 sec
c. 15 sec d. 10 sec

Answer Key

[illegible]

Explanations

1. (b) t_1 = time taken at stationary escalator

t_2 = time taken at moving escalator

t = time taken in walking up the moving escalator

d = displacement of escalator

V_1 = velocity of Preeti and V_2 = velocity of escalator

$$(V_1 + V_2) = \frac{d}{t}$$

$$V_1 = \frac{d}{t}, V_2 = \frac{d}{t_2}$$

$$\frac{d}{t_1} + \frac{d}{t_2} = \frac{d}{t}$$

$$t = \frac{t_1 t_2}{t_2 + t_1}$$

2. (b) $X_p(t) = at + bt^2$ $V_p = \frac{dX_p}{dt} = a + 2bt$

$$X_Q(t) = ft - t^2 \quad V_Q = \frac{dX_Q}{dt} = f - 2t$$

as $V_p = V_Q$

$$a + 2bt = f - 2t \Rightarrow t = \frac{f - a}{2(1 + b)}$$

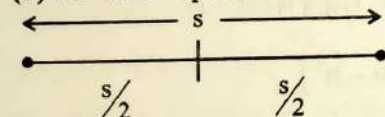
3. (c) We know, differentiation of displacement with respect to time gives velocity. $v = At + Bt^2$

$$\frac{dx}{dt} = At + Bt^2$$

$$\int_0^x dx = \int_0^t (At + Bt^2) dt$$

$$x = \frac{A}{2}(2^2 - 1^2) + \frac{B}{3}(2^3 - 1^3) = \frac{3A}{2} + \frac{7B}{3}$$

4. (b) distance = speed $\times$ time



time t_1 with speed v_1 and time t_2 with speed v_2

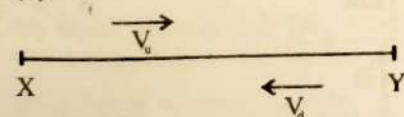
$$\text{Av. Speed} = \frac{\text{Total distance}}{\text{Total time taken}}$$

$$= \frac{s}{t_1 + t_2} = \frac{s}{\frac{s}{2v_1} + \frac{s}{2v_2}} = \frac{2v_1 v_2}{v_1 + v_2}$$

Speed V_1 and V_2 distance are not equal but time interval is equal (i.e., $t_1 = t_2$)

$$\text{So, Av. speed} = \frac{V_1 t_1 + V_2 t_2}{t_1 + t_2} = \frac{V_1 + V_2}{2}$$

5. (d)



Total distance = $d + d$

time taken to go from x to y, $t_1 = \frac{d}{V_u}$

time taken to go from y to x, $t_2 = \frac{d}{V_d}$

$$\begin{aligned} \text{Average speed} &= \frac{d + d}{\frac{d}{V_u} + \frac{d}{V_d}} \\ &= \frac{2V_u V_d}{V_d + V_u} \end{aligned}$$

6. (b) Average velocity = $\frac{\text{Total displacement}}{\text{Time taken}}$

For each circular lap $\Rightarrow \frac{0}{62.8} = 0$

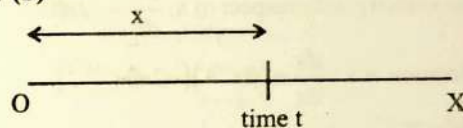
Average speed

$$= \frac{\text{Distance covered in 1 circular lap}}{\text{Time taken}}$$

$$\text{Distance} = 2\pi r = 2 \times 3.14 \times 100 = 628 \text{ m}$$

$$\text{Average speed} = \frac{628}{62.8} = 10 \text{ m/s}$$

7. (b)



x = distance from O, at time t

$$x = 40 + 12t - t^3$$

$$\text{Velocity} = \frac{dx}{dt}$$

$$= \frac{d(40 + 12t - t^3)}{dt} \Rightarrow (12 - 3t^2) \text{ m/s}$$

When particle will come to rest,

$$v = 0 \text{ now } 12 - 3t^2 = 0$$

$$\Rightarrow t = 2 \text{ s}$$

$$\text{Distance at } t = 2 \text{ sec} = 40 + 12 \times 2 - (2)^3$$

$$= 40 + 24 - 8 = 56 \text{ m}$$

$$\text{Dist. at } t = 0 = 40 + 12 \times 0 - (0)^3$$

$$= 40 \text{ m}$$

Distance travelled by particle before coming to rest,

$$56 - 40 = 16 \text{ m}$$

8. (d) Here, displacement of the particle, $x = ae^{-at} + be^{bt}$

$$v = \frac{dx}{dt} = -a e^{-at} + b e^{bt}$$

$$v = -a e^{-at} + b e^{bt}$$

As t increases e^{-at} decreases and v increases. So v increases with time.

9. (c) $t = \sqrt{x} + 3 \Rightarrow x = (t-3)^2$

$$v = \frac{dx}{dt} = 2(t-3) = 0$$

at $t = 3$, $x = (3-3)^2 = 0$

10. (c) Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$\Rightarrow V_{av} = \frac{x}{\frac{x}{3 \times 10} + \frac{x}{3 \times 20} + \frac{x}{3 \times 60}}$$

$$V_{av} = \frac{1}{\frac{1}{10} + \frac{1}{20} + \frac{1}{60}} = 18 \text{ km/h}$$

11. (b) $V_{avg} = \frac{2xy}{x+y}$

$$48 = \frac{2 \times 40 \times V}{40 + V} \Rightarrow V = 60 \text{ km/h}$$

12. (b) Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$V_{av} = \frac{x}{\frac{x}{2 \times 40} + \frac{x}{2 \times 60}} = \frac{1}{\frac{1}{80} + \frac{1}{120}} = \frac{120 \times 2}{5}$$

$$V_{as} = \frac{240}{5} = 48 \text{ km/h}$$

13. (a) Velocity = βx^{-2n}

Differentiate velocity with respect to x , $\frac{dv}{dx} = -2n\beta x^{-2n-1}$

Now Acceleration $a = v \frac{dv}{dx} = (\beta x^{-2n})(-2n\beta x^{-2n-1})$

$$\Rightarrow a = -2n\beta^2 x^{-4n-1}$$

14. (d) Velocity, $\bar{v} = \frac{dx}{dt} = 0 + 12 - 3t^2 = 0 \Rightarrow t = 2 \text{ s}$

At $t = 2 \text{ s}$

$$\text{Retardation} = -\frac{dv}{dt} = -(-6t) = 12 \text{ ms}^{-2}$$

15. (b) distance $x = \frac{1}{t+5} \Rightarrow v = \frac{dx}{dt} = -\frac{1}{(t+5)^2}$

Acceleration,

$$a = \frac{dv}{dt} = \frac{2}{(t+5)^3} \Rightarrow a \propto (\text{velocity})^{3/2}$$

16. (b) Since there is a constant force, acceleration will be constant.

Distance covered in first 10 seconds

$$(S_1) = \frac{1}{2} a (10)^2$$

Distance covered in first 20 seconds

$$(S_2) = \frac{1}{2} a (20)^2 \Rightarrow S_2 = 4S_1$$

17. (a) Distance travelled in the 3rd second

= Distance travelled in 3 s - distance travelled in 2 s.

As, $u = 0$

$$S(3^{\text{rd}} \text{ s}) = \frac{1}{2} a \cdot 3^2 - \frac{1}{2} a \cdot 2^2 = \frac{1}{2} a \cdot 5$$

Here $a = \frac{4}{3} \text{ ms}^{-2}$ and put the value of a in eq. (i)

$$\therefore S(3^{\text{rd}} \text{ s}) = \frac{1}{2} \times \frac{4}{3} \times 5 = \frac{10}{3} \text{ m}$$

18. (b) $v^2 - u^2 = 2as$

Given $v = 20 \text{ ms}^{-1}$, $u = 10 \text{ ms}^{-1}$, $s = 135 \text{ m}$

$$\therefore a = \frac{400 - 100}{2 \times 135} = \frac{300}{270} = \frac{10}{9} \text{ ms}^{-2}$$

$$v = u + at \Rightarrow t = \frac{v - u}{a} = \frac{10}{\frac{10}{9}} = 9 \text{ s}$$

19. (c) Given

Acceleration $(f) = f_0 \left(1 - \frac{t}{T}\right)$

[$\because f_0 = \text{constant}$]

$T = \text{constant}$

At $t = 0$, $v = 0$

When $f = 0$, $f_0 \left(1 - \frac{t}{T}\right) = 0 \Rightarrow t = T$

And velocity = v_x

$$f = \frac{dv}{dt} \Rightarrow dv = f dt$$

$$\Rightarrow \int_0^v dv = \int_0^T f_0 \left(1 - \frac{t}{T}\right) dt$$

$$\Rightarrow v = \int_0^T f_0 dt - \int_0^T \frac{f_0 t}{T} dt = [f_0 t]_0^T - \left[\frac{f_0 t^2}{2T}\right]_0^T$$

$$= f_0 T - \frac{f_0 T^2}{2} = \frac{1}{2} f_0 T$$

20. (a) Here, position of the particle: $x = 9t^2 - t^3$

when, it achieves maximum speed along the +x direction, the acceleration will become zero.

$$\Rightarrow x = 9t^2 - t^3$$

$$\Rightarrow \frac{dx}{dt} = v = \frac{d}{dt}(9t^2 - t^3) \Rightarrow v = 18t - 3t^2$$

$$\text{Now, } a = \frac{dv}{dt} = \frac{d}{dt}(18t - 3t^2) = 18 - 6t$$

Since, 'a' will be zero,

$$\Rightarrow 18 - 6t = 0 \Rightarrow t = 3 \text{ s}$$

Position of particle (x) = $(9 \times 3^2) - 3^3 = 54 \text{ m}$

21. (b) $v = \frac{ds}{dt} = \frac{d}{dt}(3t^3 + 7t^2 + 14t + 8)$

$$V = 9t^2 + 14t + 14$$

For acceleration we differentiate Velocity with respect to time

$$a = \frac{dv}{dt} = 18t + 14$$

At, $t = 1 \text{ sec} \Rightarrow a = 32 \text{ m/s}^2$

22. (c) $\langle v \rangle_{\text{time}} = \frac{\int_0^T v dt}{\int_0^T dt} = \frac{\int_0^T at dt}{\int_0^T dt} = \frac{aT}{2}$

$$\begin{aligned} \langle v \rangle_{\text{space}} &= \frac{\int v ds}{\int ds} = \frac{\int v \frac{ds}{dt} dt}{\int \frac{ds}{dt} dt} \\ &= \frac{\int_0^T v^2 dt}{\int_0^T v dt} = \frac{\int_0^T a^2 t^2 dt}{\int_0^T at dt} = \frac{2}{3} aT \\ \frac{\langle v \rangle_{\text{space}}}{\langle v \rangle_{\text{time}}} &= \frac{2aT/3}{aT/2} = \frac{4}{3} \end{aligned}$$

23. (d) Initial velocity $u = 0$,

Final velocity = $144 \text{ km/h} \times \frac{5}{18} = 40 \text{ m/s}$ and time = 20 sec.

Using $v = u + at = a \rightarrow v/t = 2 \text{ m/s}^2$

Again, $s = ut + \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times (20)^2 = 400 \text{ m}$

24. (a) Displacement $(x) = at^2 - bt^3$

Therefore velocity $(v) = \frac{dx}{dt} = \frac{d}{dt}(at^2 - bt^3)$
 $= 2at - 3bt^2$

Acceleration $a = \frac{dv}{dt} = \frac{d}{dt}(2at - 3bt^2) = 2a - 6bt = 0$ or

$$t = \frac{2a}{6b} = \frac{a}{3b}$$

25. (c) Acceleration $\propto bt$, i.e., $\frac{d^2x}{dt^2} = a \propto bt$

Integrating, $\frac{dx}{dt} = \frac{bt^2}{2} + C$

Initially, $t = 0$, $dx/dt = v_0$

Therefore, $\frac{dx}{dt} = \frac{bt^2}{2} + v_0$

Integrating again, $x = \frac{bt^3}{6} + v_0t + C$

When $t = 0$, $x = 0 \rightarrow C = 0$.

i.e., distance travelled by the particle in time t

$$= v_0t + \frac{bt^3}{6}$$

26. (a) Initial velocity $(u) = 20 \text{ km/h}$;

Final velocity $(v) = 60 \text{ km/h}$ and time $(t) = 4$ hours.

Velocity $(v) = 60 = u + at = 20 + (a \times 4)$

Or $a = \frac{\Delta v}{\Delta t} = \frac{60 - 20}{4} = 10 \text{ km/h}^2$.

Therefore distance travelled in 4 hours is

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 = (20 \times 4) + \frac{1}{2} \times 10 \times (4)^2 \\ &= 160 \text{ km.} \end{aligned}$$

27. (d) Initial velocity $(u) = 0$; Acceleration in the first phase = α ; Deceleration in the second phase = β and total time = t .

When car is accelerating then,

Final velocity $(v) = u + at = 0 + at_1$

Or $t_1 = \frac{v}{\alpha}$ and when car is decelerating, then final velocity
 $0 = v - \beta t_2$ or $t_2 = \frac{v}{\beta}$.

Therefore total time $(t) = t_1 + t_2 = \frac{v}{\alpha} + \frac{v}{\beta}$

$$t = v \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) = v \left(\frac{\beta + \alpha}{\alpha\beta} \right) \text{ or } v = \frac{\alpha\beta t}{\alpha + \beta}$$

28. (c) Displacement $(s) = t^3 - 6t^2 + 3t + 4$ metres.

Velocity $(v) = \frac{ds}{dt} = 3t^2 - 12t + 3$... (i)

Acceleration $(a) = \frac{dv}{dt} = 6t - 12$.

When $a = 0$, we get $t = 2$ seconds.

Put $t = 2$ in eq. (i) we get $V = -9 \text{ m/s}$

29. (a) Distance covered in n^{th} second is given by

$$s_n = u + \frac{a}{2}(2n - 1)$$

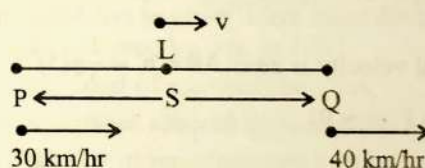
Since body from rest, so, $u = 0$

$$\therefore S_4 = 0 + \frac{a}{2}(2 \times 4 - 1) = \frac{7a}{2}$$

$$S_3 = 0 + \frac{a}{2}(2 \times 3 - 1) = \frac{5a}{2}$$

$$\therefore \frac{S_4}{S_3} = \frac{7}{5}$$

30. (c)



Let $PQ = s$ and L is the midpoint of PQ and v be velocity of the car at point L .

Using third equation of motion,

$$(40)^2 - (30)^2 = 2as$$

$$\text{Or } a = \frac{(40)^2 - (30)^2}{2s} = \frac{350}{s} \quad \dots (i)$$

$$\text{Also, } v^2 - (30)^2 = 2a \frac{s}{2}$$

$$\text{Or } v^2 - (30)^2 = 2 \times \frac{350}{s} \times \frac{s}{2} \Rightarrow v = 25\sqrt{2} \text{ km/hr}$$

31. (a) $\therefore$ Velocity is equal to slope of x - t graph

$$\Rightarrow v \propto \tan \theta$$

$$\frac{v_1}{v_2} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}}$$

32. (d) Since the slope is highest at C , $v = \frac{ds}{dt}$ is maximum.

33. (a) The velocity $(v) = \frac{ds}{dt}$.

The slope of the curve is -ve at point E

Therefore, instantaneous velocity at point E is negative.

34. (b) In one dimensional motion, the body can have only one value of velocity at a time.

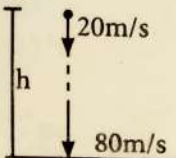
35. (d) distance travelled during nth second is given by

$$S_{nth} = u + \frac{a}{2}(2n-1)$$

$$= 0 + \frac{a}{2}(2n-1) \text{ [As the body is starting from rest, } \therefore u = 0]$$

$$S_{nth} = (2n-1) \Rightarrow S_{1st} : S_{2nd} : S_{3rd} : S_{4th} \\ = [2(1)-1] : [2(2)-1] : [2(3)-1] : [2(4)-1] = 1 : 3 : 5 : 7$$

36. (c)



v is the velocity of the ball before it strikes the earth and u in the initial velocity

$$v^2 = u^2 + 2gh$$

$$80^2 = 20^2 + 2 \times 10h$$

$$h = 300\text{m}$$

37. (a) From equation of motion

$$S = ut + \frac{1}{2}at^2$$

here $S = 1.5\text{ m}$ and $t = 0.1\text{ s}$

$$1.5 = u(0.1) + \frac{1}{2}(10)(0.1)(0.1) \Rightarrow u = 14.5\text{ m/s}$$

38. (c) Here initial velocity is zero $AB = h_1 = \frac{1}{2}g(5)^2$

$$\Rightarrow h_1 = 125\text{ m } (\because u = 0)$$

$$h_2 = BC = \frac{1}{2}g[10^2 - 5^2] \Rightarrow h_2 = 375\text{ m}$$

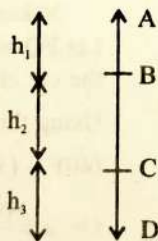
$$h_3 = CD = \frac{1}{2}g[15^2 - 10^2]$$

$$h_3 = 625\text{ m}$$

$$h_1 : h_2 : h_3$$

$$125 : 375 : 625 = 1 : 3 : 5$$

$$\Rightarrow h_1 = \frac{h_2}{3} = \frac{h_3}{5}$$



39. (b) Since, particle start from rest, $u = 0$

$$v^2 - u^2 = 2gh \Rightarrow v^2 - 0 = 2gh$$

$$v = \sqrt{2gh} \Rightarrow v = 20\text{ m/s}$$

40. (a) Let the two balls meet after t s at distance x from the platform.

For the first ball, $u = 0$, $t = 18\text{ s}$, $g = 10\text{ m/s}^2$

$$\text{Using } x = ut + \frac{1}{2}gt^2$$

$$\therefore x = \frac{1}{2} \times 10 \times 18^2$$

Initial velocity of the second ball = V and $t = 12\text{ s}$, $g = 10\text{ m/s}^2$

$$\therefore x = v \times 12 + \frac{1}{2} \times 10 \times 12^2$$

From equations (i) and (ii), we get

$$\frac{1}{2} \times 10 \times 18^2 = 12v + \frac{1}{2} \times 10 \times (12)^2$$

$$\text{or } 12v = \frac{1}{2} \times 10 \times [(18)^2 - (12)^2]$$

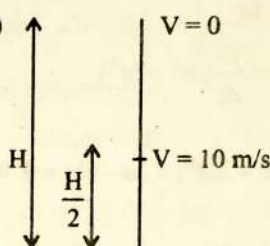
$$12v = \frac{1}{2} \times 10 \times 30 \times 6 \text{ or } v = \frac{1 \times 10 \times 30 \times 6}{2 \times 12} = 75\text{ m/s}$$

41. (d) If a body is dropped from height h , time taken to reach

the ground = $\sqrt{\frac{2h}{g}}$ and it is independent of mass.

$$\Rightarrow T \propto \sqrt{h} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{h_1}{h_2}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

42. (b)



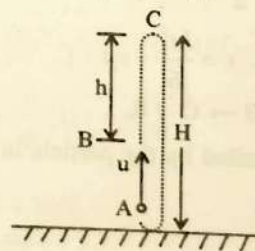
$$v^2 = u^2 + 2gH \Rightarrow 0 = 100 - 2g \times \frac{H}{2}$$

$$\Rightarrow 100 = gH \Rightarrow H = 10\text{m} \quad (\because g = 10\text{ m/s}^2)$$

43. (b) Let time of flight be T , then $T = \frac{u}{g}$

Let h be the distance covered during last ' t ' second

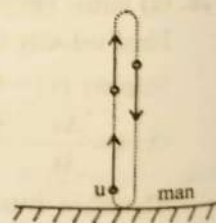
Velocity at point B = $V_B = u - g(T - t)$



$$V_B = u - g\left(\frac{u}{g} - t\right) = gt$$

$$h = v_B t - \frac{1}{2}gt^2$$

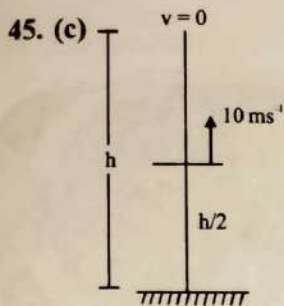
$$\Rightarrow h = gt^2 - \frac{1}{2}gt^2 = \frac{1}{2}gt^2$$



44. (a) In this case time of flight of a ball $\geq 2 \times 2 = 4\text{ sec}$
 $\therefore$ Time of flight

$$= \frac{2u}{g} \geq 4 \Rightarrow u \geq 2g$$

$$\Rightarrow u \geq 19.6\text{ m/s } (\because g = 9.8\text{ m/s}^2)$$



According to third law of motion:

$$v^2 = u^2 - 2g \frac{h}{2}$$

$$0 = (10)^2 - 10h \Rightarrow h = 10 \text{ m}$$

46. (b) Here, $u = 0$, so $h = \frac{1}{2}gt^2$

$$\frac{h}{2} = \frac{1}{2}g(t-1)^2 \quad \dots(i)$$

Substituting the value of h in Eq. (i), we get

$$\frac{1}{4}gt^2 = \frac{1}{2}g(t-1)^2$$

$$\frac{t}{\sqrt{2}} = t-1 \Rightarrow t\left(1 - \frac{1}{\sqrt{2}}\right) = 1$$

$$t = \frac{\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = \sqrt{2}(\sqrt{2}+1) = 2 + \sqrt{2}$$

47. (a) Initial velocity of first body (u_1) = 0;

Final velocity (v_1) = 3 m/s and initial velocity of second body (u_2) = 4 m/s.

$$\text{Height } (h) = \frac{v_1^2}{2g} = \frac{(3)^2}{2 \times 9.8} = 0.46 \text{ m}$$

Therefore velocity of the second body,

$$v^2 = \sqrt{u_2^2 + 2gh} = \sqrt{(4)^2 + 2 \times 9.8 \times 0.46} = 5 \text{ m/s.}$$

48. (a) Height of tap = 5 m. For the first drop,

$$5 = ut + \frac{1}{2}gt^2 = \frac{1}{2} \times 10t^2 = 5t^2 \text{ or } t^2 = 1$$

or $t = 1$ sec.

It means that the third drop leaves after one second of the first drop, or each drop leaves after every 0.5 sec. Distance covered by the second drop in 0.5 sec

$$= \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (0.5)^2 = 1.25 \text{ m}$$

Therefore distance of the second drop above the ground = $5 - 1.25 = 3.75$ m.

49. (b) Let h be height of the tower and t is the time taken by the body to reach the ground.

Here, $u = 0$, $a = g$

$$\text{Or } h = \frac{1}{2}gt^2 \quad \dots(i)$$

Distance covered in last two second is

$$40 = \frac{1}{2}gt^2 - \frac{1}{2}g(t-2)^2 \quad (\text{Here, } u = 0)$$

$$\frac{1}{2}gt^2 - \frac{1}{2}g(t-2)^2 = 40$$

$$5t^2 - 5(t-2)^2 = 40$$

$$t^2 - (t-2)^2 = 8 \Rightarrow (2t-2) \times 2 = 8$$

$$2t = 6 \Rightarrow t = 3 \text{ sec}$$

From Eq. (i), we get $h = \frac{1}{2} \times 10 \times (3)^2$ or $h = 45$ m.

50. (b) Distance covered in n^{th} second is given by

$$s_n = u + \frac{a}{2}(2n-1)$$

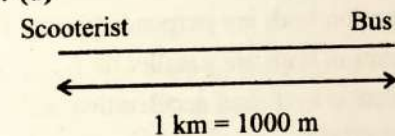
Since body is dropped from rest so, $u = 0$, and $a = g$

$$\therefore s_4 = \frac{g}{2}(2 \times 4 - 1) = \frac{7g}{2}$$

$$s_5 = \frac{g}{2}(2 \times 5 - 1) = \frac{9g}{2}$$

$$\therefore \frac{s_4}{s_5} = \frac{7}{9}$$

51. (d)



Speed of bus = 10 ms^{-1}

Scooterist has to cover 1 km more distance with respect to distance covered by bus, in 100 s

Let the speed of scooterist be $x \text{ ms}^{-1}$

Then, distance covered by scooterist

= 1000 m + distance covered by bus

$$\Rightarrow x \times 100 = 1000 + 10 \times 100$$

$$\Rightarrow 100x = 2000 \Rightarrow x = 20 \text{ ms}^{-1}$$

Hence, the speed of scooterist is 20 ms^{-1}

52. (d) Velocity of train $v_T = +10 \text{ ms}^{-1}$

Velocity of parrot $v_p = -5 \text{ ms}^{-1}$

Both are travelling along the straight line.

Relative velocity of parrot with respect to train

$$= v_p - v_T = (-5 \text{ ms}^{-1}) - (+10 \text{ ms}^{-1}) = -15 \text{ ms}^{-1}$$

i.e., parrot appears to move with a speed of 15 ms^{-1} from North to South.

$\therefore$ Time taken by parrot to cross the train

$$= \frac{150 \text{ m}}{15 \text{ ms}^{-1}} = 10 \text{ s}$$

Vectors

1. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is: (2016 - I)

a. 1° b. 90°
c. 45° d. 180°

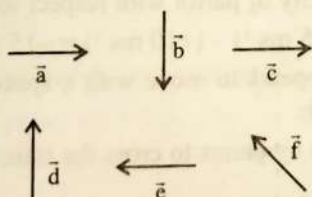
2. A particle moves so that its position vector is given by $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ where ω is a constant. Which of the following is true? (2016 - I)

a. Velocity and acceleration both are perpendicular to $\vec{r}$
b. Velocity and acceleration both are parallel to $\vec{r}$
c. Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin
d. Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin

3. If vectors $\vec{A} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ and $\vec{B} = \cos \frac{\omega t}{2} \hat{i} + \sin \frac{\omega t}{2} \hat{j}$ are functions of time, then the value of t at which they are orthogonal to each other are: (2015 Re)

a. $t = 0$ b. $t = \frac{\pi}{4\omega}$
c. $t = \frac{\pi}{2\omega}$ d. $t = \frac{\pi}{\omega}$

4. Six vectors, $\vec{a}$ to $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements are true? (2010 Pre)



a. $\vec{b} + \vec{e} = \vec{f}$ b. $\vec{b} + \vec{c} = \vec{f}$
c. $\vec{d} + \vec{e} = \vec{f}$ d. $\vec{d} + \vec{c} = \vec{f}$

5. The vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$. The angle between the two vectors is (2006)

a. 45° b. 90°
c. 60° d. 75°

6. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$ then the value of α is (2005)

a. $1/2$ b. $-1/2$
c. 1 d. -1

7. If the angle between $\vec{A}$ and $\vec{B}$ is θ , the value of the product $(\vec{B} \times \vec{A}) \cdot \vec{A}$ is equal to: (2005)

a. $BA^2 \sin \theta$ b. $BA^2 \cos \theta$
c. $BA^2 \sin \theta \cos \theta$ d. Zero

8. If $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$ then the value of $|\vec{A} + \vec{B}|$ is: (2004)

a. $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$ b. $A + B$
c. $\left(A^2 + B^2 + \sqrt{3}AB\right)^{1/2}$ d. $(A^2 + B^2 + AB)^{1/2}$

9. The vector sum of two forces is perpendicular to their vector differences. In that case, the forces: (2003)

a. Are equal to each other
b. Are equal to each other in magnitude
c. Are not equal to each other in magnitude
d. Cannot be predicted

10. If $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$ then angle between A and B will be: (2001)

a. 90° b. 120°
c. 0° d. 60°

11. Identify the vector quantity among the following: (1997)

a. Distance b. Angular momentum
c. Heat d. Energy

12. The position vector of a particle is $\vec{r} = (a \cos \omega t) \hat{i} + (a \sin \omega t) \hat{j}$. The velocity of the particle is: (1995)

a. Directed towards the origin
b. Directed away from the origin
c. Parallel to the position vector
d. Perpendicular to the position vector.

13. The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be: (1994)

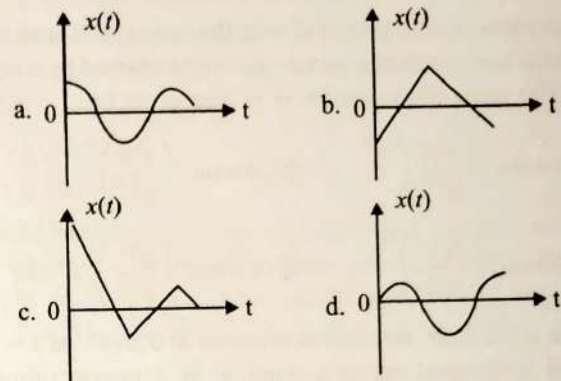
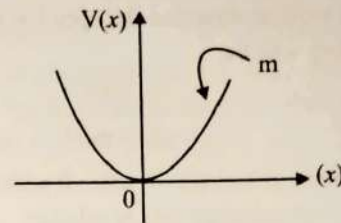
a. 90° b. 180°
c. Zero d. 45°

14. The resultant of $\vec{A} \times \vec{0}$ will be equal to: (1992)
 a. Zero b. $\vec{A}$
 c. Zero vector d. Unit vector.
15. The magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively. If $\vec{A} + \vec{B} = \vec{C}$, the angle between $\vec{A}$ and $\vec{B}$ is: (1988)
 a. $\pi/2$ b. $\cos^{-1}(0.6)$
 c. $\tan^{-1}(7/5)$ d. $\pi/4$

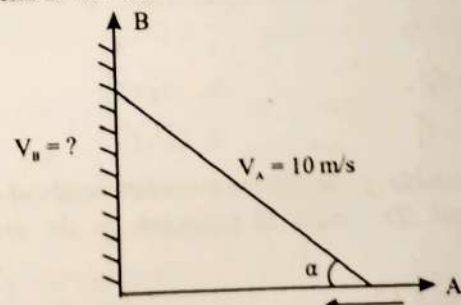
Motion in a Plane with Contant Acceleration

16. The 'x' and 'y' coordinates of the particle at any time are ' $x = 5t - 2t^2$ ' and ' $y = 10t$ ', respectively, where 'x' and 'y' are in metres and 't' in seconds. The acceleration of the particle at $t = 2$ s is: (2017-Delhi)
 a. 5 m/s^2 b. -4 m/s^2
 c. -8 m/s^2 d. 0
17. The position vector of a particle $\vec{R}$ as a function of time is given by:
 $\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$
 Where R is in metres, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x and y-direction, respectively. Which one of the following statements is wrong for the motion of particle? (2015)
 a. Path of the particle is a circle of radius 4 metre
 b. Acceleration vectors is along $-\vec{R}$
 c. Magnitude of acceleration vector is $\frac{v^2}{R}$ where v is the velocity of particle.
 d. Magnitude of the velocity of particle is 8 metre/second
18. A particle is moving such that its position coordinates (x, y) are:
 (2 m, 3 m) at time $t = 0$,
 (6 m, 7 m) at time $t = 2$ s and
 (13 m, 14 m) at time $t = 5$ s
 Average velocity vector ($\vec{v}_a$) from $t = 0$ to $t = 5$ s is: (2014)
 a. $\frac{1}{5}(13\hat{i} + 14\hat{j})$ b. $\frac{7}{3}(\hat{i} + \hat{j})$
 c. $2(\hat{i} + \hat{j})$ d. $\frac{11}{5}(\hat{i} + \hat{j})$
19. A particle has initial velocity $(2\hat{i} + 3\hat{j})$ and acceleration $(0.3\hat{i} + 0.2\hat{j})$. The magnitude of velocity after 10 sec will be: (2012 Pre)
 a. $9\sqrt{2}$ units b. $5\sqrt{2}$ units
 c. 5 units d. 9 units
20. A body is moving with velocity 30 m/s towards east. After 10 seconds its velocity becomes 40 m/s towards north. The average acceleration of the body is: (2011 Pre)
 a. 1 m/s^2 b. 7 m/s^2
 c. 7 m/s^2 d. 5 m/s^2

21. A particle of mass m is released from rest and follows a parabolic path as shown. Assuming that the displacement of the mass from the origin is small, which graph correctly depicts the position of the particle as a function of time? (2011 Pre)



22. A particle has initial velocity $3\hat{i} + 4\hat{j}$ and has acceleration $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is: (2010 Pre)
 a. 10 units b. 7 units
 c. $7\sqrt{2}$ units d. 8.5 units
23. A particle starting from the origin (0, 0) moves in a straight line in the (x, y) plane. Its coordinates at a later time are $(\sqrt{3}, 3)$. The path of the particle makes with the x-axis an angle of: (2007)
 a. 45° b. 60°
 c. 0° d. 30°
24. Two boys are standing at the ends A and B of a ground, where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1 . The boy at A starts running simultaneously with velocity v and catches the other boy in a time t, where t is: (2005)
 a. $\frac{a}{\sqrt{(v^2 + v_1^2)}}$ b. $\frac{a}{\sqrt{(v^2 - v_1^2)}}$
 c. $a/(v - v_1)$ d. $a/(v + v_1)$
25. A rigid rod is placed against the wall as shown in figure. When its velocity of lower end is 10 ms^{-1} and its base makes an angle $\alpha = 60^\circ$ with horizontal, then the vertical velocity of its end B will be: (1998)



- a. $10\sqrt{3}$ b. $10/\sqrt{3}$
 c. $5\sqrt{3}$ d. $5/\sqrt{3}$

Projectile Motion

26. A ball is projected with a velocity, 10 ms^{-1} , at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be: (2022)

a. 10 ms^{-1} b. Zero
c. $5\sqrt{3} \text{ ms}^{-1}$ d. 5 ms^{-1}

27. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle ' θ ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ , is then given by: (2021)

a. $\theta = \cos^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$ b. $\theta = \sin^{-1} \left(\frac{\pi^2 R}{gT^2} \right)^{1/2}$
c. $\theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R} \right)^{1/2}$ d. $\theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{1/2}$

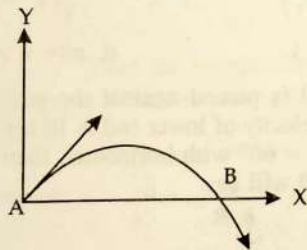
28. A car starts from rest and accelerates at 5 m/s^2 . At $t = 4 \text{ s}$, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6 \text{ s}$? (2021)

a. $20 \text{ m/s}, 0$ b. $20\sqrt{2} \text{ m/s}, 0$
c. $20\sqrt{2} \text{ m/s}, 10 \text{ m/s}^2$ d. $20 \text{ m/s}, 5 \text{ m/s}^2$

29. A projectile is fired from the surface of the earth with a velocity of 5 ms^{-1} and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 ms^{-1} at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in ms^{-2}) is: (given $g = 9.8 \text{ ms}^{-2}$) (2014)

a. 3.5 b. 5.9
c. 16.3 d. 110.8

30. The velocity of a projectile at the initial point A ($2\hat{i} + 3\hat{j}$) m/s. Its velocity (in m/s) at point B is: (2013)



a. $2\hat{i} + 3\hat{j}$ b. $-2\hat{i} - 3\hat{j}$
c. $-2\hat{i} + 3\hat{j}$ d. $2\hat{i} - 3\hat{j}$

31. The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is: (2012 Pre)

a. $\theta = \tan^{-1} \left(\frac{1}{4} \right)$ b. $\theta = \tan^{-1} (4)$
c. $\theta = \tan^{-1} (2)$ d. $\theta = 45^\circ$

32. A missile is fired for maximum range with an initial velocity of 20 m/s . If $g = 10 \text{ m/s}^2$, the range of the missile is: (2011 Pre)

a. 40 m b. 50 m
c. 60 m d. 20 m

33. A projectile is fired at an angle of 45° with the horizontal. Elevation angle of the projectile at its highest point as seen from the point of projection, is: (2011 Mains)

a. 60° b. $\tan^{-1} \frac{1}{2}$
c. $\tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$ d. 45°

34. The speed of a projectile at its maximum height is half of its initial speed. The angle of projection is: (2010 Mains)

a. 60° b. 15°
c. 30° d. 45°

35. A particle of mass m is projected with velocity v making an angle of 45° with the horizontal. When the particle lands on the level ground the magnitude of the change in its momentum will be (2008)

a. $\sqrt{2}mv$ b. Zero
c. $2mv$ d. $\frac{mv}{\sqrt{2}}$

36. For angles of projection of a projectile at angles $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of: (2006)

a. 1 : 1 b. 2 : 3
c. 1 : 2 d. 3 : 2

37. A particle (A) is dropped from a height and another particle (B) is projected in horizontal direction with speed of 5 m/s from the same height, then correct statement is: (2002)

a. Particle (A) will reach at ground first with respect to particle (B)
b. Particle (B) will reach at ground first with respect to particle (A)
c. Both particles will reach at ground simultaneously
d. Both particles will reach at ground with same speed

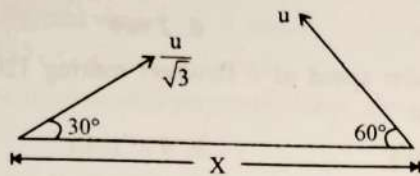
38. A particle is projected making angle 45° with horizontal having kinetic energy K . The kinetic energy at highest point will be: (2001)

a. $\frac{K}{\sqrt{2}}$ b. $\frac{K}{2}$
c. $2K$ d. K

39. Two projectiles of same mass and with same velocity are thrown at an angle 60° and 30° with the horizontal, then which quantity will remain same: (2000)

a. Time of flight
b. Horizontal range of projectile
c. Max height acquired
d. All of them

40. Two particles separated at a horizontal distance x as shown in fig. they projected at the same line as shown in fig. with different initial speeds. The time after which the Horizontal distance between them become zero: (1999)



- a. $\frac{x}{u}$
b. $\frac{u}{2x}$
c. $\frac{2u}{x}$
d. None of these
41. Two particles are projected with same initial velocity one makes angle θ with horizontal while other makes an angle θ with vertical. If their common range is R then product of their time of flight is directly proportional to: (1999)
- a. R
b. R^2
c. $\frac{1}{R}$
d. R^0
42. If a body A of mass M is thrown with velocity v at an angle of 30° to the horizontal and another body B of the same mass is thrown with the same speed at an angle of 60° to the horizontal, the ratio of horizontal range of A to B will be: (1992, 90)
- a. 1:3
b. 1:1
c. $1:\sqrt{3}$
d. $\sqrt{3}:1$
43. The maximum range of a gun of horizontal terrain is 16 km. If $g = 10 \text{ ms}^{-2}$, then muzzle velocity of a shell must be: (1990)
- a. 160 ms^{-1}
b. $200\sqrt{2} \text{ ms}^{-1}$
c. 400 ms^{-1}
d. 800 ms^{-1}

Relative Velocity in 2-D

44. The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by: (2019)
- a. 30° west
b. 0°
c. 60° west
d. 45° west
45. A ship A is moving Westwards with a speed of 10 km/h and a ship B 100 km South of A , is moving Northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest, is: (2015)
- a. 5 h
b. $5\sqrt{2}$ h
c. $10\sqrt{2}$ h
d. 0 h

46. The width of river is 1 km. The velocity of boat is 5 km/hr. The boat covered the width of river with shortest possible path in 15 min. Then the velocity of river stream is: (2000)
- a. 3 km/hr
b. 4 km/hr
c. $\sqrt{29}$ km/hr
d. $\sqrt{41}$ km/hr
47. The speed of a boat is 5 km/hr in still water. It crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of river water is: (1998)
- a. 3 km/hr
b. 4 km/hr
c. 5 km/hr
d. 2 km/hr
48. A boat is sent across a river with a velocity of 8 kmh^{-1} . If the resultant velocity of boat is 10 kmh^{-1} , then velocity of river is: (1994, 93)
- a. 12.8 kmh^{-1}
b. 6 kmh^{-1}
c. 8 kmh^{-1}
d. 10 kmh^{-1}
49. A bus is moving on a straight road towards north with a uniform speed of 50 km/hour then it turns left through 90° . If the speed remains unchanged after turning, the increase in the velocity of bus in the turning process is: (1989)
- a. 70.7 km/hr along south-west direction
b. Zero
c. 50 km/hr along west
d. 70.7 km/hr along north-west direction

Uniform Circular Motion

50. Two particles A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speed v_A and v_B respectively. Their time period of rotation is the same. The ratio of angular speed of A to that of B will be: (2019)
- a. $r_A : r_B$
b. $v_A : v_B$
c. $r_B : r_A$
d. 1 : 1
51. When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be: (2019)
- a. $1:\sqrt{2}$
b. $\sqrt{2}:1$
c. $1:\sqrt{3}$
d. $1:2\sqrt{3}$
52. A particle moves in a circle of radius 5 cm with constant speed and time period 0.2π s. The acceleration of the particle is (2011 Pre)
- a. 15 m/s^2
b. 25 m/s^2
c. 36 m/s^2
d. 5 m/s^2
53. A particle moves in x - y plane according to rule $x = a \sin \omega t$ and $y = a \cos \omega t$. The particle follows: (2010 Mains)
- a. An elliptical path
b. A circular path
c. A parabolic path
d. A straight line path inclined equally to x and y -axes

54. A particle moves along a circle of radius $\left(\frac{20}{\pi}\right)\text{m}$ with constant tangential acceleration. If the velocity of the particle is 80 m/s at the end of the second revolution after motion has begun, the tangential acceleration is: (2003)

- a. 40 ms^{-2} b. $640\pi\text{ m}^{-2}$
c. $160\pi\text{ ms}^{-2}$ d. $40\pi\text{ ms}^{-2}$

55. Two particles having mass 'M' and 'm' are moving in a circular path having radius R and r respectively. If their time period are same then the ratio of angular velocity will be: (2001)

- a. $\frac{r}{R}$ b. $\frac{R}{r}$
c. 1 d. $\sqrt{\frac{R}{r}}$

56. A body is whirled in a horizontal circle of radius 20 cm. It has an angular velocity of 10 rad/s. What is its linear velocity at any point on its circular path? (1996)

- a. 20 m/s b. $\sqrt{2}\text{ m/s}$
c. 10 m/s d. 2 m/s

57. The angular speed of a flywheel making 120 revolution/minute is: (1995)

- a. $4\pi\text{ rad/s}$ b. $4\pi^2\text{ rad/s}$
c. $\pi\text{ rad/s}$ d. $2\pi\text{ rad/s}$

58. An electric fan has blades of length 30 cm measured from the axis of rotation. If the fan is rotating at 120 rpm, the acceleration of a point on the tip of the blade is: (1990)

- a. 1600 ms^{-2} b. 47.4 ms^{-2}
c. 23.7 ms^{-2} d. 50.55 ms^{-2}

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	c	d	d	b	b	d	d	b	c	b	d	a	c	a	b	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	b	d	a	c	b	b	b	c	c	c	a	d	b	a	b	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
a	a	c	b	b	a	a	b	c	a	a	a	a	b	a	d	c
52	53	54	55	56	57	58										
d	b	a	c	d	a	b										

Explanations

1. (b) Let the two vectors be $\vec{A}$ and $\vec{B}$ with magnitudes A and B respectively.

$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$$

$$A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta$$

$$4AB \cos \theta = 0$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

2. (c) $\vec{r} = \cos \omega t \hat{x} + \sin \omega t \hat{y}$

$$\hat{v} = -(\omega \sin \omega t) \hat{x} + (\omega \cos \omega t) \hat{y}$$

$$\vec{a} = -(\omega^2 \cos \omega t) \hat{x} + \omega^2 (-\sin \omega t) \hat{y}$$

$$= -\omega^2 \vec{r}$$

$$\vec{r} \cdot \vec{v} = 0 \text{ hence } \vec{r} \perp \vec{v}$$

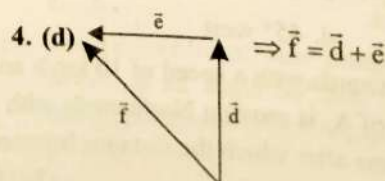
3. (d) Dot product of the two vectors $\vec{A}$ and $\vec{B}$ will be zero, because they are orthogonal to each other.

$$\vec{A} \cdot \vec{B} = 0$$

$$\left(\cos \omega t \cos \frac{\omega t}{2} \right) + \left(\sin \omega t \sin \frac{\omega t}{2} \right) = 0$$

$$\cos \left(\omega t - \frac{\omega t}{2} \right) = 0 \Rightarrow \cos \frac{\omega t}{2} = 0$$

$$\Rightarrow \frac{\omega t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{\omega}$$



5. (b) Let θ be angle between $\vec{A}$ and $\vec{B}$

$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \text{ then } |\vec{A} + \vec{B}|^2 = |\vec{A} - \vec{B}|^2$$

$$\text{or } (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B})$$

$$\text{or } \vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B} = \vec{A} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B}$$

$$\text{or } 4AB \cos \theta = 0 \text{ or } \cos \theta = 0 \text{ or } \theta = 90^\circ$$

6. (b) Let the two vectors be $\vec{a}$ and $\vec{b}$

$$\vec{a} = 2\hat{i} + 3\hat{j} + 8\hat{k}, \vec{b} = -4\hat{i} + 4\hat{j} + \alpha\hat{k}$$

Since the two vector is perpendicular so, dot product is zero.

$$\text{If } \vec{a} \perp \vec{b}$$

$$(2\hat{i} + 3\hat{j} + 8\hat{k}) \cdot (-4\hat{i} + 4\hat{j} + \alpha\hat{k}) = 0$$

$$\text{or, } -8 + 12 + 8\alpha = 0 \Rightarrow 4 + 8\alpha = 0$$

$$\Rightarrow \alpha = -1/2$$

7. (d) Since $(\vec{B} \times \vec{A})$ will be perpendicular to $\vec{A}$, their dot product will be zero. As we know that cross product of two vectors is perpendicular to both vectors.

$$8. (d) |\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$$

$$\Rightarrow AB \sin \theta = \sqrt{3} AB \cos \theta$$

$$\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$$

$$\text{therefore } |\vec{A} + \vec{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$= \sqrt{A^2 + B^2 + 2AB \cos 60^\circ} = \sqrt{A^2 + B^2 + AB}$$

$$= (A^2 + B^2 + AB)^{1/2}$$

9. (b) $(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 0$

$$\Rightarrow A^2 - \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} - B^2 = 0 \quad (\because \vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A})$$

$$\Rightarrow A = B$$

10. (c) $|\vec{A} + \vec{B}|^2 = A^2 + B^2 + 2AB$

$$\Rightarrow A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 + 2AB$$

$$\Rightarrow 2AB \cos \theta = 2AB$$

$$\therefore \cos \theta = 1$$

$$\Rightarrow \theta = 0^\circ$$

11. (b) Since the angular momentum has both magnitude and direction, it is a vector quantity.

12. (d) Position vector of the particle

$$(\vec{r}) = (a \cos \omega t) \hat{i} + (a \sin \omega t) \hat{j}$$

Velocity vector

$$\vec{v} = \frac{d\vec{r}}{dt} = (-a\omega \sin \omega t) \hat{i} + (a\omega \cos \omega t) \hat{j}$$

$$\vec{v} = \omega \left[(-a \sin \omega t) \hat{i} + (a \cos \omega t) \hat{j} \right]$$

$$\vec{v} \cdot \vec{r} = \omega \left[(-a \sin \omega t) \hat{i} + (a \cos \omega t) \hat{j} \right] \cdot \left[(a \cos \omega t) \hat{i} + (a \sin \omega t) \hat{j} \right]$$

$$= \omega \left[-a^2 \sin \omega t \cos \omega t + a^2 \cos \omega t \sin \omega t \right] = 0$$

Hence velocity vector is perpendicular to the position vector.

13. (a) $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|}$$

$$= \frac{(3\hat{i} + 4\hat{j} + 5\hat{k}) \cdot (3\hat{i} + 4\hat{j} - 5\hat{k})}{\left[\sqrt{(3)^2 + (4)^2 + (5)^2} \right] \times \left[\sqrt{(3)^2 + (4)^2 + (5)^2} \right]}$$

$$= \frac{9 + 16 - 25}{50} = 0$$

$$\text{Or } \theta = 90^\circ$$

14. (c) $\vec{A} \times \vec{B}$ is a vector, with its direction perpendicular to both $\vec{A}$ and $\vec{B}$. If side B is zero then $\vec{A} \times 0$ is zero.

$$\therefore \text{Scalar} \times \text{vector} \Rightarrow \text{vector.}$$

$$\text{Hence } \vec{A} \times 0 \text{ is zero vector.}$$

15. (a) Let θ be angle between $\vec{A}$ and $\vec{B}$

$$\text{Given: } A = |\vec{A}| = 3 \text{ units}$$

$$B = |\vec{B}| = 4 \text{ units}$$

$$C = |\vec{C}| = 5 \text{ units}$$

$$\vec{A} + \vec{B} = \vec{C}$$

$$\therefore (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B}) = \vec{C} \cdot \vec{C}$$

$$\vec{A} \cdot \vec{A} + \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B} = \vec{C} \cdot \vec{C}$$

$$A^2 + 2AB \cos \theta + B^2 = C^2$$

$$9 + 2AB \cos \theta + 16 = 25 \text{ or } 2AB \cos \theta = 0$$

$$\text{Or } \cos \theta = 0$$

$$\therefore \theta = 90^\circ = \pi/2$$

16. (b) $x = 5t - 2t^2$ and $y = 10t$

$$v = \frac{dx}{dt} = 5 - 4t, v = \frac{dy}{dt} = 10$$

$$a_x = \frac{dv}{dt} = -4 \text{ ms}^{-2}, a_y = 0$$

$$a = -4 \text{ m/s}^2$$

17. (d) $\vec{R} = 4 \sin(2\pi t) \hat{i} + 4 \cos(2\pi t) \hat{j}$

$$\vec{v} = \frac{d\vec{R}}{dt} = 8\pi \cos 2\pi t \hat{i} - 8\pi \sin 2\pi t \hat{j}$$

$$|\vec{v}| = 8 \text{ m/s}$$

$$|\vec{v}| = \sqrt{[8\pi \cos(2\pi t)]^2 + [-8\pi \sin(2\pi t)]^2}$$

$$= \sqrt{64\pi^2}$$

$$= 8\pi \text{ m/s}$$

$\therefore$ Statement in option (d) is wrong

18. (d) $\vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t} = \frac{(13-2)\hat{i} + (14-3)\hat{j}}{5-0} = \frac{11}{5}(\hat{i} + \hat{j})$

19. (b) $\vec{v} = \vec{u} + \vec{at} = (2\hat{i} + 3\hat{j}) + (0.3\hat{i} + 0.2\hat{j})(10) = 5\hat{i} + 5\hat{j}$

$$\text{Hence } |\vec{v}| = 5\sqrt{2} \text{ units}$$

20. (d) It is given in question, $\vec{v}_1 = 30\hat{i} \text{ m/s}$
 $\vec{v}_2 = 40\hat{j} \text{ m/s}$

$$\text{Average acceleration } (\bar{a}_v) = \frac{|\Delta \vec{v}|}{\Delta t}$$

$$= \frac{\vec{v}_2 - \vec{v}_1}{10} = \frac{40\hat{j} - 30\hat{i}}{10}$$

$$= \frac{50}{10} = 5 \text{ m/s}^2$$

21. (a) From the graph, it is clear that

$$U = Kx^2$$

$$\text{Therefore, Force } F = -\frac{dU}{dx} = -2Kx$$

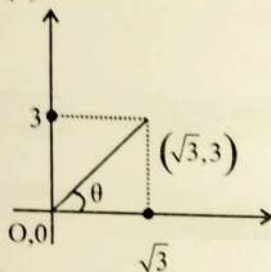
As F is proportional to negative x therefore this represents SHM whose graph is represented by the graph in option (a).

22. (c) $\vec{v} = \vec{u} + \vec{a}t = (3\hat{i} + 4\hat{j}) + (0.4\hat{i} + 0.3\hat{j})(10)$

$$= 7\hat{i} + 7\hat{j}$$

$$\text{So, speed} = |\vec{v}| = 7\sqrt{2} \text{ units}$$

23. (b)

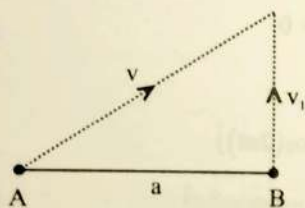


Let the particle moving in a straight line makes an angle θ with x -axis.

$$\text{slope} = \tan \theta = \frac{y}{x} = \frac{3}{\sqrt{3}}$$

$$\text{Since, } \tan \theta = \sqrt{3}, \theta = 60^\circ$$

24. (b)

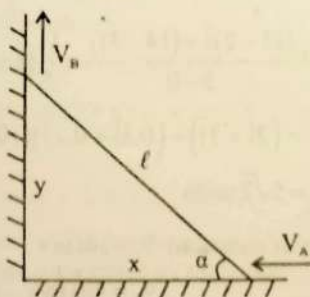


Since, the boy at A has to cover distance 'a' along A to B and component of velocity along AB

$$(v_1) = \sqrt{v^2 - v_1^2}$$

$$\text{So, time taken} = \frac{\sqrt{a^2}}{\sqrt{v^2 - v_1^2}} = \frac{a}{\sqrt{v^2 - v_1^2}}$$

25. (b) $x^2 + y^2 = \ell^2 = \text{constant}$



$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \Rightarrow \frac{dx}{dt} = -V_A = -10$$

$$\frac{dy}{dt} = V_B \quad \& \quad \frac{y}{x} = \tan \alpha = \tan 60^\circ = \sqrt{3}$$

$$10 = \sqrt{3}V_B \Rightarrow V_B = \frac{10}{\sqrt{3}}$$

26. (c) At highest point only horizontal component of velocity remains

$\therefore$ angle with the vertical is 60° , therefore angle with horizontal is 30°

$$\Rightarrow u_x = u \cos \theta$$

$$= 10 \cos 30^\circ$$

$$= 5\sqrt{3} \text{ ms}^{-1}$$

27. (c) According to the question, the time period is given by,

$$T = \frac{2\pi R}{V}$$

$$\Rightarrow V = \frac{2\pi R}{T}$$

...(i)

$$H_{\max} = \frac{V^2 \sin^2 \theta}{2g} = \frac{\left(\frac{2\pi R}{T}\right)^2 \sin^2 \theta}{2g}$$

$$\Rightarrow 4R = \frac{4\pi^2 R^2 \sin^2 \theta}{2T^2 g}$$

$$\Rightarrow 2T^2 g = \pi^2 R \sin^2 \theta \Rightarrow \sin^2 \theta = \frac{2T^2 g}{\pi^2 R}$$

$$\Rightarrow \sin^2 \theta = \left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}} \Rightarrow \theta = \sin^{-1} \left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

28. (c) Initial velocity, $u = 0$, Final velocity, v
acceleration, $a = 5 \text{ m/s}^2$

$$\text{time, } t = 4 \text{ sec}$$

Now, using equation of motion,

$$v = u + at \Rightarrow v = 0 + (5)(4)$$

$$\Rightarrow v = 20 \text{ m/s}$$

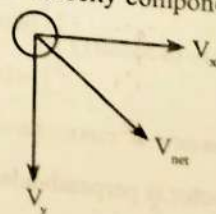
$$\text{At } t = 6 \text{ sec}$$

For stone

$$\text{acceleration due to gravity, } a = g = 10 \text{ m/s}^2$$

As car is having, $v = 20 \text{ m/s}$. So, stone also does. Horizontal velocity component, $V_x = 20 \text{ m/s}$

Vertical downward velocity component, V_y



$$V_y = u + at$$

$$V_y = 0 + g(6 - 4)$$

$$V_y = 10 \times 2 = 20 \text{ m/s}$$

$$V_{\text{net}} = \sqrt{V_x^2 + V_y^2}$$

$$= \sqrt{(20)^2 + (20)^2} = \sqrt{400 + 400} = \sqrt{800}$$

$$V_{\text{net}} = 20\sqrt{2} \text{ m/s}$$

29. (a) If the trajectory is same for both the particles their, maximum height will be the same i.e.,

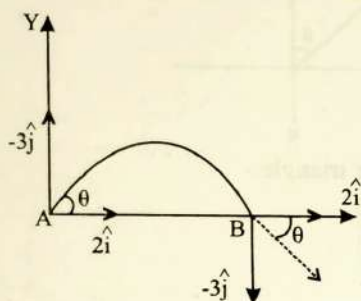
$$(H_{\text{max}})_1 = (H_{\text{max}})_2$$

$$\Rightarrow \frac{u_1^2 \sin^2 \theta}{2g_1} = \frac{u_2^2 \sin^2 \theta}{2g_2}$$

$$\Rightarrow \frac{u_1^2}{u_2^2} = \frac{g_1}{g_2} \Rightarrow g_2 = \frac{9.8 \times 9}{25} \Rightarrow g_2 = 3.5 \text{ m/s}^2$$

30. (d) In a projectile vertical component of velocity keeps on changing with time.

While horizontal velocity component remains constant



$\therefore$ Velocity is $2\hat{i} - 3\hat{j}$

31. (b) $R = H_{\text{max}} \Rightarrow \frac{2u^2 \sin \theta \cos \theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$

$$\Rightarrow \tan \theta = 4 \Rightarrow \theta = \tan^{-1}(4)$$

32. (a) Here, $u = 20 \text{ m/s}$,

For maximum range, angle of projection (θ) should be 45°

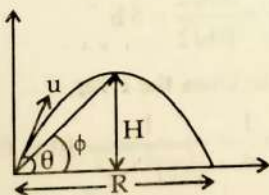
$$R = \frac{v^2 \sin 2\theta}{g}$$

$$R = \frac{v^2 \sin 90^\circ}{g} = \frac{400}{10} = 40 \text{ m}$$

33. (b) $\theta = 45^\circ$

Let ϕ be elevation angle of projectile at its highest point.

$$\therefore \tan \phi = \frac{H}{(R/2)}$$

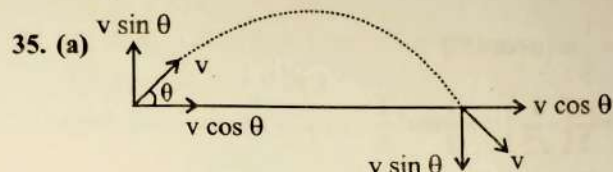


Here,

According to formula, $R \tan \theta = 4H$
 $\Rightarrow R = 4H$ [$\because \tan 45^\circ = 1$]

$$\tan \phi = \frac{H}{(R/2)} = \frac{1}{2} \Rightarrow \phi = \tan^{-1}\left(\frac{1}{2}\right)$$

34. (a) $\frac{u}{2} = u \cos \theta \Rightarrow \theta = 60^\circ$



Velocity in horizontal direction will not change because acceleration in horizontal direction is zero.

Net change in momentum will be,

$$mv \sin \theta - (-mv \sin \theta) = 2mv \frac{1}{\sqrt{2}} = \sqrt{2}mv$$

36. (a) Range = R_1 for angle $(45^\circ - \theta)$,

Range = R_2 for angle $(45^\circ + \theta)$,

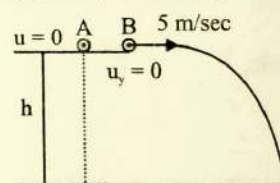
$$\frac{R_1}{R_2} = \frac{\left[\frac{u^2 \sin 2(45^\circ - \theta)}{g} \right]}{\left[\frac{u^2 \sin 2(45^\circ + \theta)}{g} \right]} = \frac{u^2 \sin(90 - 2\theta)}{u^2 \sin(90 + 2\theta)}$$

$$= \frac{\cos 2\theta}{\cos 2\theta} = 1$$

$$\therefore R_1 = R_2$$

Hence, for complementary angles, ranges will be same.

37. (c) For particle (A)



$$h = \frac{1}{2}gt^2$$

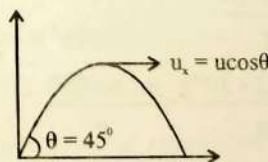
$$t_A = \sqrt{\frac{2h}{g}}$$

For particle (B) In vertical direction

Initial velocity of particle B in vertical direction is zero,

$$S = ut + \frac{1}{2}at^2 \Rightarrow h = \frac{1}{2}gt_B^2 \Rightarrow t_B = \sqrt{\frac{2h}{g}}$$

38. (b)



$$\text{At } H_{\text{max}}, v_y = u_y - gt$$

$$\Rightarrow v_y = u_y - g \frac{u_y}{g}$$

$$\Rightarrow v_y = u_y - u_y$$

$$\Rightarrow v_y = 0$$

at highest point,

$$K_1 = \frac{1}{2}mv_y^2 + \frac{1}{2}mv_x^2$$

$$K_1 = 0 + \frac{1}{2} m (u \cos 45^\circ)^2$$

$$K' = \frac{1}{2} m (u \cos 45^\circ)^2$$

$$K' = \frac{m}{2} \left(\frac{u}{\sqrt{2}} \right)^2 \Rightarrow \frac{1}{2} m u^2 \frac{1}{2}$$

$$K' = \frac{K}{2}$$

39. (b) Time period, $T = \frac{2u \sin \theta}{g}$

Maximum height, $H = \frac{u^2 \sin^2 \theta}{2g}$

Range, $R = \frac{u^2 \sin 2\theta}{g}$

All the quantities will change for $\theta = 30^\circ$ or 60° but range will remain same as $\sin 2\theta$ is same for 30° and 60° .

40. (a) Time = $\frac{\text{Relative horizontal distance}}{\text{Relative horizontal velocity}}$

$$= \frac{x}{u \cos 60^\circ + \frac{u}{\sqrt{3}} \cos 30^\circ} = \frac{x}{u}$$

41. (a) Range of the projectile,

$$R = \frac{u^2 \sin 2\theta}{g}$$

Time of flight of the first particle,

$$t_1 = \frac{2u \sin \theta}{g}$$

Time of flight of the second particle

$$t_2 = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u \cos \theta}{g}$$

$$\therefore t_1 t_2 = \frac{4u^2 \sin \theta \cos \theta}{g^2} = \frac{2u^2 \sin 2\theta}{g^2} = \frac{2R}{g}$$

Or $t_1 t_2 \propto R$

42. (b) For a projectile, the horizontal range is the same for the complementary angles.

Horizontal range, $R = \frac{u^2 \sin 2\theta}{g}$

$$\therefore \text{For body A, } R_A = \frac{u^2 \sin(2 \times 30^\circ)}{g} = \frac{u^2 \sin 60^\circ}{g}$$

For, body B, $R_B = \frac{u^2 \sin(2 \times 60^\circ)}{g}$

$$R_B = \frac{u^2 \sin 120^\circ}{g} = \frac{u^2 \sin(180^\circ - 60^\circ)}{g} = \frac{u^2 \sin 60^\circ}{g}$$

The range is the same whether the angle is θ or $90^\circ - \theta$.

$\therefore$ The ratio of ranges is 1 : 1.

43. (c) Horizontal range $R = \frac{u^2 \sin 2\theta}{g}$

For maximum horizontal range $\theta = 45^\circ$

$$\text{Or } R_m = \frac{u^2}{g}$$

u is muzzle velocity of a shell

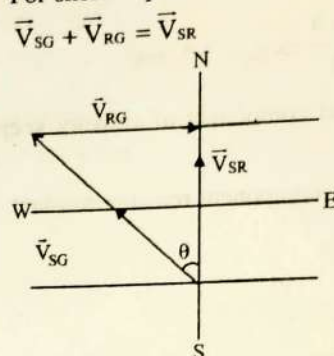
$$\therefore 16000 \text{ m} = \frac{u^2}{(10 \text{ ms}^{-2})^2}$$

$$\text{Or } u = 400 \text{ ms}^{-1}$$

44. (a) $V_{SG} = 20 \text{ m/s}$

$$V_{RG} = 10 \text{ m/s}$$

For shortest path



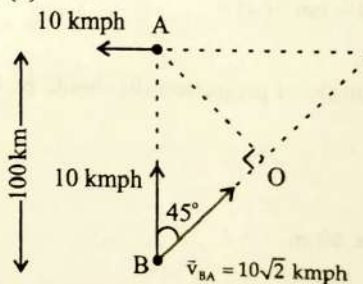
From vector triangle:-

$$\sin \theta = \frac{|V_{RG}|}{|V_{SG}|}$$

$$\sin \theta = \frac{10}{20}$$

$$\sin \theta = \frac{1}{2} = \theta = 30^\circ \text{ west}$$

45. (a)



$$|V_{BA}| = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ kmph}$$

$$\text{Distance } OB = 100 \cos 45^\circ = 50\sqrt{2} \text{ km}$$

Time taken to reach the shortest distance between

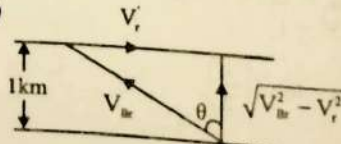
$$A \text{ \& } B = \frac{50\sqrt{2}}{|V_{BA}|} = \frac{50\sqrt{2}}{10\sqrt{2}} = 5 \text{ h}$$

46. (a) Time taken to cross the river,

$$t = \frac{d}{\sqrt{u^2 - v^2}} \Rightarrow \frac{1}{4} = \frac{1}{\sqrt{(5)^2 - v^2}} \quad [d \text{ is the width of the river}]$$

$$\frac{1}{16} = \frac{1}{25 - v^2} \Rightarrow v = 3 \text{ km/hr}$$

47. (a)



$$V_{Br} = 5 \text{ km/hr and } t = 15 \text{ min}$$

$$t = \frac{d}{\sqrt{V_{Br}^2 - V_r^2}} \Rightarrow \frac{15}{60} = \frac{1}{\sqrt{25 - V_r^2}}$$

$$\Rightarrow 4 = \sqrt{25 - V_r^2} \Rightarrow V_r^2 = 25 - 16 \Rightarrow V_r^2 = 9 \Rightarrow V_r = 3 \text{ km/hr}$$

48. (b) Let the velocity of river be v_R and velocity of boat is v_B

$$\therefore \text{Resultant velocity} = \sqrt{v_B^2 + v_R^2 + 2v_B v_R \cos \theta}$$

$$(10) = \sqrt{v_B^2 + v_R^2 + 2v_B v_R \cos 90^\circ}$$

$$(10) = \sqrt{(8)^2 + v_R^2} \text{ or } (10)^2 = (8)^2 + v_R^2$$

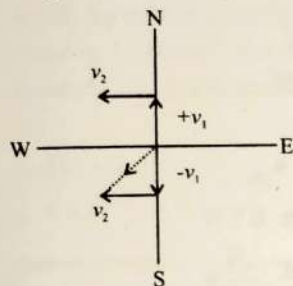
$$v_R^2 = 100 - 64 \text{ or } v_R = 6 \text{ km h}^{-1}$$

49. (a) $v_1 = 50 \text{ km/hr}$ towards north

$v_2 = 50 \text{ km/hr}$ towards west

$-v_1 = 50 \text{ km/hr}$ towards south

Magnitude of change in velocity



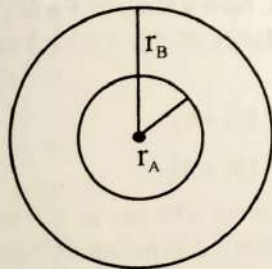
$$= |\vec{v}_2 - \vec{v}_1| = |\vec{v}_2 + (-\vec{v}_1)|$$

$$= \sqrt{v_2^2 + (-v_1)^2}$$

$$= \sqrt{(50)^2 + (50)^2}$$

$$= 70.7 \text{ km/hr along south-west direction}$$

50. (d)



$$T_A = T_B = T$$

$$\omega_A = \frac{2\pi}{T_A}$$

$$\omega_B = \frac{2\pi}{T_B}$$

$$\frac{\omega_A}{\omega_B} = \frac{T_B}{T_A} = \frac{T}{T} = 1$$

51. (c)

$$(\text{Stopping distance}) \quad x_1 = \frac{u^2}{2g \sin 60^\circ}$$

$$(\text{Stopping distance}) \quad x_2 = \frac{u^2}{2g \sin 30^\circ}$$

$$\Rightarrow \frac{x_1}{x_2} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1 \times 2}{2 \times \sqrt{3}} = 1 : \sqrt{3}$$

52. (d) Here, radius, $R = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

Time period, $T = 0.2\pi \text{ s}$

Centripetal acceleration,

$$a_c = \omega^2 R = \left(\frac{2\pi}{T}\right)^2 R = \left(\frac{2\pi}{0.2\pi}\right)^2 (5 \times 10^{-2}) = 5 \text{ ms}^{-2}$$

As particle moves with constant speed, therefore its tangential acceleration is zero. So, $a_t = 0$

The acceleration of the particle is

$$a = \sqrt{a_c^2 + a_t^2} = a_c = 5 \text{ ms}^{-2}$$

It acts towards the centre of the circle.

53. (b) $x = a \sin \omega t$, $y = a \cos \omega t$

Square and adding both sides

$$x^2 + y^2 = a^2 \sin^2 \omega t + a^2 \cos^2 \omega t = a^2$$

$$\Rightarrow x^2 + y^2 = a^2 \Rightarrow \text{circle}$$

54. (a) $r = \frac{20}{\pi} \text{ m}$, $V = 80 \text{ m/s}$

Angular displacement,

$$\theta = 4\pi \text{ (radian)} \therefore \text{for 2 revolution}$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta \quad (\omega_0 = 0)$$

$$\frac{V^2}{r^2} = \frac{2a\theta}{r}$$

$$a = \frac{V^2}{2r\theta}$$

$$a = \frac{80 \times 80}{2 \times \frac{20}{\pi} \times 4\pi} \Rightarrow a = 40 \text{ m/s}^2$$

55. (c) Angular velocity,

$$\omega = \frac{2\pi}{T}, T \text{ is same} \therefore \frac{\omega_1}{\omega_2} = 1$$

56. (d) Radius of circle (r) = $20 \text{ cm} = 0.2 \text{ m}$ and angular velocity (ω) = 10 rad/s .

Linear velocity (v) = $r\omega$

$$= 0.2 \times 10 = 2 \text{ m/s}$$

57. (a) Number of revolutions per minute (n) = 120.

Therefore angular speed (ω)

$$= \frac{2\pi n}{60} = \frac{2\pi \times 120}{60} = 4\pi \text{ rad/s}$$

58. (b) Frequency of rotation $\nu = 120 \text{ rpm} = 2 \text{ rps}$

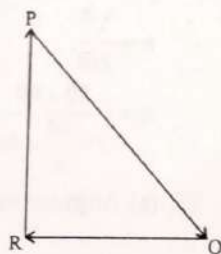
Length of blade $r = 30 \text{ cm} = 0.3 \text{ m}$

$$\text{Centripetal acceleration } a = \omega^2 r = (2\pi\nu)^2 r$$

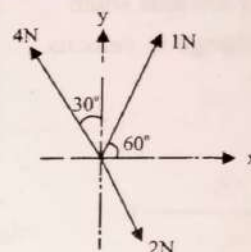
$$= 4\pi^2 \nu^2 r = 4\pi^2 (2)^2 (0.3) = 47.4 \text{ ms}^{-2}$$

Equation of Motion and Newton's Laws of Motion

1. A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. The, the ratio $\frac{S_n}{S_{n+1}}$ is: (2021)
 - a. $\frac{2n-1}{2n+1}$
 - b. $\frac{2n+1}{2n-1}$
 - c. $\frac{2n}{2n-1}$
 - d. $\frac{2n-1}{2n}$
2. A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will : (2019)
 - a. Increase
 - b. Decrease
 - c. Remain constant
 - d. Change according to the smallest force
3. A balloon with mass m is descending down with an acceleration a (where $a < g$). How much mass should be removed from it so that it starts moving up with an acceleration a ? (2014)
 - a. $\frac{2ma}{g+a}$
 - b. $\frac{2ma}{g-a}$
 - c. $\frac{ma}{g+a}$
 - d. $\frac{ma}{g-a}$
4. A body, under the action of a force $\vec{F} = 6\hat{i} - 8\hat{j} + 10\hat{k}$ acquires an acceleration of 1 m/s^2 . The mass of this body must be: (2009, 1996)
 - a. 10 kg
 - b. 20 kg
 - c. $10\sqrt{2} \text{ kg}$
 - d. $2\sqrt{10} \text{ kg}$



5. Three forces acting on a body are shown in the figure. To have the resultant force only along the y-direction, the magnitude of the minimum additional force needed is: (2008)

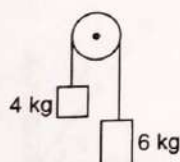


- a. $\sqrt{3} \text{ N}$
 - b. 0.5 N
 - c. 1.5 N
 - d. $\frac{\sqrt{3}}{4} \text{ N}$
6. A man weighs 80 kg. He stands on a weighing scale in a lift which is moving upwards with a uniform acceleration of 5 m/s^2 . What would be the reading on the scale? ($g = 10 \text{ m/s}^2$): (2003)
 - a. Zero
 - b. 400 N
 - c. 800 N
 - d. 1200 N
 7. An object of mass 3 kg is at rest. Now a force of $\vec{F} = 6t^2\hat{i} + 4t\hat{j}$ is applied on the object then velocity of object at $t = 3$ second is: (2002)
 - a. $18\hat{i} + 3\hat{j}$
 - b. $18\hat{i} + 6\hat{j}$
 - c. $3\hat{i} + 18\hat{j}$
 - d. $18\hat{i} + 4\hat{j}$
 8. A cricketer catches a ball of mass 150 gm in 0.1 second moving with speed 20 ms^{-1} , then he experiences force of: (2001)
 - a. 300 N
 - b. 30 N
 - c. 3 N
 - d. 0.3 N
 9. A force of 6 N acts on a body at rest and of mass 1 kg. During this time, the body attains a velocity of 30 m/s. The time for which the force acts on the body is: (1997)
 - a. 7 seconds
 - b. 5 seconds
 - c. 10 seconds
 - d. 8 seconds
 10. A 10 N force is applied on a body produce in it an acceleration of 1 m/s^2 . The mass of the body is: (1996)
 - a. 15 kg
 - b. 20 kg
 - c. 10 kg
 - d. 5 kg

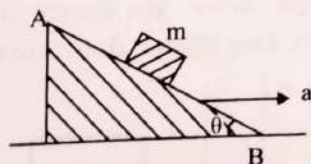
11. A block has been placed on a inclined plane with the slope angle θ , block slides down the plane at constant speed. The coefficient of kinetic friction is equal to: (1993)
- a. $\sin \theta$ b. $\cos \theta$
c. g d. $\tan \theta$
12. Physical independence of force is a consequence of: (1991)
- a. Third law of motion b. Second law of motion
c. First law of motion d. All of these laws
13. A particle of mass m is moving with a uniform velocity v_1 . It is given an impulse such that its velocity becomes v_2 . The impulse is equal to: (1990)
- a. $m[|v_2| - |v_1|]$ b. $\frac{1}{2}m[v_2^2 - v_1^2]$
c. $m[v_1 + v_2]$ d. $m[v_2 - v_1]$
14. A 600 kg rocket is set for a vertical firing. If the exhaust speed is 1000 ms^{-1} , the mass of the gas ejected per second to supply the thrust needed to overcome the weight of rocket is: (1990)
- a. 117.6 kgs^{-1} b. 58.6 kgs^{-1}
c. 6 kgs^{-1} d. 76.4 kgs^{-1}

Motion of Connected Bodies

15. Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is: (2020)

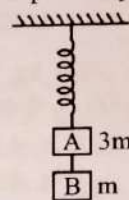


- a. $g/2$ b. $g/5$
c. $g/10$ d. g
16. A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is: (2018)

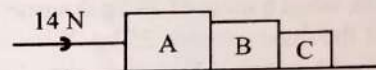


- a. $a = g \cos \theta$ b. $a = \frac{g}{\sin \theta}$
c. $a = \frac{g}{\csc \theta}$ d. $a = g \tan \theta$

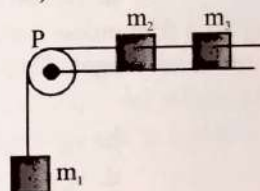
17. Two blocks A and B of masses $3m$ and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively: (2017-Delhi)



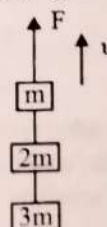
- a. $g, \frac{g}{3}$ b. g, g
c. $\frac{g}{3}, \frac{g}{3}$ d. $\frac{g}{3}, g$
18. Three blocks A, B and C of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a friction less surface, as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between A and B is: (2015)



- a. 6 N b. 8 N
c. 18 N d. 2 N
19. A system consists of three masses m_1 , m_2 and m_3 is connected by a string passing over a pulley P. The mass m_1 hangs freely and m_2 and m_3 are on a rough horizontal table (the coefficient of friction $= \mu$). The pulley is frictionless and of negligible mass. The downward acceleration of mass m_1 is: (Assume $m_1 = m_2 = m_3 = m$) (2014)

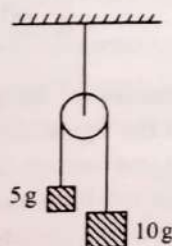


- a. $\frac{g(1 - g\mu)}{9}$ b. $\frac{2g\mu}{g}$
c. $\frac{g(1 - 2\mu)}{3}$ d. $\frac{g(1 - 2\mu)}{2}$
20. Three blocks with masses m , $2m$ and $3m$ are connected by strings, as shown in the figure. After an upward force F is applied on block m , the masses move upward at constant speed v . What is the net force on the block of mass $2m$? (g is the acceleration due to gravity) (2013)



- a. $6mg$ b. Zero
c. $2mg$ d. $3mg$

21. A person of mass 60 kg is inside a lift of mass 940 kg and presses the button on control panel. The lift starts moving upwards with an acceleration 1.0 m/s^2 . If $g = 10 \text{ ms}^{-2}$, the tension in the supporting cable is (2011 Pre)
- a. 8600 N b. 9680 N
c. 11000 N d. 1200 N
22. The mass of a lift is 2000 kg. When the tension in the supporting cable is 28000 N, then its acceleration is: (2009)
- a. 4 ms^{-2} upwards. b. 4 ms^{-2} downwards.
c. 14 ms^{-2} upwards. d. 30 ms^{-2} downwards.
23. A block of mass m is placed on a smooth wedge of inclination θ . The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (g is acceleration due to gravity) will be: (2004)
- a. $mg \sin \theta$ b. mg
c. $\frac{mg}{\cos \theta}$ d. $mg \cos \theta$
24. A monkey of mass 20 kg is holding a vertical rope. The rope will not break when a mass of 25 kg is suspended from it but will break if the mass exceeds 25 kg. What is the maximum acceleration with which the monkey can climb up along the rope? ($g = 10 \text{ m/s}^2$) (2003)
- a. 5 m/s^2 b. 10 m/s^2
c. 25 m/s^2 d. 2.5 m/s^2
25. A lift of mass 1000 kg which is moving with acceleration of 1 m/s^2 in upward direction, then the tension developed in string which is connected to lift is: (2002)
- a. 9800 N b. 10,800 N
c. 11000 N d. 10,000 N
26. A man is slipping on a frictionless inclined plane and a bag falls down from the same height. Then the speed of both is related as: (2000)
- a. $v_B > v_m$ b. $v_B < v_m$
c. $v_B = v_m$ d. v_B and v_m can't related
27. Two masses as shown are suspended from a 'massless pulley'. Calculate the acceleration of the system when masses are left free: (2000)



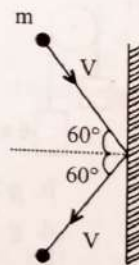
- a. $2g/3$ b. $g/3$
c. $g/9$ d. $g/7$
28. A small ball is suspended from a thread. It is lifted up with an acceleration 4.9 ms^{-2} and lowered with an acceleration 4.9 ms^{-2} then the ratio of tensions in the thread in both cases will be: (1998)
- a. 1 : 3 b. 3 : 1
c. 1 : 1 d. $1 : \sqrt{5}$

29. A monkey is descending from branch of a tree with constant acceleration. If the breaking strength is 75% of the weight of the monkey, the minimum acceleration with which monkey can slide down without branch is: (1993)

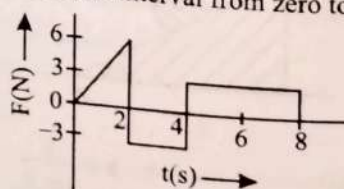
- a. g b. $\frac{3g}{4}$
c. $\frac{g}{4}$ d. $\frac{g}{2}$

Conservation of Momentum and Impulse Momentum Theory

30. A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v , the speed of the third (lighter) fragment is: (2022)
- a. $3\sqrt{2}v$ b. v
c. $\sqrt{2}v$ d. $2\sqrt{2}v$
31. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10 \text{ m/s}^2$) nearly: (2021)
- a. 4.2 kg m/s b. 2.1 kg m/s
c. 1.4 kg m/s d. 0 kg m/s
32. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure below. The value of impulse imparted by the wall on the ball will be: (2016 - II)

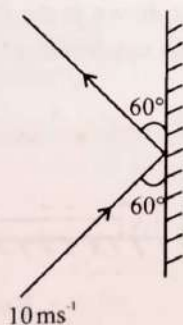


- a. $\frac{mv}{2}$ b. $\frac{mv}{3}$
c. mv d. $2mv$
33. The force F acting on a particle of mass m is indicated by the force-time graph shown. The change in momentum of the particle over the time interval from zero to 8 s is: (2014)



- a. 24 Ns b. 20 Ns
c. 12 Ns d. 6 Ns

34. A stone is dropped from a height h . It hits the ground with a certain momentum P . If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by: (2012 Mains)
- 68%
 - 41%
 - 200%
 - 100%
35. A body of mass m hits normally a rigid wall with velocity v and bounces back with the same velocity. The impulse experienced by the body is: (2011 Pre)
- mv
 - $1.5 mv$
 - $2 mv$
 - Zero
36. An explosion blows a rock into three parts. Two parts go off at right angles to each other. These two are, 1 kg first part moving with a velocity of 12 ms^{-1} and 2 kg second part moving with a velocity of 8 ms^{-1} . If the third part flies off with a velocity of 4 ms^{-1} , its mass would be: (2009)
- 7 kg
 - 17 kg
 - 3 kg
 - 5 kg
37. A particle of mass m is projected with velocity v making an angle of 45° with the horizontal. When the particle lands on the level ground the magnitude of the change in its momentum will be: (2008)
- Zero
 - $2 mv$
 - $6 mv$
 - $\sqrt{2} mv$
38. A 0.5 kg ball moving with a speed of 12 m/s strikes a hard wall at an angle of 30° with the wall. It is reflected with the same speed at the same angle. If the ball is in contact with the wall for 0.25 seconds, the average force acting on the wall is (2006)
- 96 N
 - 48 N
 - 24 N
 - 12 N
39. A 1 kg stationary bomb is exploded in three parts having mass 1 : 1 : 3 respectively. Parts having same mass move in perpendicular direction with velocity 30 ms^{-1} , then the velocity of bigger part will be: (2001)
- $10\sqrt{2} \text{ ms}^{-1}$
 - $\frac{10}{\sqrt{2}} \text{ ms}^{-1}$
 - $15\sqrt{2} \text{ ms}^{-1}$
 - $\frac{15}{\sqrt{2}} \text{ ms}^{-1}$
40. A body of mass 3 kg hits a wall at an angle of 60° and returns at the same angle. The impact time was 0.2 s. Calculate the force exerted on the wall: (2000)

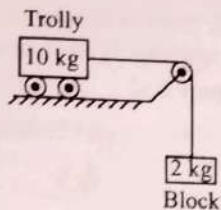


- $150\sqrt{3} \text{ N}$
- $50\sqrt{3} \text{ N}$
- 100 N
- $75\sqrt{3} \text{ N}$

41. A mass of 1 kg is thrown up with a velocity of 100 m/s . After 5 seconds, it explodes into two parts. One part of mass 400 g comes down with a velocity 25 m/s . Calculate the velocity of other part: (2000)
- 40 m/s upward
 - 40 m/s downward
 - 100 m/s upward
 - 60 m/s downward
42. A particle is projected with velocity ' u ' makes an angle θ w.r.t. horizontal. Now it breaks in two identical parts at highest point of trajectory. If one part is retrace its path, then velocity of other part is: (1999)
- $3u \cos \theta$
 - $2u \cos \theta$
 - $u \cos \theta$
 - u
43. For a rocket propulsion velocity of exhaust gases relative to rocket is 2 km/s . If mass of rocket system is 1000 kg , then the rate of fuel consumption for a rocket to rise up with acceleration 4.9 m/s^2 will be: (1998)
- 12.25 kg/s
 - 17.5 kg/s
 - 7.35 kg/s
 - 5.2 kg/s
44. If force $F = 500 - 100t$, then function of impulse with time will be: (1998)
- $500t - 50t^2$
 - $50t - 10$
 - $50 - t^2$
 - $100t^2$
45. A man fires a bullet of mass 200 gm at a speed of 5 m/s . The gun is of one kg mass. By what velocity the gun rebounds backward? (1996)
- 1 m/s
 - 0.01 m/s
 - 0.1 m/s
 - 10 m/s
46. A satellite in force free space sweeps stationary interplanetary dust at a rate of $dM/dt = \alpha v$, where M is mass and v is the speed of satellite and α is a constant. The acceleration of satellite is: (1995)
- $\frac{-\alpha v^2}{2M}$
 - $-\alpha v^2$
 - $\frac{-2\alpha v^2}{M}$
 - $\frac{-\alpha v^2}{M}$
47. In a rocket, fuel burns at the rate of 1 kg/s . This fuel is ejected from the rocket with a velocity of 60 km/s . This exerts a force on the rocket equal to: (1994)
- 6000 N
 - 60000 N
 - 60 N
 - 600 N
48. A body of mass 5 kg explodes at rest into three fragments with masses in the ratio 1 : 1 : 3. The fragments with equal masses fly in mutually perpendicular directions with speeds of 21 m/s . The velocity of heaviest fragment in m/s will be: (1989)
- $7\sqrt{2}$
 - $5\sqrt{2}$
 - $3\sqrt{2}$
 - $\sqrt{2}$

Friction

49. Calculate the acceleration of the block and trolley system shown in the figure. The coefficient of kinetic friction between the trolley and the surface is 0.05. ($g = 10 \text{ m/s}^2$, mass of the string is negligible and no other friction exists). (2020-Covid)



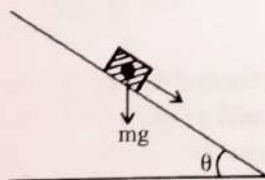
- a. 1.50 m/s^2 b. 1.66 m/s^2
c. 1.00 m/s^2 d. 1.25 m/s^2
50. Which one of the following statements is incorrect? (2018)

- a. Frictionless force opposes the relative motion.
b. Limiting value of static friction is directly proportional to normal reaction.
c. Rolling friction is smaller than sliding friction.
d. Coefficient of sliding friction has dimensions of length.

51. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and table is μ_k . When the block A is sliding on the table, the tension in the string is: (2015)

- a. $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$ b. $\frac{m_1 m_2 (1 + \mu_k)g}{(m_1 + m_2)}$
c. $\frac{m_1 m_2 (1 - \mu_k)g}{(m_1 + m_2)}$ d. $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$

52. A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30° , the box starts to slip and slides 4.0 m down the plank in 4.0 s. The coefficients of static and kinetic friction between the box and the plank will be, respectively: (2015 - Re)



- a. 0.4 and 0.3 b. 0.6 and 0.6
c. 0.6 and 0.5 d. 0.5 and 0.6

53. The upper half of an inclined plane of inclination θ is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and lower half of the plane is given by (2013)

- a. $\mu = 2 \tan \theta$ b. $\mu = \tan \theta$
c. $\mu = \frac{1}{\tan \theta}$ d. $\mu = \frac{2}{\tan \theta}$

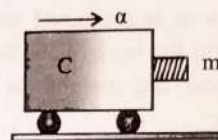
54. A conveyor belt is moving at a constant speed of 2 m/s. A box is gently dropped on it. The coefficient of friction between them is $\mu = 0.5$. The distance that the box will move relative to belt before coming to rest on it, taking $g = 10 \text{ ms}^{-2}$ (2011 Mains)

- a. 1.2 m b. 0.6 m
c. Zero d. 0.4 m

55. A gramophone record is revolving with an angular velocity ω . A coin is placed at a distance r from the center of the record. The static coefficient of friction is μ . The coin will revolve with the record if: (2010 Pre)

- a. $r \geq \frac{\mu g}{\omega^2}$ b. $r = \mu g \omega^2$
c. $r < \frac{\omega^2}{\mu g}$ d. $r \leq \frac{\mu g}{\omega^2}$

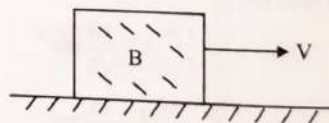
56. A block of mass m is in contact with the cart C as shown in the figure.



The coefficient of static friction between the block and the cart is μ . The acceleration α of the cart that will prevent the block from falling satisfies: (2010 Pre)

- a. $\alpha < \frac{g}{\mu}$ b. $\alpha > \frac{g}{\mu}$
c. $\alpha > \frac{g}{\mu m}$ d. $\alpha \geq \frac{g}{\mu}$

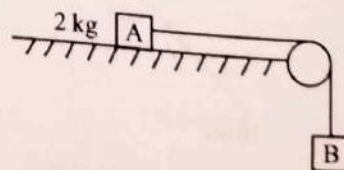
57. A block B is pushed momentarily along a horizontal surface with an initial velocity V . If μ is the coefficient of sliding friction between B and the surface, block B will come to rest after a time: (2007)



- a. $g\mu/V$ b. g/V
c. V/g d. $V/\mu g$

58. The coefficient of static friction, μ_s , between block A of mass 2 kg and the table as shown in the figure is 0.2. What would be the maximum mass value of block B so that the two blocks do not move?

The string and the pulley are assumed to be smooth and massless, ($g = 10 \text{ m/s}^2$): (2004)

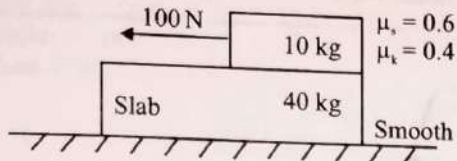


- a. 4.0 kg b. 0.2 kg
c. 0.4 kg d. 2.0 kg

59. A block of mass 10 kg placed on rough horizontal surface having coefficient of friction $\mu = 0.5$, if a horizontal force of 100 N acting on it then acceleration of the block will be: (2002)
- a. 10 m/s^2 b. 5 m/s^2
c. 15 m/s^2 d. 0.5 m/s^2

60. On the horizontal surface of a truck a block of mass 1 kg is placed ($\mu = 0.6$) and truck is moving with acceleration 5 m/s^2 then the frictional force on block will be: (2001)
- a. 5 N b. 6 N
c. 5.88 N d. 8 N

61. If 100 N force is applied to 10 kg block as shown in diagram, then acceleration produced for slab: (1999)



- a. 1.65 m/s^2 b. 0.98 m/s^2
c. 1.2 m/s^2 d. 0.25 m/s^2
62. Consider a car moving along a straight horizontal road with a speed of 72 km/h. If the coefficient of static friction between the tyres and the road is 0.5, the shortest distance, in which the car can be stopped is (taking $g = 10 \text{ m/s}^2$): (1992)

- a. 30 m b. 40 m
c. 72 m d. 20 m
63. A heavy uniform chain lies on horizontal table top. If the coefficient of friction between the chain and the table surface is 0.25, then the maximum fraction of the length of the chain that can hang over one edge of the table is: (1991)
- a. 20% b. 25%
c. 35% d. 15%

64. Starting from rest, a body slides down a 45° inclined plane in twice the time it takes to slide down the same distance in the absence of friction. The coefficient of friction between the body and the inclined plane is: (1988)
- a. 0.80 b. 0.75
c. 0.25 d. 0.33

Circular Motion, Banking of Road

65. A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be : ($g = 10 \text{ m/s}^2$) (2019)
- a. $\sqrt{10} \text{ rad/s}$ b. $\frac{10}{2\pi} \text{ rad/s}$
c. 10 rad/s d. $10\pi \text{ rad/s}$

66. One end of string of length l is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v', the net force on the particle (directed towards center) will be: (T represents the tension in the string) (2017-Delhi)

- a. $T + \frac{mv^2}{l}$ b. $T - \frac{mv^2}{l}$
c. Zero d. T

67. A car is negotiating a curved road of radius R. The road is banked at an angle θ . The coefficient of friction between the tyres of the car and the road is μ_s . The maximum safe velocity on this road is: (2016 - I)

- a. $\sqrt{gR^2 \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$
b. $\sqrt{gR \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$
c. $\sqrt{\frac{g\mu_s + \tan \theta}{R(1 - \mu_s \tan \theta)}}$
d. $\sqrt{\frac{g}{R^2} \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$

68. Two stones of masses m and 2 m are whirled in horizontal circles, the heavier one in a radius $r/2$ and the lighter one in radius r. The tangential speed of lighter stone is n times that of the value of heavier stone when they experience same centripetal forces. The value of n is: (2015 Re)

- a. 1 b. 2
c. 3 d. 4

69. A car of mass m is moving on a level circular track of radius R. If μ_s represents the static friction between the road and tyres of the car, the maximum speed of the car in circular motion is given by: (2012 Mains)

- a. $\sqrt{\mu_s m R g}$ b. $\sqrt{\frac{R g}{\mu_s}}$
c. $\sqrt{m R g / s}$ d. $\sqrt{\mu_s R g}$

70. A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45° , the speed of the car is: (2012 Pre)

- a. 20 m/s b. 30 m/s
c. 5 m/s d. 10 m/s

71. A roller coaster is designed such that riders experience "weightlessness" as they go round the top of a hill whose radius of curvature is 20 m. The speed of the car at the top of the hill is between: (2008)

- a. 13 m/s and 14 m/s
b. 14 m/s and 15 m/s
c. 15 m/s and 16 m/s
d. 16 m/s and 17 m/s

72. A tube of length L is filled completely with an incompressible liquid of mass M and closed at both the ends. The tube is then rotated in a horizontal plane about one of its ends with a uniform angular velocity ω . The force exerted by the liquid at the other end is: (2006)

a. $(ML\omega^2)/(2)$ b. $(ML^2\omega)/(2)$
c. $2ML\omega^2$ d. $(ML^2\omega^2)/(2)$

73. A stone tied to the end of a string of 1 m long is whirled in a horizontal circle with a constant speed. If the stone makes 22 revolutions in 44 seconds, what is the magnitude and direction of acceleration of the stone? (2005)

a. $\pi^2 \text{ ms}^{-2}$ and direction along the radius towards the centre
b. $\pi^2 \text{ ms}^{-2}$ and direction along the radius away from the centre

c. $\pi^2 \text{ ms}^{-2}$ and direction along the tangent to the circle
d. $\pi^2 \text{ ms}^{-2}$ and direction along the radius towards the centre

74. A particle of mass m is tied to a string of length L and whirled into a horizontal plane. If tension in the string is T then the speed of the particle will be: (1994)

a. $\sqrt{\frac{T\ell}{m}}$ b. $\sqrt{\frac{2T\ell}{m}}$
c. $\sqrt{\frac{3T\ell}{m}}$ d. $\sqrt{\frac{T}{\ell m}}$

75. When milk is churned, cream gets separated due to: (1995)

a. Centripetal force b. Centrifugal force
c. Frictional force d. Gravitational force

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	c	a	c	b	d	d	b	b	c	d	c	d	c	b	d	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	c	b	c	a	c	d	b	a	b	b	c	d	a	c	c	b
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
c	d	d	c	a	a	c	a	c	a	a	d	b	a	d	d	b
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
c	a	d	d	d	d	c	b	a	b	b	a	b	c	d	b	b
69	70	71	72	73	74	75										
d	b	b	a	a	a	b										

Explanations

1. (a) Using equation of uniform motion,

$$S_n = u + \frac{a}{2}(2n-1) \text{ where,}$$

S_n is the distance covered in n^{th} sec
[from $t = (n-1)$ to $t = n$]

S_{n+1} is the distance covered in $(n+1)^{\text{th}}$ sec
[from $t = n$ to $t = (n+1)$]

Now; $u = 0$

$$S_n = \frac{a}{2}(2n-1) \quad \dots(1)$$

$$S_{n+1} = \frac{a}{2}[2(n+1)-1] = \frac{a}{2}[2n+2-1]$$

$$S_{n+1} = \frac{a}{2}[2n+1] \quad \dots(2)$$

Dividing eqⁿ (1) and eqⁿ (2)

$$\Rightarrow \frac{S_n}{S_{n+1}} = \frac{\left(\frac{a}{2}\right)(2n-1)}{\left(\frac{a}{2}\right)(2n+1)} \Rightarrow \frac{S_n}{S_{n+1}} = \frac{2n-1}{2n+1}$$

2. (c) All these forces are forming closed loop in same order. So, net force is zero.

$$\text{Force} = ma$$

$$\Rightarrow m \frac{d\vec{v}}{dt} = 0 \Rightarrow \vec{v} = \text{constant}$$

3. (a) For downward motion

$$mg - F_a = ma \Rightarrow F_a = mg - ma$$

If some mass Δm is removed, then it starts accelerating upwards

$$F_a - (m - \Delta m)g = (m - \Delta m)a$$

$$mg - ma - mg + g\Delta m = ma - a\Delta m$$

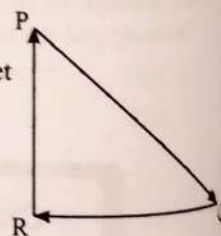
$$g\Delta m - ma = ma - a\Delta m \Rightarrow \Delta m [g + a] = 2ma$$

$$\Delta m = \frac{2ma}{g+a}$$

4. (c) Magnitude of force $(6\hat{i} - 8\hat{j} + 10\hat{k})\text{N}$

$$= \sqrt{(6)^2 + (8)^2 + (10)^2} = 10\sqrt{2}\text{N}$$

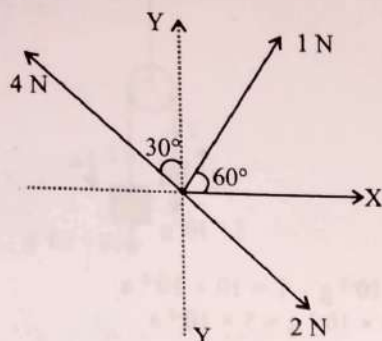
$$\text{Acceleration} = 1 \text{ ms}^{-2}$$



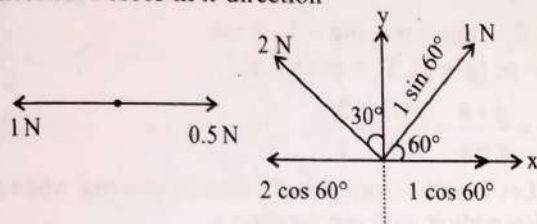
$$\text{So, Mass} = \frac{\text{Force}}{\text{Acceleration}} = \frac{10\sqrt{2}}{1}$$

$$\text{Mass} = 10\sqrt{2} \text{ kg}$$

5. (b)



To have the resultant force only along y-direction, the magnitude of components of force along x-axis must be cancelled. Forces in x-direction



So, 0.5 N additional force (in + x direction) needed to have resultant force along y-direction.

6. (d) Reading of weighing scale = $m(g + a) = 80(10 + 5) = 1200 \text{ N}$

7. (d) $\vec{a} = \frac{\vec{F}}{m} = 2t^2\hat{i} + \frac{4}{3}t\hat{j}$

Integrate on both sides

$$d\vec{v} = \left(2t^2\hat{i} + \frac{4}{3}t\hat{j} \right) dt$$

$$\text{At, } t = 3$$

$$d\vec{v} = 18\hat{i} + 4\hat{j}$$

8. (b) Impulse = Change in momentum

$$F = \frac{\Delta P}{\Delta t} = \frac{150 \times 10^{-3} \times 20}{0.1} = 30 \text{ N}$$

9. (b) Force (F) = 6 N; Initial velocity (u) = 0;
Mass (m) = 1 kg and final velocity (v) = 30 m/s.

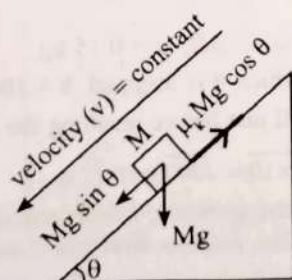
$$\text{Therefore acceleration } (a) = \frac{F}{m} = \frac{6}{1} = 6 \text{ m/s}^2$$

$$\text{Final velocity } (v) = 30 = u + at = 0 + 6 \times t \text{ or } t = 5 \text{ seconds.}$$

10. (c) Force (F) = 10 N and acceleration $a = 1 \text{ m/s}^2$

$$\text{Mass } (m) = \frac{F}{a} = \frac{10}{1} = 10 \text{ kg}$$

11. (d)



From the figure,

The acceleration is nullified by force of kinetic friction.

$Mg \sin \theta$ is downwards force.

μ_k is the coefficient of kinetic friction.

$\mu_k Mg \cos \theta$ is upwards acting force.

$$\therefore Mg \sin \theta - \mu_k Mg \cos \theta = \text{mass} \times \text{acceleration.}$$

Acceleration = 0 as v is constant

$$\therefore \mu_k = \frac{\sin \theta}{\cos \theta} = \tan \theta.$$

12. (c) Newton's first law of motion is related to physical independence of force.

13. (d) Impulse is a vector quantity and is equal to change in momentum of the body thus, (same as $F \times t$ where t is short)
Impulse = $mv_2 - mv_1 = m(v_2 - v_1)$

14. (c) The force with which the rocket moves upward is given by

$$F = u \frac{dm}{dt}$$

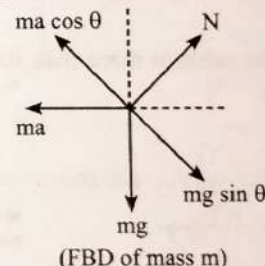
Mass of the gas ejected per second to supply the needed thrust to overcome the weight of the rocket is

$$\frac{dm}{dt} = \frac{F}{u} = \frac{m \times a}{u} \text{ or } \frac{dm}{dt} = \frac{600 \times 10}{1000} = 6 \text{ kgs}^{-1}$$

15. (b) $a = \frac{(m_2 - m_1)g}{m_1 + m_2}$

$$a = \frac{(6 - 4)g}{6 + 4} = \frac{2g}{10} = \frac{g}{5}$$

16. (d) From wedge reference frame

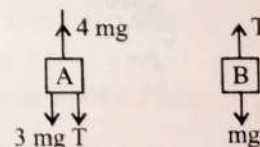


For block to be at rest

$$ma \cos \theta = mg \sin \theta$$

$$a = g \tan \theta$$

17. (d)



When the string is cut $T = 0$

for block A

$$3m a = 4mg - 3mg$$

$$a = \frac{g}{3}$$

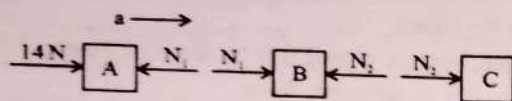
for block B

$$mg = ma \Rightarrow a = g$$

18. (a) Acceleration of system

$$\frac{F_{\text{net}}}{M_{\text{total}}}$$

$$= \frac{14}{4 + 2 + 1} = 2 \text{ ms}^{-2}$$



N_1 = Normal force between A and B

$$14 - N_1 = m_A a$$

$$14 - N_1 = 4 \times 2$$

$$14 - N_1 = 8 \Rightarrow N_1 = 6 \text{ N}$$

19. (c) Acceleration = $\frac{\text{Net force in the direction of motion}}{\text{Total mass of system}}$

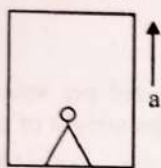
$$= \frac{m_1 g - \mu(m_2 + m_3)g}{m_1 + m_2 + m_3} = \frac{g(1 - 2\mu)}{3} [\because m_1 = m_2 = m_3]$$

20. (b) As block of mass $2m$ moves with constant velocity so net force on it is zero.

21. (c) Here, Mass of a person, $m = 60 \text{ kg}$

Mass of lift, $M = 940 \text{ kg}$, $a = 1 \text{ m/s}^2$, $g = 10 \text{ m/s}^2$

Let T be the tension in the supporting cable.



$$\therefore T - (M + m)g = (M + m)a$$

$$T = (M + m)(a + g) = (940 + 60)(1 + 10) = 11000 \text{ N}$$

22. (a) Tension in the supporting cable = 28000 N

Mass of a Lift = 2000 kg

$$\text{So, } T - mg = ma$$

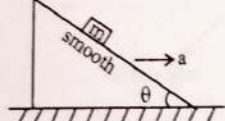
$$\Rightarrow 28000 - 20000 = 2000a$$

$$8000 = 2000a$$

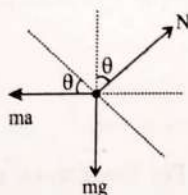
$$\Rightarrow a = 4 \text{ ms}^{-2}$$

As tension in the cable is more than the weight so, motion will be upwards.

23. (c)



FBD of m



$$\text{Here } N \cos \theta = mg \Rightarrow N = \frac{mg}{\cos \theta}$$

24. (d) $T_{\max} = 25 \text{ g}$

$$ma = T_{\max} - mg$$

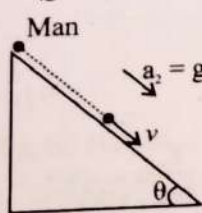
$$20a = 25g - 20g$$

$$20a = 5g$$

$$\Rightarrow a = \frac{g}{4} = \frac{10}{4} = 2.5 \text{ m/s}^2$$

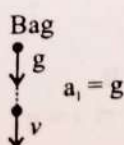
25. (b) $T = m(g + a) = 1000(9.8 + 1) = 10,800 \text{ N}$

26. (a)



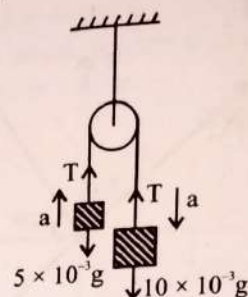
$$g > g \sin \theta \quad (0 < \theta < 90)$$

$$\therefore a_1 > a_2$$



The body with acceleration a_1 will have more speed than with acceleration a_2 . So velocity of bag will be more than velocity of man.

27. (b)



$$10 \times 10^{-3} g - T = 10 \times 10^{-3} a$$

$$T - 5 \times 10^{-3} g = 5 \times 10^{-3} a$$

$$\Rightarrow 5g = 15a$$

$$a = \frac{g}{3}$$

28. (b) $T_1 - mg = ma$, $mg - T_2 = ma$

$$T_1 = m(g + a), T_2 = m(g - a)$$

$$\frac{T_1}{T_2} = \frac{g + a}{g - a} = \frac{14.7}{4.9} = \frac{3}{1}$$

29. (c) Let T be the tension in the branch of a tree when monkey is descending with acceleration a .

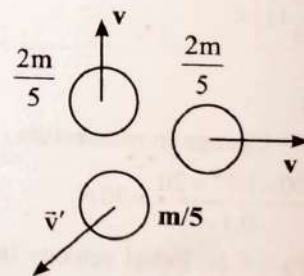
$$\text{Thus, } mg - T = ma$$

Also, $T = 75\%$ of weight of monkey

$$T = \left(\frac{75}{100}\right)mg = \frac{3}{4}mg$$

$$\therefore ma = mg - \left(\frac{3}{4}\right)mg = \frac{1}{4}mg \text{ or } a = \frac{g}{4}$$

30. (d)



As the shell was initially at rest, therefore initial momentum is zero.

Using conservation of momentum :

Initial momentum = Final momentum

$$0 = \frac{2m}{5}(-v\hat{i}) + \frac{2m}{5}(-v\hat{i}) + \frac{m}{5}\bar{v}' \Rightarrow \bar{v}' = 2v\hat{i} + 2v\hat{j}$$

$$\Rightarrow \bar{v}' = \sqrt{(2v)^2 + (2v)^2} = 2\sqrt{2}v$$

31. (a) Given, mass of ball, $m = 0.15 \text{ kg}$

Height from which it is dropped, $h = 10 \text{ m}$

Velocity of ball just before reaching the ground, $v = \sqrt{2gh}$

$$\Rightarrow v = \sqrt{2 \times 10 \times 10} = \sqrt{200} = 10\sqrt{2} \text{ m/s}$$

According to the question, height remaining the same. So, velocity will also, only its direction changes.

$$\text{i.e., } v_i = 10\sqrt{2}(-\hat{j}) \text{ and } v_f = 10\sqrt{2}(\hat{j})$$

Required magnitude of Impulse imported to the ball = change in linear momentum

$$|\vec{I}| = |\Delta \vec{P}|$$

$$|\vec{I}| = |m\vec{v}_f - m\vec{v}_i|$$

$$= |m\vec{v}_f - m(-\vec{v}_f)| = |m\vec{v}_f + m\vec{v}_f|$$

$$= |2 m\vec{v}_f|$$

$$= 2 \times 0.15 \times 10\sqrt{2}$$

$$\vec{I} = 3\sqrt{2} \text{ kg m/s}$$

$$\vec{I} = 4.2 \text{ kg m/s}$$

32. (c) Impulse = $|\Delta \vec{P}| = |m\Delta \vec{v}| = m(2v \cos 60^\circ) = mv$

33. (c) Change in momentum,

$$\int \Delta p = \int F dt = \text{Area of } F - t \text{ graph}$$

$$= \frac{1}{2} \times 2 \times 6 - 3 \times 2 + 4 \times 3 = 12 \text{ Ns}$$

34. (b) $P = mv = m\sqrt{2gh} \Rightarrow P \propto \sqrt{h}$

$$\frac{P_f - P_i}{P_i} \times 100\% = \frac{\sqrt{h_2} - \sqrt{h_1}}{\sqrt{h_1}} \times 100$$

$$= \left[\sqrt{\frac{h_2}{h_1}} - 1 \right] \times 100$$

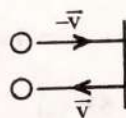
$$\left\{ \sqrt{\left(\frac{200}{100} \right)} - 1 \right\} \times 100$$

$$= (\sqrt{2} - 1) \times 100 \Rightarrow 0.414 \times 100 = 41\%$$

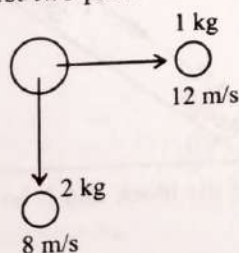
35. (c) Impulse = Change in linear momentum

$$= mv - (-mv)$$

$$= 2mv$$



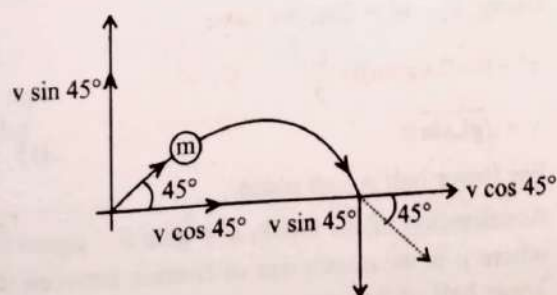
36. (d) Momentum of third part will be equal to vector sum of momentum of first two part.



$$\Rightarrow M(4) = \sqrt{(12)^2 + (16)^2}$$

$$4M = 20 \Rightarrow M = 5 \text{ kg}$$

37. (d)



$$p_i = mv \cos 45^\circ \hat{i} + mv \sin 45^\circ \hat{j}$$

$$p_f = mv \cos 45^\circ \hat{i} - mv \sin 45^\circ \hat{j}$$

[Velocity component in a horizontal direction remains same]

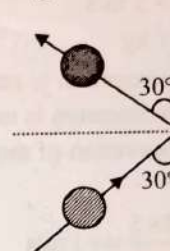
$$\Delta p = p_i - p_f$$

$$= mv \sin 45^\circ \hat{j} + mv \cos 45^\circ \hat{i} - (mv \cos 45^\circ \hat{i} - mv \sin 45^\circ \hat{j})$$

$$= 2mv \sin 45^\circ \hat{j}$$

$$|\Delta p| = (\sqrt{2}mv) \hat{j} \text{ magnitude}$$

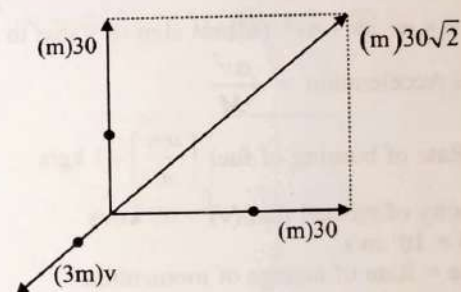
38. (c) Here, $F \times t = \text{change in momentum} = 2mv \sin \theta$



$$\therefore F = \frac{2mv \sin \theta}{t} = \frac{2 \times 0.5 \times 12 \times \sin 30^\circ}{0.25} = 48 \times \frac{1}{2} = 24 \text{ N}$$

39. (a) Apply conservation of linear momentum

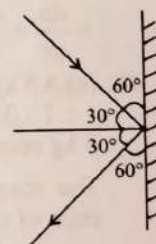
$$\Rightarrow 3mv = 30\sqrt{2}m \Rightarrow v = 10\sqrt{2} \text{ ms}^{-1}$$



40. (a) $F = \frac{\Delta p}{\Delta t} = \frac{mv \cos 30^\circ - (-mv \cos 30^\circ)}{\Delta t}$

$$= \frac{2mv \cos 30^\circ}{\Delta t} = \frac{2 \times 3 \times 10 \times \sqrt{3}}{0.2 \times 2}$$

$$= 150\sqrt{3} \text{ N}$$



41. (c) Velocity after 5 second

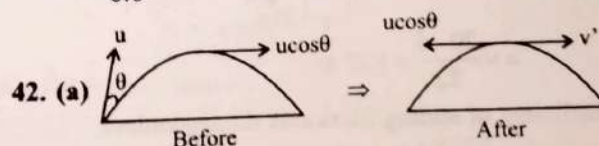
$$v = 100 - 10 \times 5 = 50 \text{ m/s.}$$

From conservation of momentum

$$1 \times 50 \hat{j} = 0.4 \times 25(-\hat{j}) + 0.6 \vec{v}$$

$$50 \hat{j} + 10 \hat{j} = 0.6 \vec{v}$$

$$\vec{v} = \frac{60 \hat{j}}{0.6} = 100 \hat{j} = 100 \text{ m/s } \hat{j}$$



$$mu \cos \theta = -\frac{mu}{2} \cos \theta + \frac{m}{2} v' \Rightarrow v' = 3u \cos \theta$$

43. (c) Here, $v \frac{dm}{dt} = m(4.9 + 9.8) = (14.7) m$

$v = 2 \text{ km/s}, \quad m = 1000$

$\therefore \frac{dm}{dt} = \frac{14.7}{2} = 7.35 \text{ kg/s}$

44. (a) $\therefore F = \frac{dP}{dt} \Rightarrow Fdt = dP$

$\Delta P = \text{Impulse} = \int Fdt = \int (500 - 100t)dt = 500t - 50t^2$

45. (a) Mass of bullet (m_1) = 200 gm = 0.2 kg

Speed of bullet (v_1) = 5 m/s

Mass of gun (m_2) = 1 kg

Before firing, total momentum is zero.

$\therefore$ After firing total momentum is $m_1 v_1 + m_2 v_2$

From the law of conservation of momentum

$m_1 v_1 + m_2 v_2 = 0$

or $v_2 = \frac{-m_1 v_1}{m_2} = \frac{-0.2 \times 5}{1} = -1 \text{ m/s}$

46. (d) Rate of change of mass $\frac{dM}{dt} = \alpha v$.

Retarding force = Rate of change of momentum

= Velocity $\times$ Rate of change in mass = $-v \times \frac{dM}{dt}$

= $-v \times \alpha v \Rightarrow -\alpha v^2$. (Minus sign of v due to deceleration)

Thus Acceleration = $-\frac{\alpha v^2}{M}$.

47. (b) Rate of burning of fuel $\left(\frac{dm}{dt}\right) = 1 \text{ kg/s}$

Velocity of ejected fuel (v) = 60 km/s

= $60 \times 10^3 \text{ m/s}$.

Force = Rate of change of momentum

= $\frac{dp}{dt} = \frac{d(mv)}{dt} = v \frac{dm}{dt} = (60 \times 10^3) \times 1 = 60000 \text{ N}$

48. (a) A 5 kg body explodes into three fragments with mass ratio 1 : 1 : 3 thus, masses of fragments will be 1 kg, 1 kg and 3 kg respectively.

The magnitude of resultant momentum of two fragments each of mass 1 kg.

$m'v' = \sqrt{(m_1 v_1)^2 + (m_2 v_2)^2}$

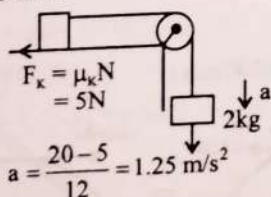
$m'v' = \sqrt{(21)^2 + (21)^2} = 21\sqrt{2} \text{ kg m/s}$

According to law of conservation of linear momentum

$m_3 v_3 = m'v' = 21\sqrt{2}$ or $3v_3 = 21\sqrt{2}$

Or $v_3 = 7\sqrt{2} \text{ m/s}$

49. (d)

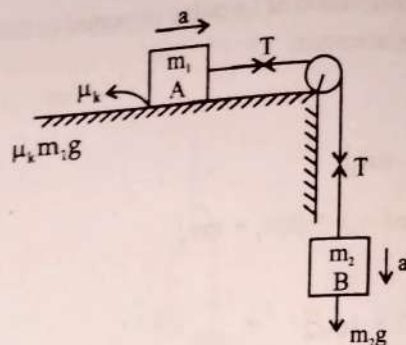


50. (d) Coefficient of sliding friction is dimensionless.

$\therefore F = \mu N \Rightarrow \mu = \frac{F}{N}$

(F and N have same unit i.e., Newton)

51. (b)



For the motion of both blocks

$m_2 g - T = m_2 a$

$T - \mu_k m_1 g = m_1 a \Rightarrow a = \frac{(m_2 - \mu_k m_1) g}{m_1 + m_2}$

For the block of mass ' m_2 '

$m_2 g - T = m_2 \left[\frac{m_2 - \mu_k m_1}{m_1 + m_2} \right] g$

$T = m_2 g - \left[\frac{m_2 - \mu_k m_1}{m_1 + m_2} \right] m_2 g = m_2 g \left[\frac{m_1 + \mu_k m_1}{m_1 + m_2} \right]$

$\Rightarrow T = \frac{m_1 m_2 (1 + \mu_k) g}{m_1 + m_2}$

52. (c) Coefficient of static friction,

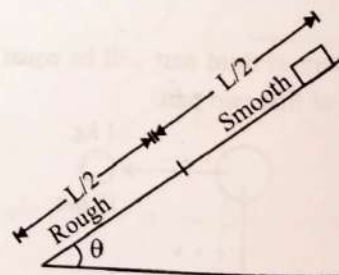
$\mu_s = \tan 30^\circ = \frac{1}{\sqrt{3}} = 0.6$

$a = g \sin 30^\circ - \mu_k g \cos 30^\circ$

$S = ut + \frac{1}{2} at^2$ [$\because u = 0$]

$\Rightarrow 4 = \frac{1}{2} \left[\frac{g}{2} - \frac{\mu_k g \sqrt{3}}{2} \right] \times 16 \Rightarrow \mu_k = 0.5$

53. (a)



Let m be mass of the block and L be length of the inclined plane.

For upper half smooth plane

Acceleration of the block, $a = g \sin \theta$

Here, $u = 0$

Using, $v^2 - u^2 = 2as$, we have

$v^2 - 0 = 2 \times g \sin \theta \times \frac{L}{2}$

$v = \sqrt{gL \sin \theta}$

...(i)

For lower half rough plane

Acceleration of the block, $a' = g \sin \theta - \mu g \cos \theta$

where μ is the coefficient of friction between the block and lower half of the plane

Here, $u = v = \sqrt{gL \sin \theta}$

$v = 0$

($\because$ block comes to rest)

$a = a' = g \sin \theta - \mu g \cos \theta$, $s = \frac{L}{2}$

Again, using $v^2 - u^2 = 2as$, we have

$0 - (\sqrt{gL \sin \theta})^2 = 2 \times (g \sin \theta - \mu g \cos \theta) \times \frac{L}{2}$

$-gL \sin \theta = (g \sin \theta - \mu g \cos \theta)L$

$-\sin \theta = \sin \theta - \mu \cos \theta$

$\mu \cos \theta = 2 \sin \theta$

$\Rightarrow \mu = 2 \tan \theta$

54. (d) Acceleration due to friction

$\frac{F}{m} = \frac{\mu mg}{m} = \mu g$

$= 0.5 \times 10 = 5 \text{ m/s}^2$

Here, μmg is force due to friction.

Using equation,

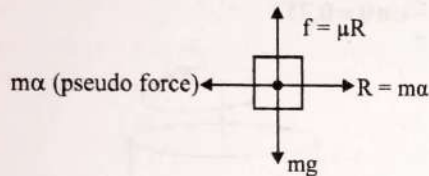
$v^2 - u^2 = 2as \Rightarrow 0^2 - 2^2 = 2(-5) \times s$

$\Rightarrow s = 0.4 \text{ m}$

55. (d) Here friction force provides centripetal force so $f = m\omega^2 r$ but $f \leq \mu mg$

So $m\omega^2 r \leq \mu mg \Rightarrow r \leq \frac{\mu g}{\omega^2}$

56. (d)



Body will remain connected if

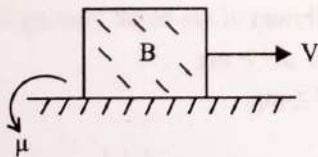
$f \geq mg$

$\Rightarrow \mu R \geq mg$

$\mu m\alpha \geq mg$

$\alpha \geq \frac{g}{\mu}$ ($\because R = m\alpha$)

57. (d)



μ - Coefficient of sliding friction

Frictional force acting on the block $= \mu \times (mg)$

$\because$ Acceleration acting in the opposite direction

of motion $= \frac{\mu mg}{m} = \mu g$

$\therefore v = u + at$ Since, $v = 0$

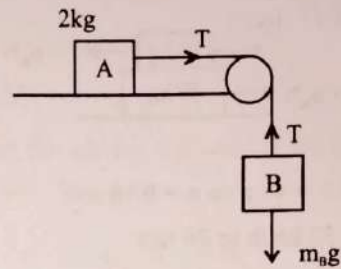
$0 = V - \mu gt$ $u = V$

$\Rightarrow V = \mu gt$ $a = -\mu g$

$t = \frac{V}{\mu g}$

Block B will come to rest after a time $\left(\frac{V}{g\mu} \right)$

58. (c)



Block B

$T - m_B g = m_B a$

Since blocks do not move

$a = 0$

$\Rightarrow T - m_B g = 0$

$T = m_B g$

Block A

$f - T = m_A a$

$a = 0$

[$\because$ blocks do not move]

$f = T$, $f = \mu m_A g$

$\mu m_A g = T$

...(ii)

From Eq. (i) and (ii)

$m_B g = \mu m_A g$

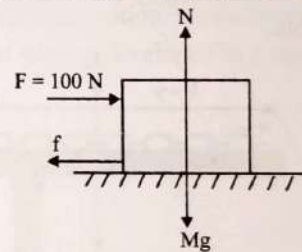
$\Rightarrow m_B = \mu m_A$

$m_B = 0.2 \times 2$

$\therefore \mu = 0.2$

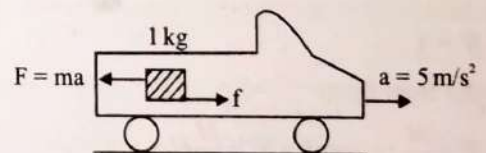
$m_B = 0.4 \text{ kg}$

59. (b) $f_{\max} = \mu N = \mu Mg = (0.5)(10)(10) = 50 \text{ N}$



$\Rightarrow a = \frac{\text{net force}}{\text{mass}}$
 $= \frac{100 - 50}{10} = 5 \text{ m/s}^2$

60. (a)



$f_L = \mu_s N$

$= \mu_s mg$

$= 0.6 \times 1 \times 10 = 6 \text{ N}$

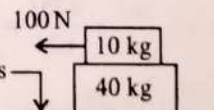
$F = \text{pseudo force} = ma = 1 \times 5$

$F = 5 \text{ N}$

$F < f_L$ block does not move so, static friction = applied force

$\therefore f_L = 5 \text{ N}$

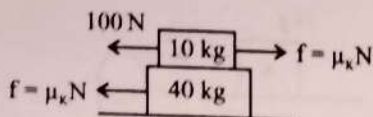
61. (b) Friction less



Let the net acceleration of the slab be "a" limiting friction

$F_s = \mu mg = 0.6 \times 10 \times 9.8 = 58.8 \text{ N}$

As $100 \text{ N} > 58.8 \text{ N}$



$$f = 40 \text{ a}$$

$$0.4 \times 10 \times 9.8 = 40 \text{ a} \Rightarrow a = 0.98 \text{ m/s}^2$$

62. (b) Speed = 72 km/h or 20 m/s

$$\mu = 0.5$$

$$ma = \mu mg \Rightarrow a = \mu g$$

$$a = 0.5 \times 10 \Rightarrow a = 5 \text{ m/s}^2$$

$$\text{for retardation } a = -5 \text{ m/s}^2$$

$$v^2 - u^2 = 2as$$

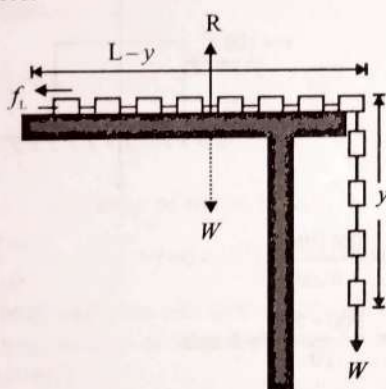
$$\text{Car is stopped } v = 0$$

$$s = -\frac{u^2}{2a}$$

$$s = -\frac{20 \times 20}{(-5) \times 2} \Rightarrow S = 40 \text{ metres}$$

63. (a) Let M is the mass of the chain in length L . If y is the maximum length of chain that can hang outside the table without sliding.

Then for equilibrium of the chain, the weight of hanging part must be balanced by the force of friction on the portion on the table.



$$W = f_L$$

But from figure

$$W = \frac{M}{L} yg \text{ and } R = W' = \frac{M}{L} (L-y)g$$

So that

$$f_L = \mu R = \mu \frac{M}{L} (L-y)g$$

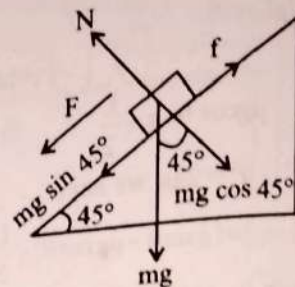
Substituting these values of W and f_L in Eq. (i), we get

$$\mu \frac{M}{L} (L-y)g = \frac{M}{L} yg$$

$$\text{Or } \mu (L-y) = y \text{ or } y = \frac{\mu L}{\mu + 1} = \frac{0.25L}{1.25} = \frac{L}{5}$$

$$\frac{y}{L} = \frac{1}{5} = 20\%$$

64. (b) Forces acting on the body have been shown in the figure. The force on the body down the inclined plane in presence of friction μ is



$$F = mg \sin \theta - f = mg \sin \theta - \mu N = ma$$

$$a = g \sin \theta - \mu g \cos \theta$$

$$\text{initial velocity } u = 0$$

$\therefore$ Time taken to slide down the plane

$$t_1 = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2s}{g \sin \theta - \mu g \cos \theta}}$$

In absence of friction time taken will be

$$t_2 = \sqrt{\frac{2s}{g \sin \theta}}$$

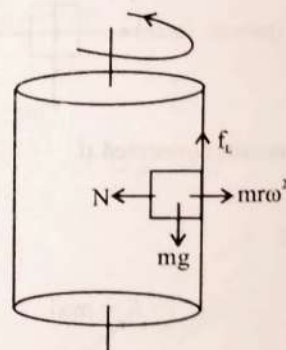
$$\text{Given : } t_1 = 2t_2$$

$$\therefore t_1^2 = 4t_2^2 \text{ or } \frac{2s}{g(\sin \theta - \mu \cos \theta)} = \frac{2s \times 4}{g(\sin \theta)}$$

$$\text{Or } \sin \theta = 4 \sin \theta - 4\mu \cos \theta$$

$$\text{or } \mu = \frac{3}{4} \tan \theta = 0.75$$

65. (c)



For equilibrium of the block limiting friction

$$f_L \geq mg \Rightarrow \mu N \geq mg$$

$$\Rightarrow \mu mr\omega^2 \geq mg$$

$$\omega \geq \sqrt{\frac{g}{r\mu}}$$

Therefore,

$$\omega_{\min} = \sqrt{\frac{g}{r\mu}}$$

$$\omega_{\min} = \sqrt{\frac{10}{0.1 \times 1}} = 10 \text{ rad/s}$$

66. (d) In uniform circular motion, tension provides the necessary centripetal force required to keep particle in motion.

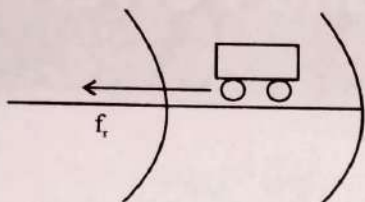
So, net force on particle = T .

67. (b) We know that $\frac{v^2}{gR} = \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)$

$$v = \sqrt{gR \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}$$

$$68. (b) (F_c)_{\text{heavier}} = (F_c)_{\text{lighter}} \Rightarrow \frac{2mV^2}{(r/2)} = \frac{m(nV)^2}{r} \Rightarrow n^2 = 4 \Rightarrow n = 2$$

69. (d)



$$f_r = F_{CP} = \frac{Mv^2}{R}$$

$$\text{But } f_r \leq \mu Mg$$

$$\text{So, } v^2 \leq \mu_s Rg$$

$$v_{\text{max}} = \sqrt{\mu_s Rg}$$

$$70. (b) \text{ Speed of car} = \sqrt{Rg \tan \theta} = \sqrt{(90)(10)(\tan 45^\circ)} \\ = \sqrt{900} = 30 \text{ m/s}$$

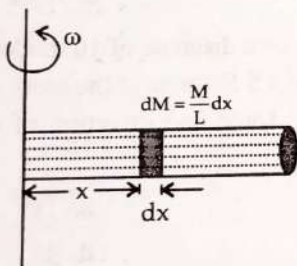
71. (b) To experience weightlessness,

$$\frac{mv^2}{R} = mg \Rightarrow v = \sqrt{Rg}$$

$$= \sqrt{20 \times 10} = \sqrt{200} \text{ ms}^{-1} = 14.14 \text{ ms}^{-1}$$

The speed of the car at the top of the hill is between 14 m/s and 15 m/s.

72. (a)



Let us consider a small mass element dm at a distance x from the axis of rotation.

Force exerted by liquid at the other ends

$$= \int df$$

$$= \int df(\omega)x$$

We will get the net force exerted if we integrate the force from $x = 0$ to $x = L$

$$F = \int_0^L dm(\omega^2 x) \quad \text{Since, } dm = \frac{M}{L} dx$$

$$= \int_0^L \frac{M}{L} \omega^2 x dx$$

$$= \left[\frac{M}{L} \omega^2 \frac{x^2}{2} \right]_0^L$$

$$= \frac{M\omega^2 L^2}{L \times 2} - \frac{M\omega^2 (0)^2}{L \times 2} = \frac{M\omega^2 L}{2} = (M\omega^2 L) / (2)$$

73. (a) Frequency = Number of revolution in 1 sec

$$f = \frac{22}{44}$$

$$f = \frac{1}{2} \text{ s}^{-1}$$

$$a = \omega^2 r = 4\pi^2 f^2 r = 4\pi^2 \times \frac{1}{4} \times 1 = \pi^2 \text{ ms}^{-2}$$

74. (a) In a uniform circular motion, necessary centripetal force is provided by the tension developed in a string.

$$F_{\text{cent}} = T$$

$$\frac{mv^2}{l} = T$$

$$v = \sqrt{\frac{Tl}{m}}$$

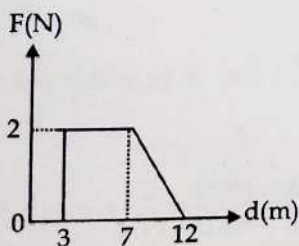
75. (b) Particles of cream are lighter so they get deposited near the centre of circular path.

When milk is churned, cream gets separated due to centrifugal force.

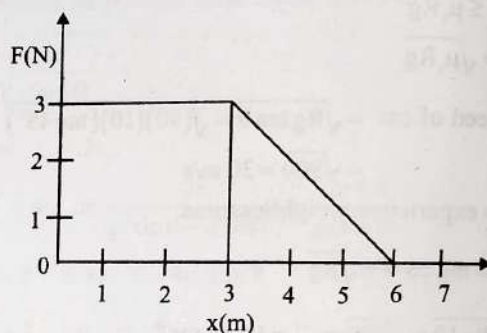
Work, Energy and Power

Work Done

1. A force $F = 20 + 10y$ acts on a particle in y direction where F is in newton and y in meter. Work done by this force to move the particle from $y = 0$ to $y = 1$ m is (2019)
 - a. 30 J
 - b. 5 J
 - c. 25 J
 - d. 20 J
2. A particle moves from a point $(-2\hat{i} + 5\hat{j})$ to $(4\hat{j} + 3\hat{k})$ when a force of $(4\hat{i} + 3\hat{j})$ N is applied. How much work has been done by the force? (2016 - II)
 - a. 5 J
 - b. 2 J
 - c. 8 J
 - d. 11 J
3. A uniform force of $(3\hat{i} + \hat{j})$ newton acts on a particle of mass 2 kg. Hence the particle is displaced from position $(2\hat{i} + \hat{k})$ metre to position $(4\hat{i} + 3\hat{j} - \hat{k})$ metre. The work done by the force on the particle is: (2013)
 - a. 15 J
 - b. 9 J
 - c. 6 J
 - d. 13 J
4. Force F on a particle moving in a straight line varies with distance d as shown in the figure. The work done on the particle during its displacement of 12 m is: (2011 Pre)



5. A force F acting on an object varies with distance x as shown here. The force is in N and x is in m. The work done by the force in moving the object from $x = 0$ to $x = 6$ m is: (2003)



- a. 10.5 J
 - b. 13.5 J
 - c. 8.5 J
 - d. 6.5 J
6. A body moves a distance of 10 m along a straight line under the action of a 5 N force. If the work done is 25 J, then angle between the force and direction of motion of the body is: (1997)
 - a. 60°
 - b. 75°
 - c. 30°
 - d. 45°
 7. A body, constrained to move in y -direction, is subjected to a force given by $\vec{F} = (-2\hat{i} + 15\hat{j} + 6\hat{k})$ N. The work done by this force in moving the body through a distance of 10 m along y -axis, is: (1994)
 - a. 150 J
 - b. 20 J
 - c. 190 J
 - d. 160 J
 8. A position dependent force, $F = (7 - 2x + 3x^2)$ N acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5$ m. The work done in joule is: (1994, 92)
 - a. 135
 - b. 270
 - c. 35
 - d. 70
 9. When a body moves with a constant speed along a circle: (1994)
 - a. No work is done on it
 - b. No acceleration is produced in it
 - c. Its velocity remains constant
 - d. No force acts on it

- a. 18 J
- b. 21 J
- c. 26 J
- d. 13 J

Energy and Conservation of Mechanical Energy

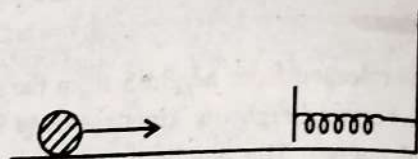
10. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively: (2021)
- $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
 - $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$
 - $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$
 - $\frac{S}{4}, \frac{3gS}{2}$
11. A body of mass $(4m)$ is lying in x - y plane at rest. It suddenly explodes into three pieces. Two pieces each of mass (m) move perpendicular to each other with equal speeds (v) . The total kinetic energy generated due to explosion is: (2014)
- mv^2
 - $3/2mv^2$
 - $2mv^2$
 - $4mv^2$
12. An explosion breaks a rock into three parts in a horizontal plane. Two of them go off at right angles to each other. The first part of mass 1 kg moves with a speed of 12 ms^{-1} and the second part of mass 2 kg moves with 8 ms^{-1} speed. If the third part flies off with 4 ms^{-1} speed, then its mass is: (2013, 2009)
- 17 kg
 - 3 kg
 - 5 kg
 - 7 kg
13. The potential energy of particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$ where A and B are positive constants and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is: (2012 Pre)
- $B/2A$
 - $2A/B$
 - A/B
 - B/A
14. The potential energy of a system increases if work is done: (2011 Pre)
- Upon the system by a nonconservative force
 - By the system against a conservative force
 - By the system against a nonconservative force
 - Upon the system by a conservative force
15. A shell of mass 200 gm is ejected from a gun of mass 4 kg by an explosion that generates 1.05 kJ of energy. The initial velocity of the shell is (2008)
- 40 m/s
 - 120 m/s
 - 100 m/s
 - 80 m/s
16. A bomb of mass 30 kg at rest explodes into two pieces of masses 18 kg and 12 kg . The velocity of 18 kg mass is 6 ms^{-1} . The kinetic energy of the other mass is: (2005)
- 243 J
 - 486 J
 - 564 J
 - 388 J
17. A particle of mass m_1 is moving with a velocity v_1 and another particle of mass m_2 is moving with a velocity v_2 . Both of them have the same momentum but their different kinetic energies are E_1 and E_2 respectively. If $m_1 > m_2$ then: (2004)
- $\frac{E_1}{E_2} = \frac{m_1}{m_2}$
 - $E_1 > E_2$
 - $E_1 = E_2$
 - $E_1 < E_2$
18. A ball of mass 2 kg and another of mass 4 kg are dropped together from a 60 feet tall building. After a fall of 30 feet each towards earth, their respective kinetic energies will be in the ratio of: (2004)
- $1 : 4$
 - $1 : 2$
 - $1 : \sqrt{2}$
 - $\sqrt{2} : 1$
19. A stationary particle explodes into two particles of masses m_1 and m_2 which move in opposite directions with velocities v_1 and v_2 . The ratio of their kinetic energies E_1/E_2 is: (2003)
- m_2/m_1
 - m_1/m_2
 - 1
 - m_1v_2/m_2v_1
20. If kinetic energy of a body is increased by 300% then percentage change in momentum will be: (2002)
- 100%
 - 150%
 - 265%
 - 73.2%
21. A child is sitting on a swing. Its minimum and maximum heights from the ground is 0.75 m and 2 m respectively, its maximum speed will be: (2001)
- 10 m/s
 - 5 m/s
 - 1 m/s
 - 15 m/s
22. The K.E. of a person is just half of K.E. of a boy whose mass is just half of that person. If person increases its speed by 1 m/s , then its K.E. equals to that of boy then initial speed of person was: (1999)
- $(\sqrt{2}+1)\text{ m/s}$
 - $(2+\sqrt{2})\text{ m/s}$
 - $2(\sqrt{2}+2)\text{ m/s}$
 - None
23. Two bodies of masses m and $4m$ are moving with equal kinetic energies. The ratio of their linear momenta is: (1998, 97, 89)
- $1 : 2$
 - $1 : 4$
 - $4 : 1$
 - $1 : 1$
24. The potential energy between two atoms, in a molecule, is given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$ where a and b are positive constants and x is the distance between the atoms. The atom is in stable equilibrium, when: (1995)
- $x = \left(\frac{2a}{b}\right)^{1/6}$
 - $x = \left(\frac{11a}{5b}\right)^{1/6}$
 - $x = 0$
 - $x = \left(\frac{a}{2b}\right)^{1/6}$

- 25.** The kinetic energy acquired by a mass m in travelling distance d , starting from rest, under the action of a constant force is directly proportional to:
- (1994)
- a. m b. m^0
- c. $\sqrt{m}$ d. $1/\sqrt{m}$
- 26.** Two masses of 1 g and 9 g are moving with equal kinetic energies. The ratio of the magnitudes of their respective linear momenta is:
- (1993)
- a. 1 : 9 b. 9 : 1
- c. 1 : 3 d. 3 : 1
- 27.** A particle of mass M is moving in a horizontal circle of radius R with uniform speed v . When it moves from one point to a diametrically opposite point, its:
- (1992)
- a. Kinetic energy change by $Mv^2/4$
- b. Momentum does not change
- c. Momentum change by $2Mv$
- d. Kinetic energy changes by Mv^2

Spring

28. Two similar springs P and Q have spring constants K_p and K_q such that $K_p > K_q$. They stretched first by the same amount (case a), then by the same force (case b). The work done by the springs W_p and W_q are related as in case (a) and case (b), respectively: (2015)
- a. $W_p = W_q$; $W_p = W_q$ b. $W_p > W_q$; $W_q > W_p$
c. $W_p < W_q$; $W_q < W_p$ d. $W_p = W_q$; $W_p > W_q$
29. A block of mass M is attached to the lower end of a vertical spring. The spring is hung from a ceiling and has force constant value k . The mass is released from rest with the spring initially unstretched. The maximum extension produced in the length of the spring will be: (2009)
- a. $2 Mg/k$ b. $4 Mg/k$
c. $Mg/2k$ d. Mg/k
30. A vertical spring with force constant k is fixed on a table. A ball of mass m at a height h above the free upper end of the spring falls vertically on the spring so that the spring is compressed by a distance d . The net work done in the process is: (2007)
- a. $mg(h+d) - \frac{1}{2}kd^2$ b. $mg(h-d) - \frac{1}{2}kd^2$
c. $mg(h-d) + \frac{1}{2}kd^2$ d. $mg(h+d) + \frac{1}{2}kd^2$
31. The potential energy of a long spring when stretched by 2 cm is U . If the spring is stretched by 8 cm the potential energy stored in it is: (2006)
- a. $4U$ b. $U/8$
c. $16U$ d. $U/4$

32. A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50 \text{ N/m}$. The maximum compression of the spring would be:



- a. 0.12 m
b. 1.5 m
c. 0.5 m
d. 0.15 m
33. When a long spring is stretched by 2 cm, its potential energy is U . If the spring is stretched by 10 cm, the potential energy stored in it will be: (2003)
- a. $U/5$
b. $5U$
c. $10U$
d. $25U$
34. Two springs A and B having spring constant K_A and K_B ($K_A = 2K_B$) are stretched by applying force of equal magnitude. If energy stored in spring A is E then energy stored in B will be: (2001)
- a. $2E$
b. $\frac{E}{4}$
c. $\frac{E}{2}$
d. $4E$
35. Two springs A and B ($K_A = 2K_B$) are stretched by same suspended weights then ratio of work done in stretching is (1999)
- a. $1 : 2$
b. $2 : 1$
c. $1 : 1$
d. $1 : 4$
36. When a spring is subjected to 4 N force its length is a metre, and if 5 N is applied length is b metre. If 9 N is applied its length is: (1999)
- a. $4b - 3a$
b. $5b - a$
c. $5b - 4a$
d. $5b - 2a$

Work Energy Theorem

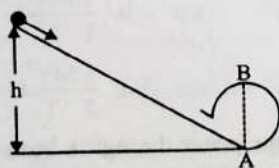
37. Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value 10 m/s². The work done by the (i) gravitational force and the (ii) resistive force of air is: (2017-Dehra)
- a. (i) 1.25 J (ii) -8.25 J
b. (i) 100 J (ii) 8.75 J
c. (i) 10 J (ii) -8.75 J
d. (i) -10 J (ii) -8.25 J
38. A bullet of mass 10 g moving horizontally with a velocity of 400 ms⁻¹ strikes a wooden block of mass 2 kg which is suspended by a light inextensible string of length 5 m. As a result, the center of gravity of the block is found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be: (2016 - 10)
- a. 120 ms⁻¹
b. 160 ms⁻¹
c. 100 ms⁻¹
d. 80 ms⁻¹

39. A block of mass 10 kg moving in x direction with a constant speed of 10 ms^{-1} , is subjected to a retarding force $F = -0.1x \text{ J/m}$ during its travel from $x = 20 \text{ m}$ to 30 m . Its final K.E. will be: (2015)
 a. 450 J b. 275 J
 c. 250 J d. 475 J
40. A body of mass 1 kg is thrown upwards with a velocity 20 m/s. It momentarily comes to rest after attaining a height of 18 m. How much energy is lost due to air friction? ($g = 10 \text{ m/s}^2$) (2009)
 a. 30 J b. 40 J
 c. 10 J d. 20 J
41. 300 J of work is done in sliding a 2 kg block up an inclined plane of height 10 m. Taking $g = 10 \text{ m/s}^2$, work done against friction is: (2006)
 a. 50 J b. 100 J
 c. Zero d. 150 J
42. A body of mass 3 kg is under a constant force which causes a displacement s in meters in it, given by the relation $s = (1/3)t^2$, where t is in s. Work done by the force in 2 s is: (2006)
 a. $(17/3)\text{J}$ b. $(3/8)\text{J}$
 c. $(8/3)\text{J}$ d. $(3/17)\text{J}$
43. If $x = 3 - 4t^2 + t^3$, then work done in first 4 will be (Mass of the particle is 3 gram): (1998)
 a. 384 mJ
 b. 168 mJ
 c. 192 mJ
 d. None of these
44. A bullet of mass 10 g leaves a rifle at an initial velocity of 100 m/s and strikes the earth at the same level with a velocity of 500 m/s. The work done in joule overcoming the resistance of air will be: (1989)
 a. 375 b. 3750
 c. 5000 d. 500
- Power**
45. An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 ms^{-1} . The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is : ($g = 10 \text{ ms}^{-2}$) (2022)
 a. 23500 b. 23000
 c. 20000 d. 34500
46. The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is: (2022)
 a. $1 \times 10^5 \text{ J}$ b. $36 \times 10^7 \text{ J}$
 c. $36 \times 10^4 \text{ J}$ d. $36 \times 10^5 \text{ J}$
47. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? ($g = 10 \text{ m/s}^2$) (2021, 2008)
 a. 8.1 kW b. 12.3 kW
 c. 7.0 kW d. 10.2 kW
48. A body of mass 1 kg begins to move under the action of a time dependent force $F = (2\hat{i} + 3t^2\hat{j}) \text{ N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t ? (2016 - I)
 a. $(2t^2 + 3t^2)\text{W}$ b. $(2t^2 + 4t^4)\text{W}$
 c. $(2t^3 + 4t^4)\text{W}$ d. $(2t^3 + 3t^5)\text{W}$
49. A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is: (2015)
 a. $\sqrt{mkt}^{-1/2}$ b. $\sqrt{2mkt}^{-1/2}$
 c. $\frac{1}{2}\sqrt{mkt}^{-1/2}$ d. $\sqrt{\frac{mk}{2}}t^{-1/2}$
50. The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be $13.6 \times 10^3 \text{ kg/m}^3$ and $g = 10 \text{ m/s}^2$, then the power of heart in watt is: (2015 Re)
 a. 1.50 b. 1.70
 c. 2.35 d. 3.0
51. A car of mass m starts from rest and accelerates so that the instantaneous power delivered to the car has a constant magnitude P_0 . The instantaneous velocity of this car is proportional to: (2012 Mains)
 a. $t^2 P_0$ b. $t^{1/2}$
 c. $t^{1/2}$ d. $\frac{1}{\sqrt{m}}$
52. A body projected vertically from the earth reaches a height equal to earth's radius before returning to the earth. The power exerted by the gravitational force is greatest (2011 Pre)
 a. At the highest position of the body
 b. At the instant just before the body hits the earth
 c. It remains constant all through
 d. At the instant just after the body is projected
53. A particle of mass M starting from rest undergoes uniform acceleration. If the speed acquired in time T is V , the power delivered to the particle is: (2010 Mains)
 a. $\frac{1}{2} \frac{MV^2}{T^2}$ b. $\frac{2}{1} \frac{MV^2}{T^2}$
 c. $\frac{MV^2}{T^2}$ d. $\frac{1}{2} \frac{MV^2}{T}$
54. An engine pumps water through a hose pipe. Water passes through the pipe and leaves it with a velocity of 2 m/s. The mass per unit length of water in the pipe is 100 kg/m. What is the power of the engine? (2010 Pre)
 a. 800 W b. 400 W
 c. 200 W d. 100 W

55. An engine pumps water continuously through a hose. Water leaves the hose with a velocity v and m is the mass per unit length of the water jet. What is the rate at which kinetic energy is imparted to water? (2009)
- a. mv^3 b. $\frac{1}{2}mv^2$
c. $\frac{1}{2}m^2v^2$ d. $\frac{1}{2}mv^3$
56. 250 N force is required to raise 75 kg mass from a pulley. If rope is pulled 12 m then the load is lifted to 3 m, the efficiency of pulley system will be: (2001)
- a. 25% b. 33.3%
c. 75% d. 90%
57. If $\vec{F} = (60\hat{i} + 15\hat{j} - 3\hat{k})\text{N}$ and $\vec{V} = (2\hat{i} - 4\hat{j} + 5\hat{k})\text{m/s}$ then instantaneous power is: (2000)
- a. 195 watt b. 45 watt
c. 75 watt d. 100 watt
58. How much water a pump of 2 kW can raise in one minute to a height of 10 m? (Take $g = 10\text{ m/s}^2$) (1990)
- a. 1000 litres b. 1200 litres
c. 100 litres d. 2000 litres

Vertical Circle

59. A point mass ' m ' is moved in a vertical circle of radius ' r ' with the help of a string. The velocity of the mass is $\sqrt{7gr}$ at the lowest point. The tension in the string at the lowest point is (2020-Covid)
- a. 7 mg b. 8 mg
c. 1 mg d. 6 mg
60. A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when: (2019)
- a. The mass is at the highest point
b. The wire is horizontal
c. The mass is at the lowest point
d. Inclined at an angle of 60° from vertical
61. A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to: (2018)



- a. $\frac{7}{5}D$ b. D
c. $\frac{3}{2}D$ d. $\frac{5}{4}D$

62. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop? (2016-J)

- a. $\sqrt{gR}$ b. $\sqrt{2gR}$
c. $\sqrt{3gR}$ d. $\sqrt{5gR}$

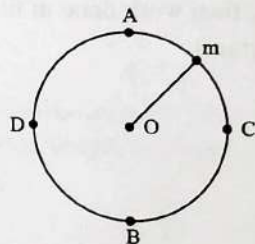
63. A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to $8 \times 10^{-4}\text{ J}$ by the end of the second revolution after the beginning of the motion? (2016-J)

- a. 0.1 m/s^2 b. 0.15 m/s^2
c. 0.18 m/s^2 d. 0.2 m/s^2

64. A stone is tied to a string of length ' ℓ ' and is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed ' u '. The magnitude of the change in velocity as it reaches a position where the string is horizontal (g being acceleration due to gravity) is: (2004)

- a. $\sqrt{u^2 - g\ell}$ b. $u - \sqrt{u^2 - 2g\ell}$
c. $\sqrt{2g\ell}$ d. $\sqrt{2(u^2 - g\ell)}$

65. A mass is performing vertical circular motion (see figure). If the average velocity of the particle is increased, then at which point the string will break: (2000)



- a. A b. B
c. C d. D

Collision

66. Body A of mass $4m$ moving with speed u collides with another body B of mass $2m$, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is: (2019)

- a. $\frac{1}{9}$ b. $\frac{8}{9}$
c. $\frac{4}{9}$ d. $\frac{5}{9}$

67. A moving block having mass m , collides with another stationary block having mass $4m$. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v , then the value of coefficient of restitution (e) will be (2018)

- a. 0.8 b. 0.25
c. 0.5 d. 0.4

68. Two identical balls A and B having velocities of 0.5 m/s and -0.3 m/s respectively collide elastically in one dimension. The velocities of B and A after the collision respectively will be: (2016, 1998, 1994, 1991)
- 0.3 m/s and 0.5 m/s
 - 0.3 m/s and 0.5 m/s
 - 0.5 m/s and 0.3 m/s
 - 0.5 m/s and -0.3 m/s
69. Two particles of masses m_1, m_2 move with initial velocities u_1 and u_2 . On collision, one of the particles get excited to higher level, after absorbing energy ϵ . If final velocities of particles be v_1 and v_2 , then we must have: (2015)
- $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \epsilon$
 - $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \epsilon = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$
 - $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 + \epsilon = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$
 - $m_1^2u_1 + m_2^2u_2 - \epsilon = m_1^2v_1 + m_2^2v_2$
70. On a frictionless surface, a block of mass M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $v/3$. The second block's speed after the collision is: (2015 Re)
- $\frac{\sqrt{3}}{2}v$
 - $\frac{2\sqrt{2}}{3}v$
 - $\frac{3}{4}v$
 - $\frac{3}{\sqrt{2}}v$
71. A ball is thrown vertically downwards from a height of 20 m with an initial velocity u_0 . It collides with the ground, loses 50 percent of its energy in collision and rebounds to the same height. The initial velocity u_0 is: (Take $g = 10 \text{ ms}^{-2}$) (2015 Re)
- 10 m/s
 - 14 m/s
 - 20 m/s
 - 28 m/s
72. Two particles A and B, move with constant motion in one dimensional with velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are $\vec{r}_1$ and $\vec{r}_2$ respectively. The condition for particle A and B for their collision is: (2015 Re)
- $\vec{r}_1 - \vec{r}_2 = \vec{v}_1 - \vec{v}_2$
 - $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|}$
 - $\vec{r}_1 \cdot \vec{v}_1 = \vec{r}_2 \cdot \vec{v}_2$
 - $\vec{r}_1 \times \vec{v}_1 = \vec{r}_2 \times \vec{v}_2$
73. Two spheres A and B of masses m_1 and m_2 respectively collide. A is at rest initially and B is moving with velocity v along x-axis. After collision B has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction. (2012 Pre)
- Same as that of B
 - Opposite of that of B
 - $\theta = \tan^{-1}(1/2)$ to the x-axis
 - $\theta = \tan^{-1}(-1/2)$ to the x-axis
74. A mass m moving horizontally (along the x-axis) with velocity v collides and sticks to a mass of $3m$ moving vertically upward (along the y-axis) with velocity $2v$. The final velocity of the combination is (2011 Mains)
- $\frac{3}{2}\hat{v}_i + \frac{1}{4}\hat{v}_j$
 - $\frac{1}{4}\hat{v}_i + \frac{3}{2}\hat{v}_j$
 - $\frac{1}{3}\hat{v}_i + \frac{2}{3}\hat{v}_j$
 - $\frac{2}{3}\hat{v}_i + \frac{1}{3}\hat{v}_j$
75. A ball moving with velocity 2 m/s collides head on with another stationary ball of double the mass. If the coefficient of restitution is 0.5 then their velocities (in m/s) after collision will be: (2010 Pre)
- 0, 2
 - 0, 1
 - 1, 1
 - 1, 0.5
76. A ball is dropped from a height of 5 m, if it rebound upto height of 1.8 m, then the ratio of velocities of the ball after and before rebound is: (1998)
- $\frac{3}{5}$
 - $\frac{2}{5}$
 - $\frac{1}{5}$
 - $\frac{4}{5}$
77. A metal ball of mass 2 kg moving with speed of 36 km/h has a head on collision with a stationary ball of mass 3 kg. If after collision, both the balls move as a single mass, then the loss in K.E. due to collision is: (1997)
- 100 J
 - 140 J
 - 40 J
 - 60 J
78. A moving body of mass m and velocity 3 km/hour collides with a rest body of mass 2 m and sticks to it. Now the combined mass starts to move. What will be the combined velocity? (1996)
- 3 km/hour
 - 4 km/hour
 - 1 km/hour
 - 2 km/hour
79. The coefficient of restitution e for a perfectly elastic collision is: (1988)
- 1
 - 0
 - ∞
 - 1

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c	a	b	d	d	a	a	a	a	c	b	c	b	b	c	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
b	a	a	b	a	a	a	b	c	c	b	a	a	c	d	d
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
a	c	c	a	d	d	b	c	a	b	d	b	a	d	d	b
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67
b	d	a	d	c	b	b	b	c	d	d	a	d	b	b	b
69	70	71	72	73	74	75	76	77	78	79					
b	b	c	b	d	b	b	a	d	c	a					

Explanations

1. (c) Work done by variable force is

$$W = \int_{y_i}^{y_f} F dy$$

Here, $y_i = 0$, $y_f = 1$ m

$$\therefore W = \int_0^1 (20 + 10y) dy = [20y + 5y^2]_0^1$$

$$= 20(1 - 0) + 5(1 - 0) = 25 \text{ J}$$

2. (a) $\vec{s} = \vec{r}_f - \vec{r}_i = 2\hat{i} - \hat{j} + 3\hat{k}$

$$W = \vec{F} \cdot \vec{s} = (4\hat{i} + 3\hat{j}) \cdot [2\hat{i} - \hat{j} + 3\hat{k}] = 8 - 3 = 5 \text{ J}$$

3. (b) $W = \vec{F} \cdot \vec{S} = (3\hat{i} + \hat{j}) \cdot [(4-2)\hat{i} + (3-0)\hat{j} + (-1-1)\hat{k}]$
 $= (3\hat{i} + \hat{j}) \cdot (2\hat{i} + 3\hat{j} - 2\hat{k}) = 3(2) + 1(3) + 0(-2) = 9 \text{ J}$

4. (d) Work done $= \int \vec{F} \cdot d\vec{x}$

= Area under $F \cdot d$ graph

$$= 2 \times (7-3) + \frac{1}{2} \times 2 \times 5 = 13 \text{ J}$$

5. (b) Work done $= \int_{x=0}^{x=6} F dx$

= the area under $F-x$ curve

$\therefore W = \text{Area of square} + \text{Area of triangle}$

$$= 3 \times 3 + \frac{1}{2} \times 3 \times 3 = 9 + 4.5 = 13.5 \text{ J}$$

6. (a) Distance (s) = 10 m; Force (F) = 5 N and work done (W) = 25 J.

Work done (W) = $Fs \cos \theta$

$$\therefore 25 = 5 \times 10 \cos \theta = 50 \cos \theta$$

$$\text{Or } \cos \theta = 25/50 = 0.5 \text{ or } \theta = 60^\circ.$$

7. (a) Work done

$$W = \vec{F} \cdot \vec{d} = (-2\hat{i} + 15\hat{j} + 6\hat{k}) \cdot (10\hat{j})$$

$$= 150 \text{ N-m} = 150 \text{ J.}$$

8. (a) Force (F) = $7 - 2x + 3x^2$;

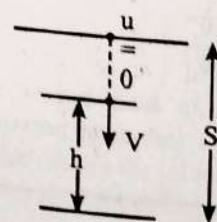
Mass (m) = 2 kg and displacement $d = 5$ m.

Therefore work done

$$(W) = \int F dx = \int_0^5 (7 - 2x + 3x^2) dx = [7x - x^2 + x^3]_0^5$$

$$= (7 \times 5) - (5)^2 + (5)^3 = 35 - 25 + 125 = 135 \text{ J.}$$

9. (a) The force applying on the body are in radial direction and moment of body is tangential (Angle between force and displacement is 90°). So the total done by forces will be zero.
10. (c) Required figure,



Particle is released, i.e., its initial velocity is zero $u = 0$ m/s.
 Let the height at any instant be 'h'.

Total energy remains conserved at any point.

$$\Rightarrow K.E + P.E = E_{\text{Total}}$$

$$\Rightarrow 3 P.E + P.E = E_{\text{Total}}$$

$$\Rightarrow 4 P.E = E_{\text{Total}}$$

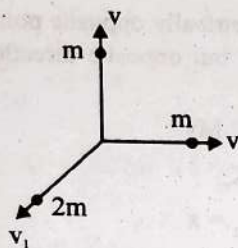
$$\Rightarrow 4(mgh) = mgS \Rightarrow h = \frac{S}{4}$$

Velocity of a particle under free fall is given by;

$$v = \sqrt{2gh}$$

$$v = \sqrt{2g\left(S - \frac{S}{4}\right)} = \sqrt{\frac{3gS}{2}}$$

11. (b)



Net momentum of equal masses = $\sqrt{2}mv$

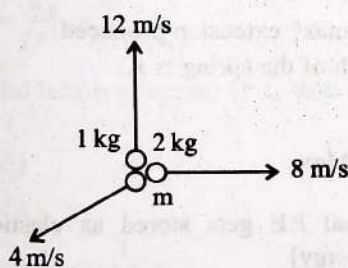
By conservation of linear momentum

$$2mv_1 = \sqrt{2}mv \Rightarrow v_1 = \frac{v}{\sqrt{2}}$$

$$\text{Total K.E generated} = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{4}(2m)v^2$$

$$= mv^2 + \frac{mv^2}{2} = \frac{3}{2}mv^2$$

12. (c)



From conservation of momentum

$$m(4) = \sqrt{(1 \times 12)^2 + (2 \times 8)^2} \Rightarrow m = 5 \text{ kg}$$

$$13. (b) F = \frac{dU}{dr} = -\frac{2A}{r^3} + \frac{B}{r^2} = 0 \Rightarrow r = \frac{2A}{B}$$

$$\text{At } r = \frac{2A}{B}; \frac{d^2U}{dr^2} = +ve, \text{ so stable equilibrium}$$

$$14. (b) \text{ As we know } \Rightarrow U = -\int \vec{F} \cdot d\vec{x}$$

where F is conservative force

$$15. (c) mv = Mv' \Rightarrow v' = \left(\frac{m}{M}\right)v$$

Total K.E. of the shell and gun

$$= \frac{1}{2}mv^2 + \frac{1}{2}Mv'^2$$

$$\text{Total K.E.} = \frac{1}{2}mv^2 + \frac{1}{2}M \cdot \frac{m^2}{M^2}v^2$$

$$\text{Total K.E.} = \frac{1}{2}mv^2 \left\{1 + \frac{m}{M}\right\}$$

$$= \left\{\frac{1}{2} \times 0.2\right\} \left\{1 + \frac{0.2}{4}\right\} v^2 = 1.05 \times 1000 \text{ J}$$

$$\Rightarrow v^2 = \frac{4 \times 1.05 \times 1000}{0.1 \times 4.2} = 100^2$$

$$\therefore v = 100 \text{ m/s}$$

16. (b) According to law of conservation of linear momentum

$$m_1v_1 + m_2v_2 = Mu \Rightarrow m_1v_1 + m_2v_2 = 0$$

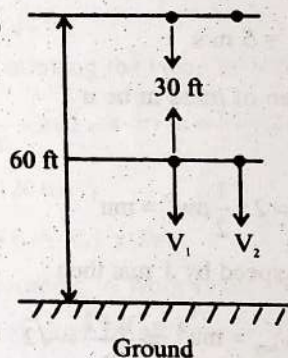
$$\Rightarrow 18 \times 6 = -12 \times v_2 \Rightarrow v_2 = -9 \text{ m/s}$$

$$\text{So, kinetic energy of other mass} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 12 \times (-9)^2 = 6 \times 81 = 486 \text{ J}$$

$$17. (d) \because E = \frac{P^2}{2m} \therefore E \propto \frac{1}{m} \text{ if } m_1 > m_2 \text{ then } E_1 < E_2$$

18. (b)



$$\frac{KE_1}{KE_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_2v_2^2} = \frac{m_1}{m_2} = \frac{2}{4} = \frac{1}{2} [\because V_1 = V_2]$$

$$19. (a) m_1v_1 = m_2v_2 (P_1 = P_2);$$

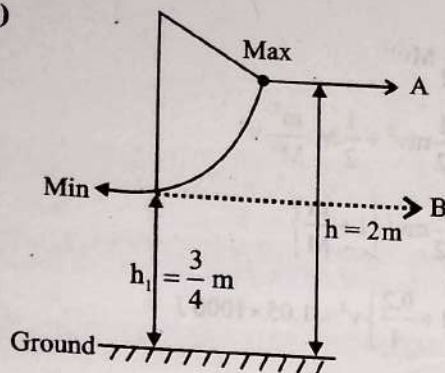
$$\frac{E_1}{E_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_2v_2^2} = \frac{\frac{P_1^2}{2m_1}}{\frac{P_2^2}{2m_2}} = \frac{m_2}{m_1} [\because P_1 = P_2]$$

$$20. (a) P_1 = \sqrt{2mE_1}$$

$$P_2 = \sqrt{2mE_2} = \sqrt{2m\left(E_1 + \frac{300}{100}E_1\right)} = \sqrt{2m(4E_1)} = 2P_1$$

$$\% \text{ change} = \frac{P_2 - P_1}{P_1} \times 100 = \frac{2P_1 - P_1}{P_1} \times 100 = 100\%$$

21. (b)



$$h = 2\text{m}$$

$$h_1 = \frac{3}{4}\text{m or } (0.75)$$

According to conservation of energy

$$P_A + K_A = P_B + K_B$$

$$mgh + 0 = mgh_1 + \frac{1}{2}mv_{\max}^2$$

$$10 \times 2 = 10 \times \frac{3}{4} + \frac{v_{\max}^2}{2}$$

$$20 = 7.5 + \frac{v_{\max}^2}{2}$$

$$\frac{v_{\max}^2}{2} = 20 - 7.5$$

$$v_{\max}^2 = 25 \Rightarrow v_{\max} = 5 \text{ m/s}$$

22. (a) Let initial speed of man of mass m be u

Then

$$K.E_{\text{man}} = \frac{1}{2}mu^2 \text{ \& } K.E_{\text{boy}} = 2 \times \frac{1}{2}mu^2 = mu^2$$

Now if man increases his speed by 1 m/s then

$$K.E_{\text{man}} = \frac{1}{2}m(u+1)^2 = K.E'_{\text{boy}} = mu^2 \Rightarrow \frac{u+1}{u} = \sqrt{2}$$

$$\Rightarrow u = \frac{1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} = (\sqrt{2}+1) \text{ m/s}$$

23. (a) Linear momentum of a body

$$P = \sqrt{2mE} \propto \sqrt{m}$$

$$\text{Therefore } \frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{m}{4m}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\text{Or } P_1 : P_2 = 1 : 2$$

$$24. (a) U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$$

$$\text{For stable equilibrium, } F = \frac{dU}{dx} = 0$$

$$\Rightarrow F = \frac{12a}{x^{13}} - \frac{6b}{x^7} = 0 \Rightarrow x^6 = \frac{2a}{b}$$

$$\therefore x = \left(\frac{2a}{b}\right)^{1/6}$$

$$25. (b) v^2 = u^2 + 2as \text{ or } v^2 - u^2 = 2as$$

$$\text{Or } v^2 - (0)^2 = 2 \times \frac{F}{m} \times s \text{ or } v^2 = \frac{2Fs}{m}$$

$$\text{And } K.E. = \frac{1}{2}mv^2 = \frac{1}{2}m \times \frac{2Fs}{m} = Fs.$$

K.E. is independent of m or directly proportional to m .

$$26. (c) P = \sqrt{m(KE)}$$

$$\text{When } KE_1 = KE_2$$

$$\frac{P_1}{P_2} = \sqrt{\frac{M_1}{M_2}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\therefore P_1 : P_2 = 1 : 3$$

27. (c) On the diametrically opposite points, the velocities have same magnitude but opposite directions. Therefore change in momentum is

$$Mv - (-Mv) = 2Mv$$

28. (b) Given $K_p > K_q$

$$\text{Case (a) : } x_1 = x_2 = x$$

$$\frac{W_p}{W_q} = \frac{\frac{1}{2}K_p x^2}{\frac{1}{2}K_q x^2} = \frac{K_p}{K_q} \Rightarrow W_p > W_q$$

$$\text{Case (b) : } F_1 = F_2 = F$$

For constant force

$$W = \frac{F^2}{2K} \Rightarrow W \propto \frac{1}{K}$$

$$\text{So, } \frac{W_p}{W_q} = \frac{K_q}{K_p} \Rightarrow W_q > W_p$$

29. (a) Let the max^m extension produced in the length of the spring is x . Then,

$$\Rightarrow \frac{1}{2}kx^2 = Mgx$$

[Gravitational P.E gets stored as elastic potential energy]

$$\Rightarrow x = \frac{2Mg}{k}$$

30. (a) Since the spring is compressed by a distance d , mass m will go downward for distance $(h+d)$, and the work done by the spring is negative.

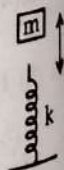
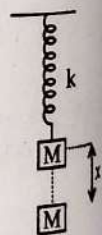
Net work done in process = Work done by gravitational force + work done by spring force.

$$\Rightarrow W = mg(h+d) - \frac{1}{2}kd^2$$

$$31. (c) \text{ Potential energy } (U) = \frac{1}{2}kx^2$$

x = displacement

Here, $x = 2 \text{ cm}$



$$\Rightarrow U = \frac{1}{2}k \times (2)^2 \Rightarrow 2k$$

$$\text{Again, } x_1 = 8 \text{ cm}$$

$$\Rightarrow U_1 = \frac{1}{2}k \times (8)^2 \Rightarrow U_1 = \frac{1}{2}k \times 64 = 32k$$

$$\Rightarrow U_1 = 16 \times (2k) [\because U = 2k]$$

$$\Rightarrow U_1 = 16U$$

32. (d) Applying conservation of energy

$$k_1 + p_1(\text{spring}) = k_2 + p_2(\text{spring})$$

$$\frac{1}{2}mv^2 + 0 = 0 + \frac{1}{2}kx^2$$

$$x = \sqrt{\frac{mv^2}{k}}$$

$$x = v \sqrt{\frac{m}{k}} = 1.5 \sqrt{\frac{0.5}{50}} = 0.15 \text{ m}$$

33. (d) $U = \frac{1}{2}K(2)^2$; $U' = \frac{1}{2}K(10)^2 = 25U$

34. (a) $E = \frac{kx^2}{2} = \frac{k^2 x^2}{2k} \Rightarrow E = \frac{f^2}{2k}$

$$\therefore \text{Force is equal } \therefore E \propto \frac{1}{K}$$

35. (a) $W = \frac{1}{2}Kx^2$, $F = -Kx$

$$W = \frac{1}{2}K \cdot \frac{F^2}{K^2} = \frac{F^2}{2K}$$

$$W \propto \frac{1}{K}$$

$$\frac{W_A}{W_B} = \frac{K_B}{K_A} = \frac{K_B}{2K_B} = \frac{1}{2}$$

36. (c) Let natural length of spring be ℓ_0 then according to question

$$4 = K(a - \ell_0)$$

$$5 = K(b - \ell_0)$$

$$\Rightarrow \ell_0 = 5a - 4b; K = \frac{1}{b-a}$$

Now if we apply 9 N force then

$$9 = K(\ell - \ell_0) \Rightarrow 9 = \frac{1}{(b-a)}[\ell - 5a + 4b] \Rightarrow \ell = 5b - 4a$$

37. (c) Work done by gravitational force = $W_g = mgh$

$$= 10^{-3} \times 10 \times 10^3 = 10 \text{ J}$$

Apply work energy theorem

$$W_{\text{all force}} = K_f - K_i$$

$$W_{(\text{conservative})} + W_{(\text{non conservative})} = K_f - K_i$$

$$W_g + W_f = \frac{1}{2}mv^2 - 0 [\because K_i = 0]$$

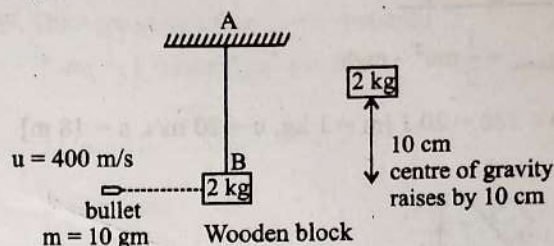
$$\Rightarrow mgh + W_f = \frac{1}{2}mv^2$$

$$\Rightarrow 10^{-3} \times 10 \times 1000 + W_f = \frac{1}{2} \times 10^{-3} \times (50)^2$$

$$\Rightarrow 10 + W_f = 1.25$$

$$\therefore \text{Work done by air resistance} = -8.75 \text{ J}$$

38. (a) $AB = 5 \text{ m}$



Apply conservation of linear momentum

$$mu + 0 = mv + MV$$

$$\frac{10}{1000} \times 400 + 0 = \frac{10}{1000}v + 2V$$

$$0.01v + 2V = 4$$

According to conservation of energy

PE at max height = KE at bottom

$$MgH = \frac{1}{2}MV^2$$

$$2 \times 10 \times \frac{10}{100} = \frac{1}{2} \times 2 \times V^2 \Rightarrow V^2 = 2$$

$$V = \sqrt{2} \text{ ms}^{-1}$$

Substituting the value of V in Eq. (1), we get

$$\frac{v}{100} + 2\sqrt{2} = 4 \Rightarrow v = (4 - 2\sqrt{2})100$$

$$\approx 120 \text{ ms}^{-1}$$

39. (d) $F = -0.1 \times \text{J/m}$

According to Work Energy theorem

$$\text{Work done by all force} = K_f - K_i$$

$$\Rightarrow \int F \cdot dx = K_f - K_i$$

$$\Rightarrow \int_{20}^{30} (-0.1x) dx = K_f - \frac{1}{2} \times mu^2$$

$$(-0.1) \left[\frac{x^2}{2} \right]_{20}^{30} = K_f - \frac{1}{2} \times 10 \times 10^2$$

$$\frac{1}{10 \times 2} [x^2]_{20}^{30} = K_f - 500$$

$$\frac{1}{20} \times [400 - 900] = K_f - 500$$

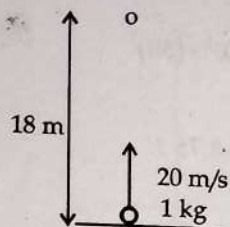
$$-\frac{500}{20} = K_f - 500$$

$$K_f = 500 - 25 = 475 \text{ J}$$

40. (d) By using Work - Energy theorem,

$$W = \Delta K.E.$$

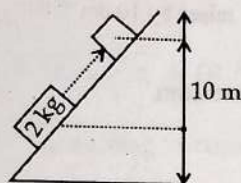
$$\Rightarrow -mgh - W_{\text{friction}} = 0 - \frac{1}{2}mu^2 \Rightarrow -W_{\text{friction}} = -\frac{1}{2}mu^2 + mgh$$



$$\Rightarrow W_{\text{friction}} = \frac{1}{2}mu^2 - mgh$$

$$= 200 - 180 = 20 \text{ J} [m = 1 \text{ kg}, u = 20 \text{ m/s}, h = 18 \text{ m}]$$

41. (b)



Total work done = $\Delta K.E.$ (Here $\Delta K.E. = 0$)

Work done by external Force = 300 J

Work done by gravity = $mgh = 2 \times 10 \times 10 = 200 \text{ J}$

Since, work done by gravity and friction is negative.

$$\therefore 300 - 200 - W_f = \Delta K.E. = 0 \Rightarrow W_f = 100 \text{ J}$$

Hence, work done against friction is 100 J

42. (c) $\therefore s = \frac{1}{3}t^2$

$$v = \frac{ds}{dt} = \frac{2}{3}t \text{ m/s}$$

$$\text{At } t = 0, v = 0 \text{ m/s}$$

$$\text{At } t = 2, v_2 = \frac{2 \times 2}{3} = \frac{4}{3} \text{ m/s}$$

According to Work - Energy theorem,

$$W = \Delta K.E.$$

$$= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$= \frac{1}{2} \times 3 \left[\left(\frac{4}{3} \right)^2 - (0)^2 \right] \quad [\text{Mass of body} = 3 \text{ kg}]$$

$$= \frac{3}{2} \times \frac{16}{9} = \frac{8}{3} \text{ J}$$

43. (a) Work energy theorem $W = \Delta K.E.$, $x = 3 - 4t^2 + t^3$

$$v = \frac{dx}{dt} = -8t + 3t^2$$

$$v_1 (t = 0) = 0$$

$$v_2 (t = 40) = 16$$

Therefore,

$$\Delta K.E. = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2} \times 3 \times 10^{-3} \times 16 \times 16 - 0$$

$$= 384 \text{ mJ}$$

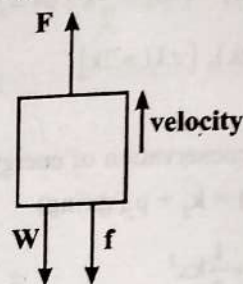
44. (b) Work done = Change in kinetic energy of the body

$$W = \frac{1}{2} \times 0.01 [(1000)^2 - (500)^2] = 3750 \text{ joule}$$

45. (d) As the velocity is constant, therefore net upward force equal to the net downward force

$$\Rightarrow F = W + f$$

$$= 20000 + 3000 = 23000 \text{ N}$$



$$\text{As, Power} = \vec{F} \cdot \vec{v}$$

$$= 23000 \times 1.5 = 34500 \text{ W}$$

46. (b) Energy = Power $\times$ time

$$= 100 \times 10^3 \times 3600 = 36 \times 10^7 \text{ J}$$

47. (a) Input power, $P_{\text{in}} = \frac{d(mgh)}{dt}$

$$P_{\text{in}} = gh \left(\frac{dm}{dt} \right)$$

$$P_{\text{in}} = gh(15) = 10 \times 60 \times 15 = 9000 \text{ watt}$$

Again,

$$P_{\text{out}} = \left(\frac{100 - 10}{100} \right) \text{ of } P_{\text{in}}$$

$$= \frac{90}{100} \times 9000 = 8100 \text{ watt} = 8.1 \text{ kW}$$

48. (d) $\vec{F} = 2t\hat{i} + 3t^2\hat{j}$

$$m \frac{d\vec{v}}{dt} = 2t\hat{i} + 3t^2\hat{j}$$

$$\int_0^t d\vec{v} = \int_0^t (2t\hat{i} + 3t^2\hat{j}) dt = t^2\hat{i} + t^3\hat{j}$$

$$\text{Power} = \vec{F} \cdot \vec{v} = (2t\hat{i} + 3t^2\hat{j}) \cdot (t^2\hat{i} + t^3\hat{j}) = (2t^3 + 3t^5) \text{ W}$$

49. (d) $P = Fv = mav \Rightarrow k = mv \frac{dv}{dt}$

$$\Rightarrow v dv = \frac{k}{m} dt$$

By integrating the equation

$$\int v dv = \int \frac{k}{m} dt$$

$$\Rightarrow \frac{v^2}{2} = \frac{k}{m} t \Rightarrow v = \sqrt{\frac{2kt}{m}}$$

$$a = \frac{dv}{dt} = \sqrt{\frac{2k}{m}} \left(\frac{1}{2} t^{-\frac{1}{2}} \right)$$

$$F = ma = m \left(\frac{1}{2} \right) \sqrt{\frac{2k}{mt}} \Rightarrow F = \sqrt{\frac{mk}{2t}} = \sqrt{\frac{mk}{2}} t^{-\frac{1}{2}}$$

50. (b) Pressure = 150 mm Hg

$$\text{Pumping rate} = \frac{dV}{dt} = \frac{5 \times 10^{-3}}{60} \text{ m}^3/\text{s}$$

$$\text{Power of heart} = P \cdot \frac{dV}{dt} = \rho gh \times \frac{dV}{dt}$$

$$= (13.6 \times 10^3 \text{ kg/m}^3)(10) \times (0.15) \times \frac{5 \times 10^{-3}}{60}$$

$$= \frac{13.6 \times 5 \times 0.15}{6} = 1.72 \text{ watt}$$

51. (b) Car starts from rest

So, $v = at$ ($\because u = 0$)

Instantaneous power $P_0 = FV$

$$P_0 = maV$$

$$P_0 = \frac{mdv}{dt} V$$

$$Vdv = \frac{P_0 dt}{m}$$

$$\int_0^v Vdv = \frac{P_0}{m} \int_0^t dt$$

$$\frac{V^2}{2} = \frac{P_0 t}{m}$$

$$V = \sqrt{\frac{2P_0 t}{m}}$$

$$V \propto \sqrt{t} \Rightarrow V \propto t^{1/2}$$

52. (b) Power, $P = \vec{F} \cdot \vec{v} = Fv \cos \theta$

Just before hitting the earth $\theta = 0^\circ$. Hence, the power exerted by the gravitational force is greatest at the instant just before the body hits the earth.

53. (d) $P_{\text{avg}} = \frac{W}{\Delta t} = \frac{\frac{1}{2}MV^2 - 0}{T} = \frac{1}{2} \frac{MV^2}{T}$

54. (a) Power = $Fv = v \left(\frac{m}{t} \right) v = v^2 (\rho Av)$

$$= \rho Av^3 = (100)(2)^3 = 800 \text{ W}$$

55. (d) Velocity of water is v , mass flowing per unit length is m

$\therefore$ Mass flowing per second = mv

$\therefore$ Rate of kinetic energy or K.E. per second

$$= \frac{1}{2} (mv)v^2 = \frac{1}{2} mv^3$$

56. (c) $\eta = \frac{\text{Useful work}}{\text{Total work}} = \frac{mgh}{F \times d} = \frac{(75 \times 10) \times 3}{250 \times 12} = 0.75$

57. (b) $P = \vec{F} \cdot \vec{v}$

$$= (60\hat{i} + 15\hat{j} - 3\hat{k}) \cdot (2\hat{i} - 4\hat{j} + 5\hat{k})$$

$$= (120 - 60 - 15) = 45 \text{ watt}$$

58. (b) Power = $\frac{\text{Work done}}{\text{Time taken}} = \frac{W}{t}$

But $W = \text{mass} \times \text{gravity} \times \text{height}$

$$\therefore P = \frac{M \times g \times h}{t}$$

$$\Rightarrow M = \frac{P \times t}{g \times h} = \frac{2000 \times 60}{10 \times 10} = 1200 \text{ kg}$$

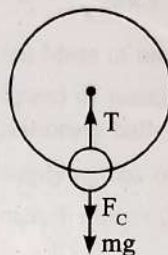
i.e. 1200 litres as one litre has a mass of 1 kg.

59. (b) According to the given question

$$T - mg = F_c (\text{centrifugal force})$$

$$T - mg = m \frac{(\sqrt{7gr})^2}{r}$$

$$\Rightarrow T = 8mg$$



60. (c) Let the mass m makes a small angle ' θ ' with vertical. Now, at point P from figure

$$T - mg \cos \theta = \frac{mv^2}{r}$$

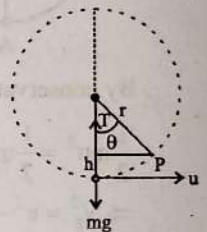
$$\therefore T = mg \cos \theta + \frac{mv^2}{r}$$

From figure, $\cos \theta = \frac{r-h}{r}$

So T will be maximum when $\cos \theta$ will be 1

$$\therefore \frac{r-h}{r} = 1 \Rightarrow h = 0$$

The tension is maximum when the mass is at the lowest position of the vertical circle, so the chance of breaking is maximum.



61. (d) To complete vertical circle minimum velocity required at lowest point of circle is $\sqrt{5gr}$. So by conservation of energy.

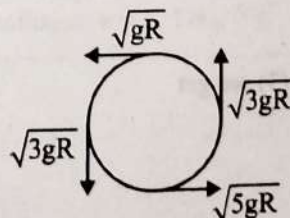
$$mgh = \frac{1}{2}mv^2$$

here, $V = \sqrt{5g \times \frac{D}{2}}$

$$\therefore mgh = \frac{1}{2}m \times 5g \frac{D}{2}$$

$$h = \frac{5D}{4}$$

62. (d) Minimum velocity required at different points to complete full vertical circle



63. (a) $\frac{1}{2}mv^2 = E \Rightarrow \frac{1}{2}\left(\frac{10}{1000}\right)v^2 = 8 \times 10^{-4}$

$$v^2 = (8 \times 10^{-4})200 = \frac{16}{100} \text{ ms}^{-1}$$

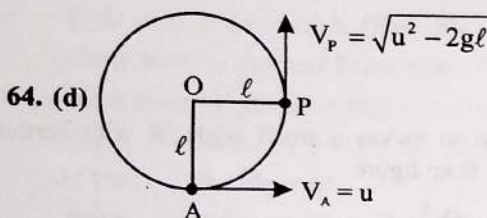
$$v = \frac{4}{10} \text{ ms}^{-1}$$

Now applying $v^2 - u^2 = 2as$

$$\left(\frac{4}{10}\right)^2 = 2a(4\pi R); [s = 2(2\pi R) = 4\pi R]$$

$$\Rightarrow \frac{16}{100} = 2a\left(4\pi \frac{6.4}{100}\right)$$

$$\therefore a = \frac{16}{100} \times \left[\frac{7 \times 100}{8 \times 22 \times 6.4}\right] = 0.1 \text{ m/s}^2$$



By conservation of energy

$$\frac{1}{2}mu^2 = \frac{1}{2}mv_p^2 + mg\ell$$

$$\Rightarrow V_p^2 = u^2 - 2g\ell$$

$$\Rightarrow V_p = \sqrt{u^2 - 2g\ell}$$

Magnitude of change in velocity $= |\vec{V}_p - \vec{V}_A|$

$$= \sqrt{V_p^2 + V_A^2} \quad (\because \text{angle b/w } \vec{V}_p \text{ and } \vec{V}_A \text{ is } 90^\circ)$$

$$= \sqrt{u^2 - 2g\ell + u^2}$$

$$= \sqrt{2(u^2 - g\ell)}$$

65. (b) Tension in the string at the lowest position B is maximum. Therefore, string will break at point B.

66. (b) From law of conservation of momentum we have

$$m_1u_1 = m_1v_1 + m_2v_2$$

$$m_2v_2 = m_1u_1 - m_1v_1 = m_1(u_1 - v_1) \quad \dots(1)$$

From the law of conservation of K.E. we have

$$\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$\Rightarrow m_2v_2^2 = m_1(u_1^2 - v_1^2) \quad \dots(2)$$

Dividing (2) with (1)

$$v_2 = u_1 + v_1 \quad \dots(3)$$

eliminating v_2 from (1) and (3) we get

$$\frac{m_1u_1 - m_1v_1}{m_2} = u_1 + v_1$$

$$m_1u_1 - m_1v_1 = m_2u_1 + m_2v_1$$

$$\Rightarrow \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 = v_1$$

Fraction of KE of m_1 carried by m_2 is

$$\frac{\frac{1}{2}m_1u_1^2 - \frac{1}{2}m_1v_1^2}{\frac{1}{2}m_1v_1^2} = 1 - \left[\frac{m_1 - m_2}{m_1 + m_2}\right]^2 = \frac{4m_1m_2}{(m_1 + m_2)^2}$$

This is also equal to the fractional transfer of KE of collision body.

Now,

$$\frac{\Delta KE}{KE} = \frac{4(m_1m_2)}{(m_1 + m_2)^2} = \frac{4(4m)2m}{(4m + 2m)^2} = \frac{32m^2}{36m^2} = \frac{8}{9}$$

This fractional transfer is equal to the fraction of energy by A.

67. (b) Since there is no external force on the system, momentum of the system will remain constant.

$$mv = 4mv' \Rightarrow v' = \frac{v}{4}$$

$$e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$$

$$\therefore e = \frac{v'}{v} = \frac{v/4}{v} = \frac{1}{4} = 0.25$$

68. (d) Since both bodies are identical and collision is elastic. Therefore velocities will be interchanged after collision.

$$v_A = -0.3 \text{ m/s and } v_B = 0.5 \text{ m/s}$$

69. (b) Energy will always be conserved so

$$K.E._{\text{initial}} = K.E._{\text{final}} + \text{Excitation energy}$$

$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \epsilon$$

70. (b) In elastic collision energy of system remains same so

$$(K.E.)_{\text{before collision}} = (K.E.)_{\text{after collision}}$$

Let speed of second body after collision is V

$$\frac{1}{2}mv^2 + 0 = \frac{1}{2}m\left(\frac{v}{3}\right)^2 + \frac{1}{2}m(V)^2 \Rightarrow v' = \frac{2\sqrt{2}}{3}v$$

71. (c) Let ball rebounds with speed V so

$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20} = 20 \text{ m/s}$$

Energy just after rebound

$$E = \frac{1}{2} \times m \times v^2 = 200 \text{ m}$$

50% energy loses in collision means just before collision energy is 400m

By using energy conservation

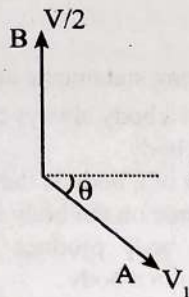
$$\frac{1}{2}mv_0^2 + mgh = 400m$$

$$\Rightarrow \frac{1}{2}mv_0^2 + m \times 10 \times 20 = 400m \Rightarrow v_0 = 20 \text{ ms}^{-1}$$

72. (b) For the collision, the final position of both the particle should be same

$$\begin{aligned} \therefore \vec{r}_1 + \vec{v}_1 t &= \vec{r}_2 + \vec{v}_2 t \\ \Rightarrow \vec{r}_1 - \vec{v}_2 t &= (\vec{v}_2 - \vec{v}_1) t \\ \Rightarrow \vec{r}_1 - \vec{r}_2 &= (\vec{v}_2 - \vec{v}_1) \frac{|\vec{r}_1 - \vec{r}_2|}{|\vec{v}_2 - \vec{v}_1|} \\ \Rightarrow \frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} &= \frac{\vec{v}_2 - \vec{v}_1}{|\vec{v}_2 - \vec{v}_1|} \end{aligned}$$

73. (d)

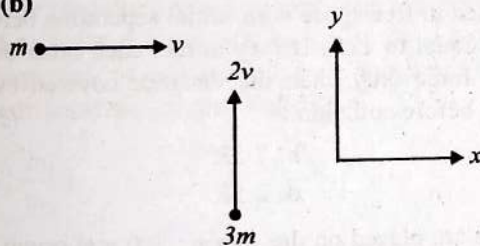


In x - direction $m_2 v = m_1 v_1 \cos \theta$... (i)

In y - direction $0 = m_2 v/2 + m_1 v_1 \sin \theta$... (ii)

From above equations $\tan \theta = \frac{-1}{2}$

74. (b)



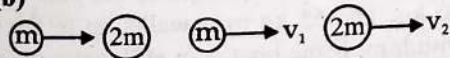
According to conservation of momentum, we get

$$m\hat{i} + (3m)2\hat{j} = (m + 3m)v'$$

where v' is the final velocity after collision

$$v' = \frac{1}{4}\hat{j} + \frac{6}{4}\hat{j} = \frac{1}{4}\hat{i} + \frac{3}{2}\hat{j}$$

75. (b)



Initial condition Final condition
By conservation of linear momentum:

$$2m = mv_1 + 2mv_2$$

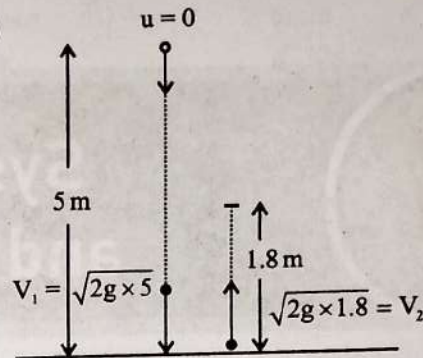
$$\Rightarrow v_1 + 2v_2 = 2$$

By definition of e : $e = \frac{1}{2} = \frac{v_2 - v_1}{2 - 0}$

$$\Rightarrow v_2 - v_1 = 1$$

$$\Rightarrow v_1 = 0 \text{ and } v_2 = 1 \text{ ms}^{-1}$$

76. (a)



$$\frac{V_2}{V_1} = \frac{\sqrt{2g \times 1.8}}{\sqrt{2g \times 5}} = \frac{\sqrt{18}}{\sqrt{50}} = \frac{\sqrt{9}}{\sqrt{25}} \Rightarrow \frac{V_2}{V_1} = \frac{3}{5}$$

77. (d) Mass of metal ball = 2 kg;

Speed of metal ball (v_1) = 36 km/h = 10 m/s and mass of stationary ball = 3 kg.

Applying law of conservation of momentum,

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$$

$$\text{or, } \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{(2 \times 10) + (3 \times 0)}{2 + 3} = \frac{20}{5} = 4 \text{ m/s.}$$

Therefore loss of energy

$$\begin{aligned} &= \left[\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right] - \frac{1}{2} \times (m_1 + m_2) v^2 \\ &= \left[\frac{1}{2} \times 2 \times (10)^2 + \frac{1}{2} \times 3 \times (0)^2 \right] - \frac{1}{2} \times (2 + 3) \times (4)^2 \end{aligned}$$

$$s = 100 - 40 = 60 \text{ J}$$

78. (c) Mass of body (m_1) = m;

Velocity of first body (u_1) = 3 km/hour; Mass of second body in rest (m_2) = 2m

Velocity of second body (u_2) = 0.

After combination, mass of the body

$$M = m + 2m = 3m$$

From the law of conservation of momentum, we get

$$Mv = m_1 u_1 + m_2 u_2$$

$$\text{Or } 3mv = (m \times 3) + (2m \times 0) = 3m \text{ Or } v = 1 \text{ km/hour}$$

79. (a) Coefficient of restitution is defined as the constant ratio of relative velocity after impact to the relative velocity of the bodies before impact when the two bodies collide head on. There velocities are in the opposite directions.

$$\therefore \frac{v_1 - v_2}{u_1 - u_2} = \text{constant} = e$$

The constant e is known as coefficient of restitution. For a perfectly elastic collision, $e = 1$ and for a perfectly inelastic collision, $e = 0$. Thus $0 \leq e \leq 1$.

System of Particles and Rotational Motion

Centre of Mass and Motion of Centre of Mass

1. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the centre of mass of the system from the 10 kg mass is : (2022)

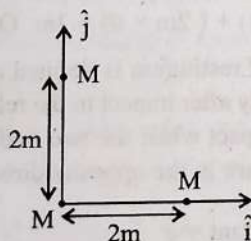
a. 5m
b. $\frac{10}{3}$ m
c. $\frac{20}{3}$ m
d. 10 m

2. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass.

The centre of mass of the system from the 5 kg particle is nearly at a distance of : (2020)

a. 50 cm
b. 67 cm
c. 80 cm
d. 33 cm

3. Three identical spheres, each of mass M , are placed at the corners of a right angle triangle with mutually perpendicular sides equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin, find the position vector of centre of mass. (2020-Covid)



a. $(\hat{i} + \hat{j})$
b. $\frac{2}{3}(\hat{i} + \hat{j})$
c. $\frac{4}{3}(\hat{i} + \hat{j})$
d. $2(\hat{i} + \hat{j})$

4. Which of the following statements are correct?
A. Centre of mass of a body always coincides with the centre of gravity of the body
B. Centre of gravity of a body is the point at which the total gravitational torque on the body is zero
C. A couple on a body produce both translational and rotational motion in a body.
D. Mechanical advantage greater than one means that small effort can be used to lift a large load (2017-Delhi)

a. A and B
b. B and C
c. C and D
d. B and D

5. Two spherical bodies of mass M and $5M$ and radii R and $2R$ are released in free space with initial separation between their centers equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body before collision is: (2015)

a. $4.5R$
b. $7.5R$
c. $1.5R$
d. $2.5R$

6. Three masses are placed on the x-axis : 300 g at origin, 500 g at $x = 40$ cm and 400 g at $x = 70$ cm. The distance of the center of mass from the origin is: (2012 Mains)

a. 40 cm
b. 45 cm
c. 50 cm
d. 30 cm

7. Two persons of masses 55 kg and 65 kg respectively, are at the opposite ends of a boat. The length of the boat is 3.0 m and weighs 100 kg. The 55 kg man walks up to the 65 kg man and sits with him. If the boat is in still water the center of mass of the system shifts by: (2012 Pre)

a. 3.0 m
b. 2.3 m
c. Zero
d. 0.75 m

8. A man of 50 kg mass is standing in a gravity free space at a height of 10 m above the floor. He throws a stone of 0.5 kg mass downwards with a speed 2 m/s. When the stone reaches the floor, the distance of the man above the floor will be (2010 Pre)

a. 9.9 m
b. 10.1 m
c. 10 m
d. 20 m

9. (1) Centre of gravity (C.G.) of a body is the point at which the weight of the body acts.
(2) Centre of mass coincides with the centre of gravity if the earth is assumed to have infinitely large radius.

- (3) To evaluate the gravitational field intensity due to any body at an external point, the entire mass of the body can be considered to be concentrated at its C.G.
- (4) The radius of gyration of any body rotating about an axis is the length of the perpendicular dropped from the C.G. of the body to the axis.

When one of the following pairs of statements is correct?
(2010 Mains)

- a. (1) and (4) b. (1) and (2)
c. (2) and (3) d. (3) and (4)
10. Two particles which are initially at rest, move towards each other under the action of their internal attraction. If their speeds are v and $2v$ at any instant, then the speed of center of mass of the system will be:
(2010 Pre)

- a. v b. $2v$
c. Zero d. $1.5v$
11. Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$, respectively. The center of mass of this system has a position vector:
(2009)
- a. $2\hat{i} - \hat{j} + \hat{k}$ b. $-2\hat{i} - \hat{j} + \hat{k}$
c. $-\hat{i} + \hat{j} + \hat{k}$ d. $-2\hat{i} + 2\hat{k}$

12. Consider a system of two particles having masses m_1 and m_2 . If the particle of mass m_1 is pushed towards the mass centre of particles through a distance 'd' by what distance would the particle of mass m_2 move so as to keep the mass centre of particles at the original position:
(2004)

- a. $\frac{m_1}{m_2}d$ b. d
c. $\frac{m_1}{m_2}$ d. $\frac{m_1}{m_1 + m_2}d$

13. A rod of length is 3 m and its mass acting per unit length is directly proportional to distance x from one of its end then its centre of gravity from that end will be at:
(2002)

- a. 1.5 m b. 2 m
c. 2.5 m d. 3.0 m

14. The centre of mass of system of particles does not depend on:
(1997)

- a. Position of the particles
b. Relative distances between the particles
c. Masses of the particles
d. Forces acting on the particle

Angular Displacement Velocity and Acceleration

15. The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is :
(2022)

- a. 104π b. 2π
c. 4π d. 12π

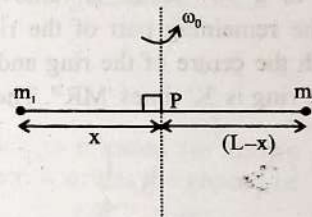
16. The angular speed of the wheel of a vehicle is increased from 360 rpm to 1200 rpm in 14 second. Its angular acceleration is.
(2020-Covid)

- a. $28\pi \text{ rad/s}^2$ b. $120\pi \text{ rad/s}^2$
c. 1 rad/s^2 d. $2\pi \text{ rad/s}^2$

17. A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its center. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2} . Its net acceleration in ms^{-2} at the end of 2.0 s is approximately:
(2016 - I)

- a. 8.0 b. 7.0
c. 6.0 d. 3.0

18. Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L , and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by:
(2015 Re)

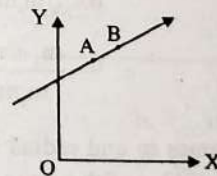


- a. $x = \frac{m_2 L}{m_1 + m_2}$ b. $x = \frac{m_1 L}{m_1 + m_2}$
c. $x = \frac{m_1}{m_2} L$ d. $x = \frac{m_2}{m_1} L$

19. A wheel has angular acceleration of 3.0 rad/sec^2 and an initial angular speed of 2.00 rad/sec . In a time of 2 sec it has rotated through an angle (in radian) of:
(2007)

- a. 10 b. 12
c. 4 d. 6

20. A particle of mass m moves in the XY plane with a velocity v along the straight line AB. If the angular momentum of the particle with respect to origin O is L_A when it is at A and L_B when it is at, then:
(2007)



- a. $L_A = L_B$
b. The relationship between L_A and L_B depends upon the slope of the line AB
c. $L_A < L_B$
d. $L_A > L_B$

21. For a body angular velocity $\vec{\omega} = \hat{i} - 2\hat{j} + 3\hat{k}$ and radius vector is $\vec{r} = \hat{i} + \hat{j} + \hat{k}$ then its velocity is: (1999)
- a. $-5\hat{i} + 2\hat{j} + 3\hat{k}$ b. $-5\hat{i} + 2\hat{j} - 3\hat{k}$
 c. $-5\hat{i} - 2\hat{j} + 3\hat{k}$ d. $-5\hat{i} - 2\hat{j} - 3\hat{k}$

Moment of Inertia, Theorem of Parallel and Perpendicular Axis and Energy in Rotation

22. The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is: (2022)

a. $1 : \sqrt{2}$ b. $2 : 1$
 c. $\sqrt{2} : 1$ d. $4 : 1$

23. From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times ' MR^2 '. Then the value of 'K' is: (2021)

a. $\frac{7}{8}$ b. $\frac{1}{4}$
 c. $\frac{1}{8}$ d. $\frac{3}{4}$

24. Three objects, A: (a solid sphere), B: (a thin circular disk) and C: (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation (2018)

a. $W_B > W_A > W_C$ b. $W_A > W_B > W_C$
 c. $W_C > W_B > W_A$ d. $W_A > W_C > W_B$

25. A light rod of length ℓ has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is: (2016 - II)

a. $(m_1 + m_2)\ell^2$ b. $\sqrt{m_1 m_2} \ell^2$
 c. $\frac{m_1 m_2}{m_1 + m_2} \ell^2$ d. $\frac{m_1 + m_2}{m_1 m_2} \ell^2$

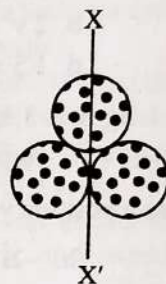
26. A solid sphere of mass m and radius R is rotating about its diameter. A solid cylinder of the same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation ($E_{\text{sphere}} / E_{\text{cylinder}}$) will be: (2016 - II)

a. $1 : 4$ b. $3 : 1$
 c. $2 : 3$ d. $1 : 5$

27. From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre? (2016)

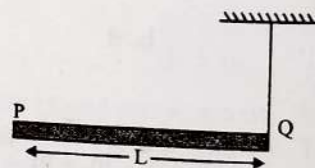
a. $15MR^2/32$
 b. $13MR^2/32$
 c. $11MR^2/32$
 d. $9MR^2/32$

28. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter to the third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is: (2015)



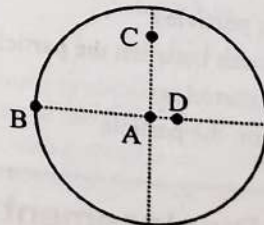
a. $3mr^2$ b. $16/5 mr^2$
 c. $4mr^2$ d. $11/5 mr^2$

29. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is: (2013)



a. $\frac{2g}{3L}$ b. $\frac{3g}{2L}$
 c. $\frac{g}{L}$ d. $\frac{2g}{L}$

30. The moment of inertia of a uniform circular disc is maximum about an axis perpendicular to the disc and passing through: (2012 Pre)



a. B
 c. D
 b. C
 d. A

31. Four identical thin rods each of mass M and length ℓ , form a square frame. Moment of inertia of this frame about an axis through the center of the square and perpendicular to its plane is: (2009)

- a. $\frac{2}{3}M\ell$ b. $\frac{13}{3}M\ell$
c. $\frac{1}{3}M\ell^2$ d. $\frac{4}{3}M\ell^2$

32. A thin rod of length L and mass M is bent at its midpoint into two halves so that the angle between them is 90° . The moment of inertia of the bent rod about an axis passing through the bending point and perpendicular to the plane defined by the two halves of the rod is: (2008)

- a. $\frac{\sqrt{2}ML^2}{24}$ b. $\frac{ML^2}{24}$
c. $\frac{ML^2}{12}$ d. $\frac{ML^2}{6}$

33. The ratio of the radii of gyration of a circular disc to that of a circular ring, each of same mass and radius, around their respective axes is: (2008)

- a. $\sqrt{2}:\sqrt{3}$ b. $\sqrt{3}:\sqrt{2}$
c. $1:\sqrt{2}$ d. $\sqrt{2}:1$

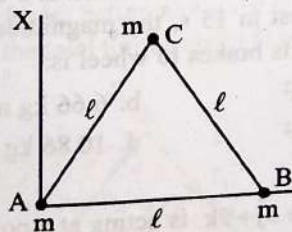
34. The moment of inertia of a uniform circular disc of radius R and mass M about an axis touching the disc at its diameter and normal to the disc is: (2006, 2005)

- a. $\frac{1}{2}MR^2$ b. MR^2
c. $\frac{2}{5}MR^2$ d. $\frac{3}{2}MR^2$

35. The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is: (2004)

- a. $2:1$ b. $\sqrt{5}:\sqrt{6}$
c. $2:3$ d. $1:\sqrt{2}$

36. Three particles, each of mass m gram, are situated at the vertices of an equilateral triangle ABC of side ℓ cm (as shown in the figure). The moment of inertia of the system, about a line AX perpendicular to AB and in the plane of ABC , in gram cm^2 units will be: (2004)



- a. $2m\ell^2$ b. $\frac{5}{4}m\ell^2$
c. $\frac{3}{2}m\ell^2$ d. $\frac{3}{4}m\ell^2$

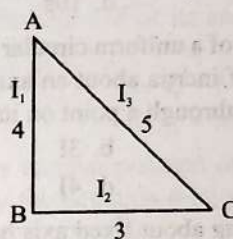
37. A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K . If radius of the ball be R , then the fraction of total energy associated with its rotational energy will be: (2003)

- a. $\frac{K^2+R^2}{R^2}$ b. $\frac{K^2}{R^2}$
c. $\frac{K^2}{K^2+R^2}$ d. $\frac{R^2}{K^2+R^2}$

38. A circular disc is to be made by using iron and aluminium so that it acquired maximum moment of inertia about geometrical axis. It is possible with: (2002)

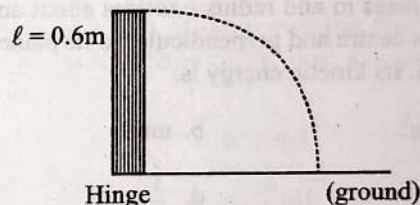
- a. Aluminium at interior and iron surround to it.
b. Iron at interior and aluminium surround to it.
c. Using iron and aluminium layers in alternate order.
d. Sheet of iron is used at both external surface and aluminium sheet as internal layers.

39. For the adjoining diagram, a triangular lamina is shown the correct relation between I_1 , I_2 and I_3 is (I - moment of inertia): (2000)



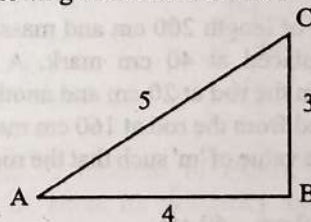
- a. $I_1 > I_2$ b. $I_2 > I_1$
c. $I_3 > I_1$ d. $I_3 > I_2$

40. When a stick is released (as shown in fig.) Its free end velocity when it strikes the ground is: (1999)



- a. 4.2 m/s b. 1.4 m/s
c. 2.8 m/s d. $\sqrt{6}$ m/s

41. The ABC is a triangular plate of uniform thickness. The sides are in the ratio shown in the figure. I_{AB} , I_{BC} and I_{CA} are the moments of inertia of the plate about AB , BC and CA respectively. Which one of the following relations is correct? (1995)

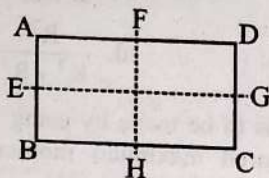


- a. $I_{AB} + I_{BC} = I_{CA}$ b. I_{CA} is maximum
c. $I_{AB} > I_{BC}$ d. $I_{BC} > I_{AB}$

42. A solid spherical ball rolls on a table. Ratio of its rotational kinetic energy to total kinetic energy is: (1994)

- a. $\frac{1}{2}$ b. $\frac{1}{6}$
c. $\frac{7}{10}$ d. $\frac{2}{7}$

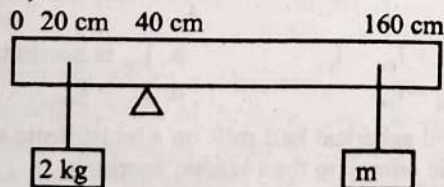
43. In a rectangle ABCD ($BC = 2AB$). The moment of inertia is minimum along axis through: (1993)



- a. BC
b. BD
c. HF
d. EG
44. The moment of inertia of a body about a given axis is 1.2 kgm^2 . Initially, the body is at rest. In order to produce a rotational kinetic energy of 1500 joule, an angular acceleration of 25 rad/sec^2 must be applied about that axis for a duration of: (1990)
- a. 4s
b. 2s
c. 8s
d. 10s
45. Moment of inertia of a uniform circular disc about a diameter is I . Its moment of inertia about an axis perpendicular to its plane and passing through a point on its rim will be: (1990)
- a. $5I$
b. $3I$
c. $6I$
d. $4I$
46. A fly wheel rotating about fixed axis has a kinetic energy of 360 joule when its angular speed is 30 rad/sec . The moment of inertia of the wheel about the axis of rotation is: (1990)
- a. 0.6 kgm^2
b. 0.15 kgm^2
c. 0.8 kgm^2
d. 0.75 kgm^2
47. A ring of mass m and radius r rotates about an axis passing through its centre and perpendicular to its plane with angular velocity ω . Its kinetic energy is: (1988)
- a. $\frac{1}{2}mr^2\omega^2$
b. $mr\omega^2$
c. $mr^2\omega^2$
d. $\frac{1}{2}mr\omega^2$

Torque, Angular Momentum and its Conservation

48. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. ($g = 10 \text{ m/s}^2$) (2021)

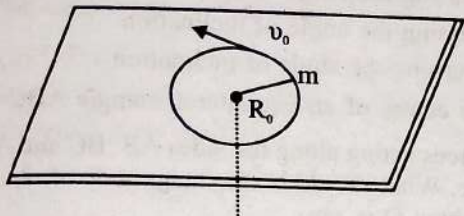


- a. $\frac{1}{3} \text{ kg}$
b. $\frac{1}{6} \text{ kg}$
c. $\frac{1}{12} \text{ kg}$
d. $\frac{1}{2} \text{ kg}$

49. Find the torque about the origin when a force of $3\hat{j} \text{ N}$ acts on a particle whose position vector is $2\hat{k} \text{ m}$. (2020)
- a. $6\hat{j} \text{ N m}$
b. $-6\hat{i} \text{ N m}$
c. $6\hat{k} \text{ N m}$
d. $6\hat{i} \text{ N m}$
50. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is (2019)
- a. $2 \times 10^{-6} \text{ N m}$
b. $2 \times 10^{-3} \text{ N m}$
c. $12 \times 10^{-4} \text{ N m}$
d. $2 \times 10^6 \text{ N m}$
51. A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere? (2018)
- a. Rotational kinetic energy
b. Moment of inertia
c. Angular velocity
d. Angular momentum
52. The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at $(2, 0, -3)$, about the point $(2, -2, -2)$ is given by (2018)
- a. $-7\hat{i} - 8\hat{j} - 4\hat{k}$
b. $-4\hat{i} - \hat{j} - 8\hat{k}$
c. $-8\hat{i} - 4\hat{j} - 7\hat{k}$
d. $-7\hat{i} - 4\hat{j} - 8\hat{k}$
53. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N? (2017-Delhi)
- a. 0.25 rad/s^2
b. 25 rad/s^2
c. 5 m/s^2
d. 25 m/s^2
54. Two rotating bodies A and B of masses m and $2m$ with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then: (2016-II)
- a. $L_B > L_A$
b. $L_A > L_B$
c. $L_A = \frac{L_B}{2}$
d. $L_A = 2L_B$
55. An automobile moves on a road with a speed of 54 km/h . The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kgm^2 . If the vehicle is brought to rest in 15 s, the magnitude of average torque transmitted by its brakes to wheel is: (2015 Rel)
- a. $2.86 \text{ kg m}^2/\text{s}^2$
b. $6.66 \text{ kg m}^2/\text{s}^2$
c. $8.58 \text{ kg m}^2/\text{s}^2$
d. $10.86 \text{ kg m}^2/\text{s}^2$
56. A force $\vec{F} = \alpha\hat{i} + 3\hat{j} + 9\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about origin is conserved is: (2015 Rel)
- a. 1
b. -1
c. 2
d. Zero
57. A rod of weight W is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The center of mass of the rod is at distance x from A. The normal reaction on A is: (2015)

- a. $\frac{Wd}{x}$
 b. $\frac{W(d-x)}{x}$
 c. $\frac{W(d-x)}{d}$
 d. $\frac{Wx}{d}$

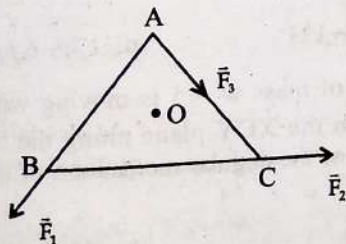
58. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0 . The mass is attached to a string which passes through a smooth hole in the plane as shown.



The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is: (2015)

- a. $\frac{1}{4}mv_0^2$
 b. $2mv_0^2$
 c. $\frac{1}{2}mv_0^2$
 d. mv_0^2
59. A solid cylinder of mass 50 kg and radius 0.5 m is free to rotate about the horizontal axis. A massless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 rev/s² is: (2014)
- a. 25 N
 b. 50 N
 c. 78.5 N
 d. 157 N
60. A circular platform is mounted on a frictionless vertical axle. Its radius $R = 2$ m and its moment of inertia about the axle is 200 kg m². It is initially at rest. A 50 kg man stands on the edge of the platform and begins to walk along the edge at the speed of 1 ms⁻¹ relative to the ground. Time taken by the man to complete one revolution is: (2012 Mains)
- a. π s
 b. $3\pi/2$ s
 c. 2π s
 d. $\pi/2$ s

61. ABC is an equilateral triangle with O as its center. $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ represent three forces acting along the sides AB, BC and AC respectively. If the total torque about O is zero the magnitude of $\vec{F}_3$ is: (2012 Pre)



- a. $F_1 + F_2$
 b. $F_1 - F_2$
 c. $F_1 - F_2$
 d. $2(F_1 + F_2)$
62. When a mass is rotating in a plane about a fixed point, its angular momentum is directed along (2012 Pre)

- a. A line perpendicular to the plane of rotation
 b. The line making an angle of 45° to the plane of rotation
 c. The radius
 d. The tangent to the orbit

63. A small mass attached to a string rotates on a frictionless table top as shown. If the tension in the string is increased by pulling the string causing the radius of the circular motion to decrease by a factor of 2, the kinetic energy of the mass will: (2011 Mains)

- a. Remain constant
 b. Increase by a factor of 2
 c. Increase by a factor of 4
 d. Decrease by a factor of 2

64. The moment of inertia of a thin uniform rod of mass M and length L about an axis passing through its midpoint and perpendicular to its length is I_0 . Its moment of inertia about an axis passing through one of its ends perpendicular to its length is: (2011 Mains)

- a. $I_0 + ML^2/2$
 b. $I_0 + ML^2/4$
 c. $I_0 + 2ML^2$
 d. $I_0 + ML^2$

65. The instantaneous angular position of a point on a rotating wheel is given by the equation $\theta(t) = 2t^3 - 6t^2$. The torque on the wheel becomes zero at: (2011 Pre)

- a. $t = 1$ s
 b. $t = 0.5$ s
 c. $t = 0.25$ s
 d. $t = 2$ s

66. From a circular disc of radius R and mass $9M$, a small disc of mass M and radius $\frac{R}{3}$ is removed concentrically. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through its center is: (2010 Mains)

- a. $\frac{40}{9}MR^2$
 b. MR^2
 c. $4MR^2$
 d. $\frac{4}{9}MR^2$

67. A thin circular ring of mass M and radius r is rotating about its axis with constant angular velocity ω . The objects each of mass m are attached gently to the opposite ends of a diameter of the ring. The ring now rotates with angular velocity given by: (2010 Mains)

- a. $\frac{(M+2m)\omega}{2m}$
 b. $\frac{2M\omega}{M+2m}$
 c. $\frac{(M+2m)\omega}{M}$
 d. $\frac{M\omega}{M+2m}$

68. A circular disk of moment of inertia I_1 is rotating in a horizontal plane, about its symmetry axis, with a constant angular speed ω_1 . Another disk of moment of inertia I_2 is dropped coaxially into the rotating disk. Initially the second disk has zero angular speed. Eventually both the disks rotate with a constant angular speed ω_f . The energy lost by the initially rotating disc due to friction is: (2010)

- a. $\frac{1}{2} \frac{I_1^2}{(I_1 + I_2)} \omega_1^2$
 b. $\frac{1}{2} \frac{I_2^2}{(I_1 + I_2)} \omega_1^2$
 c. $\frac{I_1 - I_2}{(I_1 + I_2)} \omega_1^2$
 d. $\frac{1}{2} \frac{I_1 I_2}{(I_1 + I_2)} \omega_1^2$

69. If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then: (2009)

- a. $\vec{r} \cdot \vec{F} > 0$ and $\vec{F} \cdot \vec{r} < 0$ b. $\vec{r} \cdot \vec{\tau} = 0$ and $\vec{\tau} \cdot \vec{r} = 0$
c. $\vec{r} \cdot \vec{\tau} = 0$ and $\vec{\tau} \cdot \vec{r} \neq 0$ d. $\vec{r} \cdot \vec{\tau} = 0$ and $\vec{F} \cdot \vec{r} = 0$

70. A thin circular ring of mass M and radius R is rotating in a horizontal plane about an axis vertical to its plane with a constant angular velocity ω . If two objects each of mass m be attached gently to the opposite ends of a diameter of the ring, the ring will then rotate with an angular velocity: (2009)

- a. $\frac{\omega M}{M+2m}$ b. $\frac{\omega(M+2m)}{M}$
c. $\frac{\omega M}{M+m}$ d. $\frac{\omega(M-2m)}{M+2m}$

71. A round disc of moment of inertia I_2 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_1 rotating with an angular velocity ω about the same axis. The final angular velocity of the combination of discs is: (2004)

- a. ω b. $\frac{I_1 \omega}{I_1 + I_2}$
c. $\frac{(I_1 + I_2) \omega}{I_1}$ d. $\frac{I_2 \omega}{I_1 + I_2}$

72. A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be: (2004)

- a. $\frac{\pi}{12} \text{ N-m}$ b. $\frac{\pi}{15} \text{ N-m}$
c. $\frac{\pi}{18} \text{ N-m}$ d. $\frac{2\pi}{15} \text{ N-m}$

73. A thin circular ring of mass M and radius ' r ' is rotating about its axis with a constant angular velocity ω . Four objects each of mass m , are kept gently to the opposite ends of two perpendicular diameters of the ring. The angular velocity of the ring will be: (2003)

- a. $\frac{M\omega}{4m}$ b. $\frac{M\omega}{M+4m}$
c. $\frac{(M+4m)\omega}{M}$ d. $\frac{(M+4m)\omega}{M+4m}$

74. A disc is rotating with angular speed ω . If a child sits on it, what is conserved: (2002)

- a. Linear momentum b. Angular momentum
c. Kinetic energy d. Potential energy

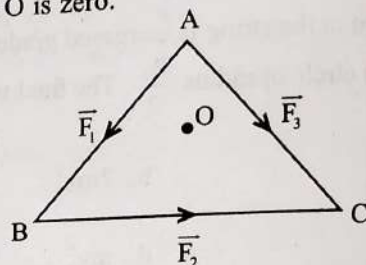
75. A circular ring of mass M and radius R is rotating about its axis with constant angular velocity ω . Two particles each of mass m are attached gently to the opposite ends of a diameter of the ring. The angular velocity of the ring will now become: (1998)

- a. $\frac{M-2m}{m\omega}$ b. $\frac{m\omega}{M-2m}$
c. $\frac{M\omega}{M+2m}$ d. $\frac{M-2m}{m}$

76. If a ladder is not in balance against a smooth vertical wall then it can be made in balance by: (1998)

- a. Decreasing the length of ladder
b. Increasing the length of ladder
c. Increasing the angle of inclination
d. Decreasing the angle of inclination

77. O is the centre of an equilateral triangle ABC . $\vec{F}_1, \vec{F}_2, \vec{F}_3$ are three forces acting along the sides AB, BC and AC as shown in figure. What should be the magnitude of $\vec{F}_3$ so that total torque about O is zero: (1998)



- a. $|\vec{F}_3| = |\vec{F}_1| + |\vec{F}_2|$ b. $|\vec{F}_3| = |\vec{F}_1| - |\vec{F}_2|$
c. $|\vec{F}_3| = \vec{F}_1 + \vec{F}_2$ d. Not possible

78. A couple produces: (1997)

- a. Linear and rotational motion
b. No motion
c. Purely linear motion
d. Purely rotational motion

79. Find the torque of a force $\vec{F} = -3\hat{i} + \hat{j} + 5\hat{k}$ acting at the point $\vec{r} = 7\hat{i} + 3\hat{j} + \hat{k}$ (1997)

- a. $-21\hat{i} + 4\hat{j} + 4\hat{k}$ b. $-14\hat{i} + 34\hat{j} - 16\hat{k}$
c. $14\hat{i} - 38\hat{j} + 16\hat{k}$ d. $4\hat{i} + 4\hat{j} + 6\hat{k}$

80. What is torque of the force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ acting at the point $\vec{r} = 3\hat{i} + 2\hat{j} + 3\hat{k}$ about origin? (1995)

- a. $-6\hat{i} + 6\hat{j} - 12\hat{k}$ b. $-17\hat{i} + 6\hat{j} + 13\hat{k}$
c. $6\hat{i} - 6\hat{j} + 12\hat{k}$ d. $17\hat{i} - 6\hat{j} - 13\hat{k}$

81. A particle of mass $w = 5$ is moving with a uniform speed $v = 3\sqrt{2}$ in the XOY plane along the line $Y = X + 4$. The magnitude of the angular momentum of the particle about the origin is: (1991)

- a. 60 units b. $40\sqrt{2}$ units
c. Zero d. 7.5 units

82. A solid homogenous sphere of mass M and radius R is moving on a rough horizontal surface, partly rolling and partly sliding. During this kind of motion of this sphere: (1988)

- a. Total kinetic energy is conserved
- b. The angular momentum of the sphere about the point of contact with the plane is conserved
- c. Only the rotational kinetic energy about the centre of mass is conserved
- d. Angular momentum about the centre of mass is conserved

Rolling Motion

83. A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it? (2019)
 - a. 3 J
 - b. 30 kJ
 - c. 2 J
 - d. 1 J
84. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is: (2018)
 - a. 10 : 7
 - b. 5 : 7
 - c. 7 : 10
 - d. 2 : 5
85. Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2 . They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is: (2017-Delhi)
 - a. $\frac{1}{4}I(\omega_1 - \omega_2)^2$
 - b. $I(\omega_1 - \omega_2)^2$
 - c. $\frac{1}{8}I(\omega_1 - \omega_2)^2$
 - d. $\frac{1}{2}I(\omega_1 - \omega_2)^2$
86. A disk and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first? (2016 - I)
 - a. Disk
 - b. Sphere
 - c. Both reach at the same time
 - d. Depends on their masses
87. The ratio of the accelerations for a solid sphere (mass m and radius R) rolling down an incline of angle ' θ ' without slipping and slipping down the incline without rolling is: (2014)
 - a. 5 : 7
 - b. 2 : 3
 - c. 2 : 5
 - d. 7 : 5
88. Small object of uniform density rolls up a curved surface with an initial velocity v . It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is: (2013)
 - a. Disc
 - b. Ring
 - c. Solid sphere
 - d. Hollow sphere
89. A solid cylinder of mass 3 kg is rolling on a horizontal surface with velocity 4 ms^{-1} . It collides with a horizontal spring of force constant 200 Nm^{-1} . The maximum compression produced in the spring will be: (2012 Pre)
 - a. 0.5 m
 - b. 0.6 m
 - c. 0.7 m
 - d. 0.2 m
90. A solid cylinder and a hollow cylinder, both of the same mass and same external diameter are released from the same height at the same time on a inclined plane. Both roll down without slipping. Which one will reach the bottom first? (2010 Mains)
 - a. Both together only when angle of inclination of plane is 45°
 - b. Both together
 - c. Hollow cylinder
 - d. Solid cylinder
91. A drum of radius R and mass M , rolls down without slipping along an inclined plane of angle θ . The frictional force: (2005)
 - a. Converts translational energy to rotational energy
 - b. Dissipates energy as heat
 - c. Decreases the rotational motion
 - d. Decreases the rotational and translational motion
92. A solid cylinder of mass M and radius R rolls without slipping down an inclined plane of length L and height h . What is the speed of its centre of mass when the cylinder reaches its bottom: (2003)
 - a. $\sqrt{2gh}$
 - b. $\sqrt{\frac{3}{4}gh}$
 - c. $\sqrt{\frac{4}{3}gh}$
 - d. $\sqrt{4gh}$
93. A point P consider at contact point of a wheel on ground which rolls on ground without slipping then value of displacement of point P when wheel completes half of rotation (If radius of wheel is 1 m): (2002)
 - a. 2m
 - b. $\sqrt{\pi^2 + 4} \text{ m}$
 - c. $\pi \text{ m}$
 - d. $\sqrt{\pi^2 + 2} \text{ m}$
94. A solid sphere of radius R is placed in smooth horizontal surface. A horizontal force ' F ' is applied, at height ' h ' from the lowest point. For the maximum acceleration of centre of mass, which is correct: (2002)
 - a. $h = R$
 - b. $h = 2R$
 - c. $h = 0$
 - d. No relation between h and R

95. A disc is rolling the velocity of its centre of mass is v_{cm} then which one will be correct: (2001)
- The velocity of highest point is $2v_{cm}$ and point of contact is zero
 - The velocity of highest point is v_{cm} and point of contact is v_{cm}
 - The velocity of highest point is $2v_{cm}$ and point of contact is v_{cm}
 - The velocity of highest point is $2v_{cm}$ and point of contact is $2v_{cm}$
96. For a hollow cylinder & a solid cylinder rolling without slipping on an inclined plane, then which of these reaches earlier on the ground: (2000)
- Solid cylinder
 - Hollow cylinder
 - Both simultaneously
 - Can't say anything

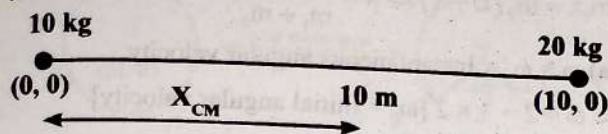
97. A solid sphere, disc and solid cylinder all of the same mass and made of the same material are allowed to roll down (from rest) on the inclined plane, then: (1993)
- Solid sphere reaches the bottom first
 - Solid sphere reaches the bottom last
 - Disc will reach the bottom first
 - All reach the bottom at the same time
98. The speed of a homogenous solid sphere after rolling down an inclined plane of vertical height h from rest without sliding is: (1992)
- $\sqrt{\frac{10}{7}gh}$
 - $\sqrt{gh}$
 - $\sqrt{\frac{6}{5}gh}$
 - $\sqrt{\frac{4}{3}gh}$
99. If a sphere is rolling, the ratio of the translational energy to total kinetic energy is given by: (1991)
- 7 : 10
 - 2 : 5
 - 10 : 7
 - 5 : 7

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	b	b	d	b	a	c	b	a	c	b	a	b	d	c	d	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
a	a	a	a	c	d	c	c	d	b	c	b	a	d	c	c	d
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
b	b	c	a	b	a	d	d	d	b	c	c	a	c	b	a	d
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
d	b	a	b	b	c	b	d	c	a	a	c	b	a	a	d	d
69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
d	a	b	b	b	b	c	c	a	d	c	d	a	b	a	b	a
86	87	88	89	90	91	92	93	94	95	96	97	98	99			
b	a	a	b	d	a	c	b	d	a	a	a	a	d			

Explanations

1. (c)

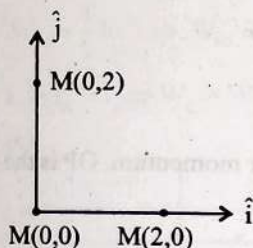


$$\therefore X_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \Rightarrow X_{CM} = \frac{20 \times 10}{20 + 10} = \frac{20}{3} \text{ m}$$

2. (b) Here, we have two masses let $m_1 = 5 \text{ kg}$, $m_2 = 10 \text{ kg}$. For two bodies system

$$\text{By using } x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$= \frac{5 \times 0 + 100 \times 10}{5 + 10} = \frac{200}{3} = 66.66 \text{ cm}$$

3. (b) We have three identical spheres, each of mass M 

$$\text{Using } X_{cm} = \frac{M_1 X_1 + M_2 X_2 + M_3 X_3}{M_1 + M_2 + M_3}$$

$$\text{Similarly, } Y_{cm} = \frac{M_1 Y_1 + M_2 Y_2 + M_3 Y_3}{M_1 + M_2 + M_3}$$

$$X_{cm} = \frac{M \times 0 + M \times 2 + M \times 0}{3M} = \frac{2}{3}$$

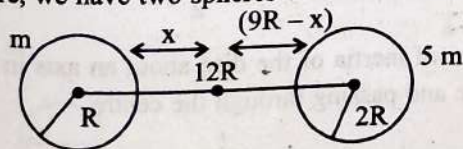
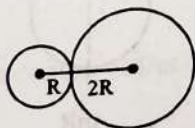
$$Y_{cm} = \frac{M \times 0 + M \times 2 + M \times 0}{3M} = \frac{2}{3}$$

$$\text{Position vector} = \frac{2}{3}\hat{i} + \frac{2}{3}\hat{j} = \frac{2}{3}(\hat{i} + \hat{j})$$

4. (d) Centre of mass may lie on centre of gravity net torque of gravitational pull is zero about centre of mass.

$$\text{Mechanical advantage} = \frac{\text{Load}}{\text{Effort}} > 1$$

Load > Effort

5. (b) Here, we have two spheres of mass m and $5m$ Initial distance between their centers = $12R$ At time of collision the distance between their centers = $3R$ So total distance travelled by both = $12R - 3R = 9R$

Since the bodies move under mutual forces, center of mass will remain stationary so

$$m_1 x_1 = m_2 x_2 \Rightarrow mx = 5m(9R - x)$$

$$x = 45R - 5x \Rightarrow 6x = 45R$$

$$x = \frac{45}{6}R = 7.5R$$

6. (a) Distance of centre of mass from the origin.

$$\text{Hence, } Y_{cm} = 0$$

$$\begin{aligned} x_{cm} &= \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} \\ &= \frac{300(0) + 500(40) + 400(70)}{300 + 500 + 400} \\ &= \frac{20000 + 28000}{1200} = \frac{48000}{1200} = 40 \text{ cm} \end{aligned}$$

7. (c) Net external force is equal to zero so there is no displacement of center of mass of the system.

8. (b) Hence, $m_1 = 50 \text{ kg}$, $m_2 = 0.5 \text{ kg}$ From conservation of linear momentum $m_1 v_1 = m_2 v_2$

$$50 \times v_1 = 0.5 \times 2$$

$$v_1 = \frac{1}{50} \text{ m/s}$$

$$\text{Time taken by stone to reach the ground} = \frac{10}{2} = 5 \text{ sec.}$$

Distance covered by man in 5 sec.

$$\Rightarrow v_1 \times t = \frac{1}{50} \times 5 = 0.1 \text{ m}$$

Distance of the man above the floor = $10 + 0.1 = 10.1 \text{ m}$

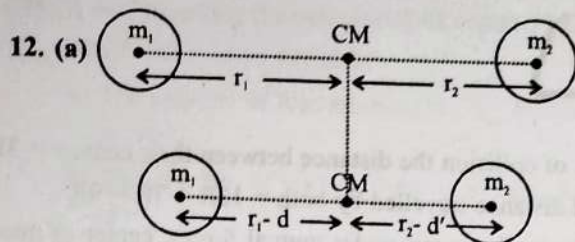
9. (a) Centre of gravity (C.G.) of a body is the point at which the weight of the body acts. The radius of gyration of any body rotating about an axis is the length of the perpendicular dropped from the C.G. of the body to the axis.

10. (c) Net external force on system is zero

$$\text{So, } \vec{v}_{cm} = \text{zero}$$

11. (b) Position of center of mass ($\vec{r}_{cm}$) = $\frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$

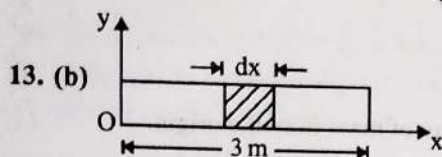
$$\begin{aligned} &= \frac{1(\hat{i} + 2\hat{j} + \hat{k}) + 3(-3\hat{i} - 2\hat{j} + \hat{k})}{1 + 3} \\ &= \frac{\hat{i} + 2\hat{j} + \hat{k} - 9\hat{i} - 6\hat{j} + 3\hat{k}}{4} = -2\hat{i} - \hat{j} + \hat{k} \end{aligned}$$



$$m_1 r_1 = m_2 r_2 \quad \dots(1)$$

$$m_1 (r_1 - d) = m_2 (r_2 - d') \quad \dots(2)$$

from Eqs. (1) and (2) We get $d' = \frac{m_1}{m_2} d$



Here $\rho = kx$ where k is a constant mass of small element of dx length is $dm = kx \cdot dx$

$$x_{cm} = \frac{\int x \cdot dm}{\int dm} = \frac{\int_0^3 (x^2 dx)}{\int_0^3 x \cdot dx} = \frac{\left[\frac{x^3}{3} \right]_0^3}{\left[\frac{x^2}{2} \right]_0^3} = \frac{\frac{27}{3}}{\frac{9}{2}} = 2 \text{ m}$$

14. (d) Their centre of mass does not depend upon the forces acting on the particles as the resultant of all forces, on any system of particles, is zero.

15. (c) $\omega = \omega_0 + \alpha t$

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$\text{where } \omega_0 = 1200 \text{ rpm} = \frac{1200}{60} \times 2\pi \text{ rad/s}$$

$$\text{and } \omega = 3120 \text{ rpm} = \frac{3120}{60} \times 2\pi \text{ rad/s}$$

$$\alpha = 4\pi \text{ rad/s}^2$$

16. (d) Initial angular speed of wheel, $\omega_0 = 2\pi f_0$

$$= 2\pi \times \frac{360}{60} \text{ rad/s} = 12\pi \text{ rad/s}$$

Final angular speed of wheel, $\omega = 2\pi f$

$$= 2\pi \times \frac{1200}{60} \text{ rad/s} = 40\pi \text{ rad/s.}$$

Hence, $t = 14$ sec, Now, $\omega = \omega_0 + \alpha t$

$$\Rightarrow \alpha = \frac{\omega - \omega_0}{t} = \frac{40\pi - 12\pi}{14} = \frac{28\pi}{14} = 2\pi \text{ rad/s}^2$$

17. (a) Particle at periphery will have both radial and tangential acceleration

$$a_t = R\alpha = 0.5 \times 2 = 1 \text{ ms}^{-2}$$

Final angular speed, $\omega = \omega_0 + \alpha t$

$$\omega = 0 + 2 \times 2 = 4 \text{ rad/sec}$$

$$a_c = \omega^2 R = (4)^2 \times 0.5 = 16 \times 0.5 = 8 \text{ ms}^{-2}$$

$$a_{\text{total}} = \sqrt{a_t^2 + a_c^2} = \sqrt{1^2 + 8^2} \approx 8 \text{ ms}^{-2}$$

18. (a) The position of point P on rod through which the axis should pass so that the work required to set the rod rotating with minimum angular velocity ω_0 is their center of mass is

$$m_1 x = m_2 (L - x) \Rightarrow x = \frac{m_2 L}{m_1 + m_2}$$

19. (a) $\omega = \omega_0$ = Instantaneous angular velocity

$$\Rightarrow \omega = 2 + 3 \times 2 \quad [\omega_0 = \text{Initial angular velocity}]$$

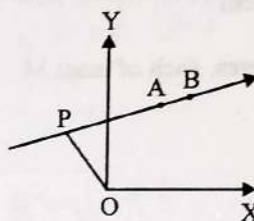
$$\omega = 2 + 6 = 8 \text{ rad/s.}$$

Angle through which it has

$$\text{Rotated } (\theta) = \frac{\omega^2 - \omega_0^2}{2\alpha}$$

$$= \frac{64 - 4}{2 \times 3} = \frac{60}{6} = 10 \text{ rad}$$

20. (a)



Moment of momentum is angular momentum. OP is the same whether the mass is at A or B.

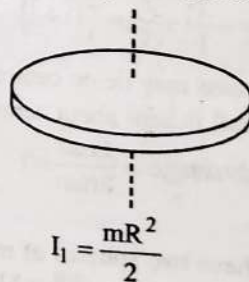
Therefore $L_A = L_B$.

$$21. (a) \vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$$

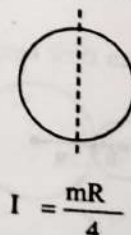
$$\hat{i}(-2-3) - \hat{j}(1-3) + \hat{k}(1+2)$$

$$= -5\hat{i} + 2\hat{j} + 3\hat{k}$$

22. (c) Moment of inertia of the disc about an axis perpendicular to the disc of passing through the centre.

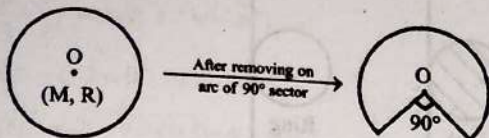


Moment of inertia of the disc about an axis in the plane of the disc and passing through the centre.



$$\Rightarrow \frac{k_1}{k_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{mR^2/2}{mR^2/4}} = \sqrt{2} : 1$$

23. (d)



Mass of new ring, $M' = M - \frac{M}{4} = \frac{3M}{4}$
 Radius, R

Required moment of Inertia, $I_R = M'R^2$

$$\Rightarrow I_R = \left(\frac{3M}{4}\right)R^2 \Rightarrow I_R = \frac{3}{4}MR^2$$

So, value of 'K' is -.

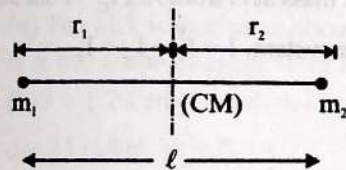
Hence, correct option is (d).

24. (c) From work energy theorem, to stop the system

$$W_{\text{ext}} = \frac{1}{2}I\omega^2 \Rightarrow W_{\text{ext}} \propto I$$

$$I_C > I_B > I_A \Rightarrow W_C > W_B > W_A$$

25. (c)



$$r_1 = \frac{m_2 \ell}{m_1 + m_2}, r_2 = \frac{m_1 \ell}{m_1 + m_2}$$

$$I_{\text{cm}} = m_1 r_1^2 + m_2 r_2^2 = \frac{m_1 m_2}{m_1 + m_2} \ell^2$$

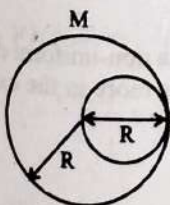
26. (d) $K.E_{\text{rotation}} = \frac{1}{2}I\omega^2$

$$E_{\text{sphere}} = \frac{1}{2}I_s \omega^2 = \frac{1}{2} \times \frac{2}{5}MR^2 \times \omega^2$$

$$E_{\text{cylinder}} = \frac{1}{2}I_c (2\omega)^2 = \frac{1}{2} \times \frac{MR^2}{2} \times 4\omega^2$$

$$\frac{E_{\text{sphere}}}{E_{\text{cylinder}}} = \frac{1}{5}$$

27. (b)



Mass per unit area of disc = $M/\pi R^2$

$$\text{Mass of removed portion of disc} = \frac{M}{\pi R^2} \times \pi \left(\frac{R}{2}\right)^2 = \frac{M}{4}$$

Moment of inertia of removed portion about an axis passing through centre of disc.

$$I'_0 = \frac{1}{2} \times \frac{M}{4} \times \left(\frac{R}{2}\right)^2 + \frac{M}{4} \times \left(\frac{R}{2}\right)^2$$

$$I'_0 = \frac{3MR^2}{32}$$

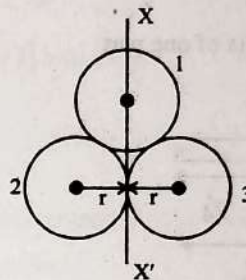
Moment of inertia of disc with removed portion is

$$I = I_0 - I'_0$$

$$I = \frac{1}{2}MR^2 - \frac{3MR^2}{32}$$

$$I = \frac{13MR^2}{32}$$

28. (c)



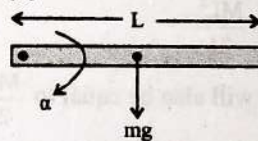
$$I_{xx'} = I_1 + I_2 + I_3$$

$$\frac{2}{3}mr^2 + \left(\frac{2}{3}mr^2 + mr^2\right) + \left(\frac{2}{3}mr^2 + mr^2\right)$$

(Using parallel axis theorem)

$$\Rightarrow I_{xx'} = 2mr^2 + 2mr^2 = 4mr^2$$

29. (b)



$$\tau = I\alpha \Rightarrow mg \left(\frac{L}{2}\right) = \left(\frac{mL^2}{3}\right)\alpha \Rightarrow \alpha = \frac{3g}{2L}$$

30. (a) Moment of inertia $I \propto M \propto R^2$

MOI of the disc

About C = $I_A + I_C$

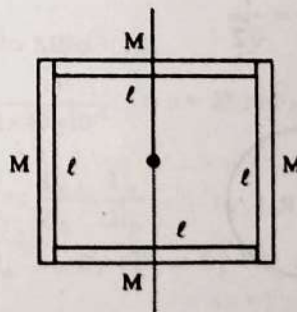
About B = $I_A + I_B$ maximum

About D = $I_A + I_D$ minimum

$R_B > R_C > R_D \Rightarrow I_B > I_C > I_D$

$\therefore$ MOI is maximum about B.

31. (d)

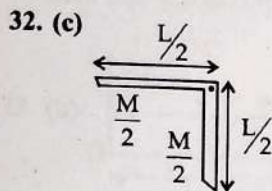


Let us find the moment of inertia of one rod and then multiply it by 4

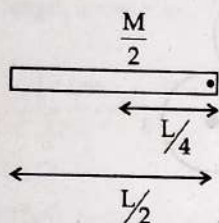
$$\begin{aligned}\text{Moment of Inertia of one rod} &= \frac{M\ell^2}{12} + M\left(\frac{\ell}{2}\right)^2 \\ &= \frac{M\ell^2}{12} + \frac{M\ell^2}{4} = \frac{M\ell^2}{3}\end{aligned}$$

So, the moment of inertia of four rods

$$= \frac{4M\ell^2}{3}$$



Moment of Inertia of one part



$$= \frac{ML^2}{3}$$

Here, $m = \frac{M}{2}$

$$L = \frac{L}{2}$$

$$I_1 = \frac{M}{2} \times \left(\frac{L}{2}\right)^2 \times \frac{1}{3} = \frac{M}{2} \times \frac{L^2}{4} \times \frac{1}{3} \Rightarrow \frac{ML^2}{24}$$

Moment of Inertia of other part will also be equal to $\frac{ML^2}{24}$

So, moment of Inertia of bent rod

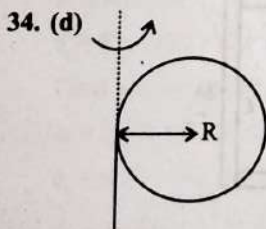
$$= I_1 + I_2 = \frac{ML^2}{24} + \frac{ML^2}{24} = \frac{ML^2}{12}$$

33. (c) For circular disc, for circular ring,

$$MK_1^2 = \frac{MR^2}{2} \Rightarrow K_1 = \frac{R}{\sqrt{2}}$$

$$\Rightarrow MK_2^2 = MR^2 \Rightarrow K_2 = R$$

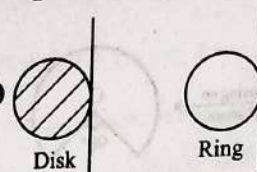
$$\text{So, } \frac{K_1}{K_2} = \frac{\frac{R}{\sqrt{2}}}{R} = \frac{1}{\sqrt{2}}$$



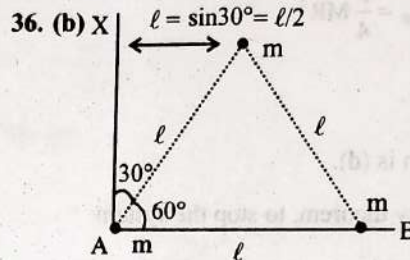
According to parallel axis theorem,

$$I = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

35. (b)



$$\begin{aligned}I_{\text{disk}} &= \frac{5}{4}MR^2 = MK_{\text{disk}}^2 \\ I_{\text{ring}} &= \frac{3}{2}MR^2 = MK_{\text{ring}}^2\end{aligned} \Rightarrow \frac{K_{\text{disk}}}{K_{\text{ring}}} = \sqrt{\frac{5}{6}}$$



Perpendicular distance of mass at A from AX = $r_A = 0$

Perpendicular distance of mass at B from AX = $r_B = l$

Perpendicular distance of mass at C from AX $r_C = l \sin 30 = l/2$

Moment of inertia of the system, $I = I_A + I_B + I_C$

$$I = m_A r_A^2 + m_B r_B^2 + m_C r_C^2$$

$$I = ml^2 + ml^2 \times \frac{1}{4}$$

$$I = \frac{5}{4}ml^2$$

37. (c) T.K.E. = $\frac{1}{2}mv^2(1 + K^2/R^2)$;

$$\text{R.K.E.} = \frac{1}{2}mv^2(K^2/R^2)$$

$$\frac{\text{R.K.E.}}{\text{T.K.E.}} = \frac{\frac{1}{2}mv^2(K^2/R^2)}{\frac{1}{2}mv^2(1 + K^2/R^2)} = \frac{K^2}{K^2 + R^2}$$

38. (a) A disc is composed of rings and moment of inertia of disc can be found by using integral of moment of inertia of concentric elemental rings. Moment of inertia of continuous body is found by using $I = \int R^{2dm}$

For increasing the moment of inertia of a non-uniform disc, it is hence desired that the mass density is more in the exterior part of the disc.

39. (b) For triangular lamina

Longest side $\rightarrow I_{\min}$

Smallest side $\rightarrow I_{\max}$

Therefore $I_2 > I_1 > I_3$

40. (a) According to law of conservation of energy

$$\frac{1}{2} I \omega^2 = mgh$$

$$\frac{1}{2} \frac{m \ell^2}{3 \ell^2} V^2 = mgh \quad (\because v = \omega \ell)$$

$$\text{Here } h = \ell / 2 = 0.3 \text{ m}$$

$$v = \sqrt{6gh} = \sqrt{6 \times 9.8 \times 0.3} = 4.2 \text{ m/s}$$

41. (d) The intersection of medians is the centre of mass of the triangle. Since the distances of centre of mass from the sides is related $x_{BC} < x_{AB} < x_{AC}$.

$$\text{Therefore } I_{BC} > I_{AB} > I_{AC} \text{ or } I_{BC} > I_{AB}.$$

42. (d) Linear K.E. of ball $= \frac{1}{2} mv^2$ and rotational K.E. of ball

$$= \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{2}{5} mr^2 \right) \omega^2 = \frac{1}{5} mv^2.$$

$$\text{Therefore total K.E.} = \frac{1}{2} mv^2 + \frac{1}{5} mv^2 = \frac{7}{10} mv^2.$$

And ratio of rotational K.E. and total K.E.

$$= \frac{(1/5)mv^2}{(7/10)mv^2} = \frac{2}{7}.$$

43. (d) The M.I. is minimum about EG because mass distribution is at minimum distance from EG.

44. (b) $I = 1.2 \text{ kgm}^2$, rotational kinetic energy, $K.E_R = 1500 \text{ J}$,

$$\alpha = 25 \text{ rad/s}^2, \omega_1 = 0, t = ?$$

$$\text{As rotational kinetic energy, } K.E_R = \frac{1}{2} I \omega^2, \omega = \sqrt{\frac{2K.E_R}{I}}$$

$$\omega = \sqrt{\frac{2 \times 1500}{1.2}} = 50 \text{ rad/sec}$$

$$\omega_2 = \omega_1 + \alpha t.$$

$$50 = 0 + 25t, \text{ or } t = 2 \text{ s.}$$

45. (c) M.I. of uniform circular disc about diameter $= I$

According to theorem of perpendicular axis.

$$\text{M.I. of disc about axis} = 2I = \frac{1}{2} mr^2$$

Applying theorem of parallel axes

Moment of inertia of disc about the given axis

$$= 2I + mr^2 = 2I + 4I = 6I.$$

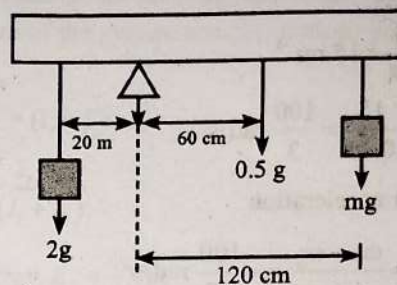
46. (c) $K.E. = \frac{1}{2} I \omega^2$

$$I = \frac{2K.E.}{\omega^2} = \frac{2 \times 360}{30 \times 30} = 0.8 \text{ kgm}^2$$

47. (a) Kinetic energy $= \frac{1}{2} I \omega^2$, and for ring $I = mr^2$

$$\therefore KE = \frac{1}{2} mr^2 \omega^2$$

48. (c)



For balanced torque, net torque on the system will be zero

$$2g \times 20 = 0.5g \times 60 + mg \times 120 \Rightarrow 40 = 0.5 \times 60 + 120m$$

$$\Rightarrow 40 - 30 = 120m \Rightarrow 10 = 120m$$

$$\Rightarrow 1 = 12m \Rightarrow m = \frac{1}{12} \text{ kg}$$

49. (b) $\vec{F} = 3\hat{j}\text{N}, \vec{r} = 2\hat{k}$

$$\vec{\tau} = \vec{r} \times \vec{F} = 2\hat{k} \times 3\hat{j} = 6(\hat{k} \times \hat{j})$$

$$= 6(-\hat{i})$$

$$\vec{\tau} = -6\hat{i} \text{ Nm}$$

50. (a) angular displacement, $\theta = 2\pi$ revolution

$$\theta = 2\pi \times 2\pi = 4\pi^2 \text{ rad}$$

$$\omega_1 = 2\pi f = 2\pi \times \frac{3}{60} \text{ rad/s} \quad (\because f = \frac{3}{60} \text{ s}^{-1})$$

$$\omega_f = 0$$

Work energy theorem.

$$W = \frac{1}{2} I (\omega_f^2 - \omega_1^2) \Rightarrow -\tau \theta = \frac{1}{2} \times \frac{1}{2} mr^2 (0^2 - \omega_1^2)$$

$$\Rightarrow -\tau = \frac{\frac{1}{2} \times \frac{1}{2} \times 2 (4 \times 10^{-2})^2 \left(-3 \times \frac{2\pi}{60} \right)^2}{4\pi^2} \Rightarrow \tau = 2 \times 10^{-6} \text{ Nm}$$

51. (d) As $\tau_{\text{ext}} = 0$ Angular momentum will remain conserved.

52. (d) Torque, $\vec{\tau} = \vec{r} \times \vec{F}$ [$\vec{r}$ is position vector and $\vec{F}$ is force]

$$\vec{\tau} = (0\hat{i} + 2\hat{j} - \hat{k}) \times (4\hat{i} + 5\hat{j} - 6\hat{k})$$

$$\vec{\tau} = -7\hat{i} - 4\hat{j} - 8\hat{k}$$

53. (b) Hollow cylinder is equivalent to ring

$$I_{\text{ring}} = MR^2$$

$$\tau = I\alpha$$

$$\tau = FR$$

$$I\alpha = FR$$

$$MR^2\alpha = FR \Rightarrow MR\alpha = F$$

$$\alpha = \frac{F}{MR} \Rightarrow \frac{30}{3 \times 40 \times 10^{-2}} \Rightarrow \alpha = 25 \text{ rad/s}^2$$

54. (a) $K_A = K_B \Rightarrow \frac{L_A^2}{2I_A} = \frac{L_B^2}{2I_B}$

$$\text{As } I_B > I_A \text{ So, } L_A^2 < L_B^2 \Rightarrow L_A < L_B \Rightarrow L_B > L_A$$

55. (b) Velocity of the automobile

$$v = 54 \times \frac{5}{18} = 15 \text{ ms}^{-1}$$

$$\omega_0 = \frac{v}{R} = \frac{15}{0.45} = \frac{100}{3} \text{ rad/s}$$

So angular acceleration

$$\alpha = \frac{\Delta\omega}{t} = \frac{\omega_f - \omega_0}{t} = -\frac{100}{45} \text{ rad/s}^2$$

$$\tau = I\alpha = 3 \times \frac{100}{45} = 6.66 \text{ kgm}^2/\text{s}^2$$

56. (b) Since, net torque is zero, so angular momentum is conserved

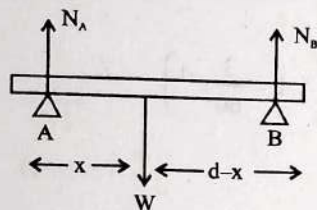
$$\vec{\tau} = \vec{r} \times \vec{F} = 0$$

$$(-12 + 12\alpha)\hat{j} + (6 + 6\alpha)\hat{k} = 0$$

$$\Rightarrow \alpha = -1$$

57. (c) By torque balancing about B

$$N_A(d) = W(d-x) \Rightarrow N_A = \frac{W(d-x)}{d}$$



58. (b) Angular momentum remains constant because of the torque of tension is zero.

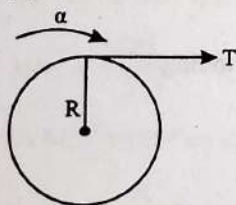
So, initial angular momentum = final angular momentum

$$\Rightarrow L_i = L_f \Rightarrow mv_0 R = mv \frac{R}{2}$$

$$\Rightarrow v = 2v_0$$

$$KE_f = \frac{1}{2} m (2v_0)^2 = 2mv_0^2$$

59. (d)



Here, mass of the cylinder, $M = 50 \text{ kg}$

Radius of the cylinder, $R = 0.5 \text{ m}$

Angular acceleration, $\alpha = 2 \text{ rev/s}^2$

$$= 2 \times 2\pi \text{ rad s}^{-2} = 4\pi \text{ rad s}^{-2}$$

Torque $\tau = TR$

Moment of inertia of the solid cylinder about its axis,

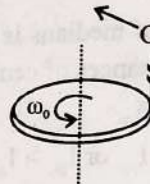
$$I = \frac{1}{2} MR^2$$

$\therefore$ Angular acceleration of the cylinder

$$\alpha = \frac{\tau}{I} = \frac{TR}{\frac{1}{2} MR^2}$$

$$T = \frac{MR\alpha}{2} = \frac{50 \times 0.5 \times 4\pi}{2} = 157 \text{ N}$$

60. (c) $v = 1 \text{ m/s}$ (w.r.t. ground)



Moment inertia of platform = 200 kgm^2 ,

Moment inertia of man = $mR^2 = 200 \text{ kgm}^2$

For system (platform + man) by using C.O.A.M. (Conservation of angular momentum)

$$I_p \omega_0 = mvR \Rightarrow \omega_0 = \frac{50 \times 1 \times 2}{200} = \frac{1}{2} \text{ rad/s}$$

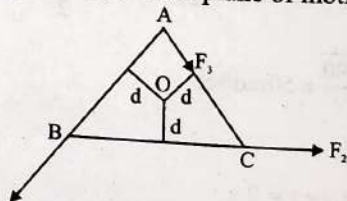
Angular velocity of man w.r.t. Platform

$$= \frac{v}{R} + \omega_0 = \frac{1}{2} + \frac{1}{2} = 1 \text{ rad/s}$$

Time taken by the man to complete one revolution

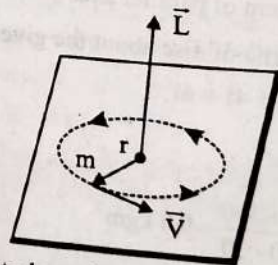
$$= \frac{2\pi \text{ rad}}{1 \text{ rad/s}} = 2\pi \text{ s}$$

61. (a) Angular momentum will be along the axis of rotation which is perpendicular to the plane of motion.



$$\Sigma \tau = 0 \Rightarrow (F_1 + F_2)d = F_3 d \Rightarrow F_3 = F_1 + F_2$$

62. (a) When a mass is rotating in a plane about a fixed point its angular momentum is directed along a line perpendicular to the plane of rotation.



63. (c) According to law of conservation of angular momentum,

$$L_i = L_f$$

$$\Rightarrow mvr = mv'r'$$

Increasing tension in string reduced the radius by a factor of 2

$$\Rightarrow vr = v' \frac{r}{2} \Rightarrow v' = 2v$$

$$\therefore \frac{K}{K'} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}m(v')^2} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}m(2v)^2} = \frac{1}{4}$$

So, $K' = 4K$

Hence, kinetic energy of the mass will increase by a factor of 4

64. (b) According to the theorem of parallel axes, the moment of inertia of the thin rod of mass M and length L about an axis passing through one of the ends is

$$I = I_{CM} + Md^2$$

Where I_{CM} is moment of inertia of the given rod about an axis passing through its centre of mass and perpendicular to its length and d is the distance between two parallel axes.

$$\text{Here } I_{CM} = I_0, d = \frac{L}{2}$$

$$\therefore I = I_0 + M\left(\frac{L}{2}\right)^2 = I_0 + \frac{ML^2}{4}$$

65. (a) $\theta(t) = 2t^3 - 6t^2$

Torque on wheel will be zero, if the angular acceleration of wheel will be zero

$$\alpha = \frac{d^2\theta}{dt^2} = \frac{d}{dt}\left[\frac{d\theta}{dt}\right]$$

$$= \frac{d}{dt}[6t^2 - 12t]$$

$$\alpha = 12t - 12$$

$$\text{So, } \alpha = 0 \Rightarrow 12t - 12 = 0$$

$$t = 1 \text{ s}$$

66. (a) Moment of Inertia of the complete disc about an axis passing through its centre O and perpendicular to its plane I_1

$$= \frac{9}{2}MR^2$$

Now, moment of inertia of the disc with removed portion, I_2

$$= \frac{1}{18}MR^2$$

Therefore, moment of inertia of remaining portion of disc

$$\text{about } O \text{ is } I = I_1 - I_2 = \frac{9MR^2}{2} - \frac{MR^2}{18} = \frac{40MR^2}{9}$$

67. (d) By conservation of angular momentum $\therefore \tau_{\text{ext}} = 0$

$$Mr^2\omega_i = (M + 2m)r^2\omega_f$$

$$\omega_f = \frac{M\omega}{(M + 2m)}$$

68. (d) As no external torque is applied to the system, the angular momentum of the system remains conserved.

$$\therefore L_i = L_f$$

$$\text{Here, } I_i\omega_i = (I_f + I_b)\omega_f$$

$$\text{or } \omega_f = \frac{I_i\omega_i}{(I_f + I_b)} \quad \dots(i)$$

$$\text{Initial energy, } E_f = \frac{1}{2}I_i\omega_i^2 \quad \dots(ii)$$

$$\text{Final energy, } E_f = \frac{1}{2}(I_f + I_b)\omega_f^2 \quad \dots(iii)$$

Substituting the value of ω_f from equation (i) in equation (iii), we get

$$\begin{aligned} \text{Final energy, } E_f &= \frac{1}{2}(I_f + I_b)\left(\frac{I_i\omega_i}{I_f + I_b}\right)^2 \\ &= \frac{1}{2} \frac{I_i^2\omega_i^2}{(I_f + I_b)} \end{aligned} \quad \dots(iv)$$

Loss of energy, $\Delta E = E_i - E_f$ (Using Eqs. (ii) and (iv))

$$\begin{aligned} &= \frac{1}{2}I_i\omega_i^2 - \frac{1}{2} \frac{I_i^2\omega_i^2}{(I_f + I_b)} \\ &= \frac{\omega_i^2}{2} \left(I_i - \frac{I_i^2}{(I_f + I_b)} \right) = \frac{\omega_i^2}{2} \left(\frac{I_i^2 + I_bI_i - I_i^2}{(I_f + I_b)} \right) \\ &= \frac{1}{2} \frac{I_bI_i}{(I_f + I_b)} \omega_i^2 \end{aligned}$$

69. (d) Since, $\vec{\tau} = \vec{r} \times \vec{F}$

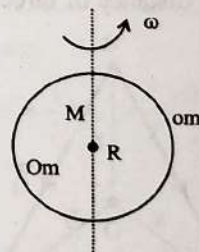
$\vec{\tau}$ is perpendicular to both $\vec{r}$ and $\vec{F}$

$$\text{So, } \vec{r} \cdot \vec{\tau} = 0 \Rightarrow \vec{F} \cdot \vec{\tau} = 0$$

70. (a) External torque, $\tau_{\text{ext}} = 0$

By conservation of angular momentum (L)

$$L_i = L_f$$



$$\Rightarrow MR^2\omega = (MR^2 + 2mR^2)\omega_1$$

$$\Rightarrow M\omega = (M + 2m)\omega_1 \Rightarrow \omega_1 = \frac{M\omega}{M + 2m}$$

[Let ω_1 be new angular velocity]

71. (b) Total angular momentum of system initially

$$L_i = I_1\omega + I_2(0) = I_1\omega$$

Total angular momentum of the system finally, $L_f = (I_1 + I_2)\omega_2$

According to conservation of linear momentum i.e. $L_i = L_f$

$$\Rightarrow I_1\omega = (I_1 + I_2)\omega_2 \Rightarrow \omega_2 = \frac{I_1\omega}{I_1 + I_2}$$